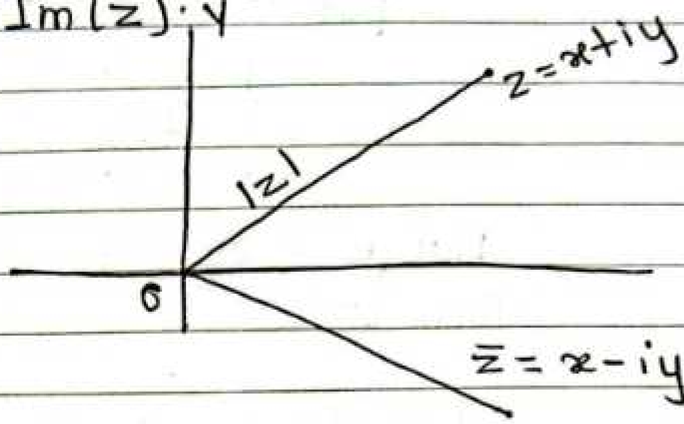
Unit 1
Complex Analysis

Complex numbers:

A number of the form $x+iy$ where x and y are real numbers and $i=\sqrt{-1}$, is called a complex number denoted by z i.e. $z=x+iy$, x is called the real part of z and is denoted by $Re(z)$ and y is called the imaginary part of z and is denoted by $Im(z)$.



$\bar{z}=x-iy$ be the complex conjugate of $z=x+iy$

The modulus of z is denoted by $|z|$ and is defined by $|z|=\sqrt{x^2+y^2}$

\# Every complex number can always be expressed in the polar form

$$\text{Let } z=x+iy$$
$$\text{put } x=r\cos\theta$$
$$y=r\sin\theta$$
$$\text{i.e. } z=x+iy$$
$$=r(\cos\theta+i\sin\theta)$$
$$\text{where } r=\sqrt{x^2+y^2}$$
$$\theta=\tan^{-1}\left(\frac{y}{x}\right)$$

is also a complex in polar form.

$$z^n = \{r(\cos\theta + i\sin\theta)\}^n$$
$$z^n = r^n(\cos n\theta + i\sin n\theta)$$

The angle θ is called the amplitude or argument $z = x + iy$ and is written as $amp(z)$ or $arg(z)$ $= \tan^{-1}\left(\frac{y}{x}\right)$.

Every complex number can always be expressed as in the exponential form.

Let $z = x + iy$

put $x = r\cos\theta$

$y = r\sin\theta$

i.e. $z = x + iy$

$= r(\cos\theta + i\sin\theta)$

But $e^{i\theta} = \cos\theta + i\sin\theta$

then $\boxed{z = re^{i\theta}}$

Powers and roots of complex numbers:

Let $z = re^{i\theta}$ be non zero complex number, the power of the complex number is given by $z^n = r^n e^{in\theta}$

By De Moivre's formula

$$\boxed{z^n = r^n(\cos n\theta + i\sin n\theta)}$$

$i^2 = -1$

Exercise

Express the following in $x+iy$, modulus and amplitude form.

i) $\dfrac{2-i\sqrt{3}}{1+i}$

Soln.

Let $z = \dfrac{2-i\sqrt{3}}{1+i}$

$$= \frac{(2-i\sqrt{3})}{1+i} \times \frac{(1-i)}{1-i}$$

$$= \frac{2-2i-i\sqrt{3}+(-1)\sqrt{3}}{1-(-1)}$$

$$= \frac{(2-\sqrt{3})-i(2+\sqrt{3})}{2}$$

$$x+iy = \frac{2-\sqrt{3}}{2} + \left\{-\frac{(2+\sqrt{3})}{2}\right\}i$$

Equating real & imaginary parts.

$$x = \frac{2-\sqrt{3}}{2}$$

$$y = -\frac{(2+\sqrt{3})}{2}$$

$$|z| = \sqrt{x^2+y^2}$$

$$= \sqrt{\left(\frac{2-\sqrt{3}}{2}\right)^2 + \left(-\frac{(2+\sqrt{3})}{2}\right)^2}$$

$$= \sqrt{\frac{4-3}{4} + \frac{4+(\sqrt{3})^2}{4}}$$

$$= \sqrt{\frac{4-\cancel{3}+4+\cancel{3}}{4}}$$

$$= \sqrt{\frac{\cancel{8}\,2}{\cancel{4}\,1}} \quad = \sqrt{2}$$

$$e^{x} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\cdots$$

$$e^{-x} = 1 + \frac{(-x)}{1!} + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!} + \cdots$$

$$= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\cdots$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{or, } \theta = \tan^{-1}\left(\frac{-\left(\frac{2+\sqrt{3}}{2}\right)}{\frac{2-\sqrt{3}}{2}}\right)$$

$$\text{or, } \theta = \tan^{-1}\left(-\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right)\right)$$

$$\text{or, } \theta = -\tan^{-1}\left(\frac{2+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}\right)$$

$$\text{or, } \theta = -\tan^{-1}\left(\frac{\cancel{2+\sqrt{3}}\ 4+3}{4-3}\right)$$

$$\therefore \theta = -\tan^{-1}(7)$$

Exponential and Circular functions of complex variables:

$$z = x + iy$$

$$e^{z} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots\cdots$$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\cdots$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots\cdots$$

Now,

$$\cos z + i \sin z$$

$$= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots + i\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\cdots\right)$$

$$= 1 + \frac{iz}{1!} + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \cdots\cdots$$

$$= e^{iz}$$

$$\boxed{\therefore e^{iz} = \cos z + \sin z} \quad \text{—— (1)}$$

$\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$

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Similarly, $\boxed{e^{-iz} = \cos z - i\sin z}$ —— ②

Adding ① and ②

$$2\cos z = e^{iz} + e^{-iz}$$

$$\boxed{\therefore \cos z = \frac{e^{iz}+e^{-iz}}{2}}$$

Subtracting ① from ②

$$\boxed{\therefore \sin z = \frac{e^{iz}-e^{-iz}}{2i}}$$

We have,

$$\boxed{\cosh z = \frac{e^{z}+e^{-z}}{2}}$$

$$\boxed{\sinh z = \frac{e^{z}-e^{-z}}{2}}$$

Relation between complex trigonometric and hyperbolic functions.

We have,

$$\sin z = \frac{e^{iz}-e^{-iz}}{2i}$$

replacing z by iz

$$\sin iz = \frac{e^{i(iz)}-e^{-i(iz)}}{2i}$$

$$\sin iz = \frac{e^{-z}-e^{z}}{2i}$$

$$= -\frac{e^{z}-e^{-z}}{2i}$$

$$= i\left(\frac{e^{z}-e^{-z}}{2}\right)$$

$$\boxed{\sin iz = i\sinh z}$$

$i^2 = -1$

$1 = -i^2$

$\frac{1}{i} = \frac{-i^2}{i}$

$\boxed{\frac{1}{i} = -i}$

→ $\cos(iz) = \cosh z$

→ $\sinh(iz) = i\sin z$

→ $\cosh(iz) = \cos z$

Complex function:

Let $\boxed{z = x+iy}$ → complex variable and $w = u(x,y) + iv(x,y)$ be two complex numbers in a certain region R of complex plane. Then a function f defined on R is a rule that assigns to every z in R a complex number w, called the value of f at z

That is $f(z) = w$

i.e. $\boxed{f(z) = u+iv}$ Complex function.

Exercise 2.2

Express the following function of z in the form $u+iv$

① $f(z) = \sin z$

Soln

Let $f(z) = u+iv$

or, $u+iv = \sin z$

$= \sin(x+iy)$

[$\sin(A+B) = \sin A\cos B + \cos A \sin B$]

$= \sin x \cos(iy) + \cos x \sin(iy)$

$= \sin x \cdot \cosh y + \cos x \cdot i\sinh y$

Equating the coefficient of like terms;

$\therefore u = \sin x \cosh y$

$v = \cos x \cdot \sinh y$

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(2) $f(z) = ze^z$

Soln

Let $f(z) = u+iv$

or, $u+iv = ze^z$

$= (x+iy)e^{x+iy}$

$= (x+iy)\, e^x . e^{iy}$ [$e^{i\theta} = \cos\theta + i\sin\theta$]

$= e^x [(x+iy)(\cos y + i\sin y)]$

$= e^x \{x\cos y + ix\sin y + iy\cos y - y\sin y\}$

$= e^x \{(x\cos y - y\sin y) + i(x\sin y + y\cos y)\}$

$= e^x (x\cos y - y\sin y) + i\, e^x (x\sin y + y\cos y)$

Equating the value of real & imaginary part,

$\therefore u = e^x (x\cos y - y\sin y)$

$v = e^x (x\sin y + y\cos y)$

(3) $f(z) = e^{z^2}$

Soln

Let $f(z) = u+iv$

$u+iv = e^{z^2}$

$= e^{(x+iy)^2}$

$= e^{(x^2 + 2xiy - y^2)}$

$= e^{(x^2-y^2)} . e^{2xiy}$

$= e^{(x^2-y^2)} [(\cos 2xy + i\sin 2xy)]$

$= e^{(x^2-y^2)} \cos 2xy + e^{(x^2-y^2)} . i\sin 2xy$

Equating the value of real & imaginary part

$\therefore u = e^{(x^2-y^2)} \cos 2xy$

$v = e^{(x^2-y^2)} \sin 2xy$

Derivative:

Let $w = f(z)$ be a function of complex variable z defined in a domain of the region R contains a neighbourhood a point z. Then $f(z)$ is said to be derivative if the limit exists

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

written as $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

VIMP # Analytic functions:

A single-valued function of the complex variable is analytic or regular holomorphic in a certain domain, if it has a derivative at each point in the domain.

If the function $f(z)$ is differentiable at a single point, then it is not said to be analytic function.

Eg: ① $f(z) = |z|^2$ is differentiable only at $z = 0$, which is not analytic.

VVIMP • The Necessary conditions for $f(z)$ to be analytic (The Cauchy Riemann Equations)

If a $f(z) = u(x,y) + iv(x,y)$ is differentiable at any point $z = x + iy$, then the first order partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ must exist and satisfy $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ i.e. $u_x = v_y$

$f(x,y)$, $\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$

$f(x) : f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{x + \Delta x - x}$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{i.e. } u_y = -v_x$$

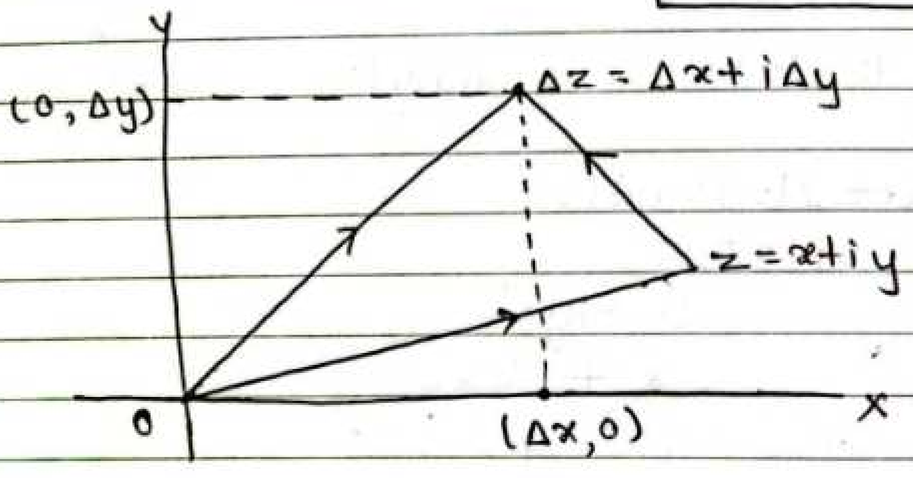
Proof:

Since $f(z) = u(x,y) + v(x,y)$ is differentiable at a point $z = x+iy$ then $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

provide the limit exist so that $\Delta z = \Delta x + i\Delta y$

$$f'(z) = \lim_{(\Delta x + i\Delta y)\to 0} \left[\frac{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y) - \{u(x,y) + iv(x,y)\}}{\Delta x + i\Delta y} \right]$$

$$f'(z) = \lim_{(\Delta x + i\Delta y)\to 0} \left[\left\{ \frac{u(x+\Delta x, y+\Delta y) - u(x,y)}{\Delta x + i\Delta y} \right\} + i\left\{ \frac{v(x+\Delta x, y+\Delta y) - v(x,y)}{\Delta x + i\Delta y} \right\} \right] \quad (1)$$



Taking Δz is purely real part so that $\Delta y = 0$, $\Delta x \to 0$

$$f'(z) = \lim_{\Delta x \to 0} \left[\frac{u(x+\Delta x, y) - u(x,y)}{\Delta x} + i \frac{\{v(x+\Delta x, y) - v(x,y)\}}{\Delta x} \right.$$

$f'(z) = u_x + iv_x$ —— (2)

and taking Δz is purely imaginary part so that $\Delta x = 0$, $\Delta y \to 0$ then eq^n (1) becomes.

$$f'(z) = \lim_{i\Delta y \to 0} \left[\frac{u(x, y+\Delta y) - u(x,y)}{i\Delta y} + \frac{\cancel{i}\{v(x, y+\Delta y) - v(x,y)\}}{\cancel{i}\Delta y} \right]$$

$$f'(z) = \frac{1}{i} u_y + v_y$$

$$f'(z) = v_y - iu_y \text{ —— (3)}$$

from (2) and (3)

$$u_x + iv_x = v_y - iu_y$$

Equating real & imaginary part,

$$\boxed{\therefore u_x = v_y}$$

$$v_x = -u_y$$

$$\boxed{\therefore u_y = -v_x}$$

Hence proved.

Que: Check $f(z) = z^2$ is analytic or not.

Sol^n:

Let $f(z) = u + iv$

or, $u + iv = z^2$

$= (x+iy)^2$

$= x^2 + i2xy - y^2$

$u + iv = (x^2 - y^2) + i2xy$

Equating real & imaginary part,

$u = x^2 - y^2$, $u_x = 2x$, $u_y = -2y$

$v = 2xy$, $v_x = 2y$, $v_y = 2x$

$\therefore u_x = v_y$ & $u_y = -2y$

$u_y = -v_x$

and hence $f(z) = z^2$ is analytic function.

$d(uv) = uv' + vu'$

Que: Check $f(z) = ze^z$ is analytic or not.

Sol^n

Let $f(z) = u+iv$

or, $u+iv = ze^z$

$= (x+iy)\, e^{(x+iy)}$

$= (x+iy)\, e^x \cdot e^{iy}$

$= e^x [(x+iy)(\cos y + i\sin y)]$

$= e^x \{x\cos y + ix\sin y + iy\cos y - y\sin y\}$

$= e^x \{(x\cos y - y\sin y) + i(x\sin y + y\cos y)\}$

$= e^x (x\cos y - y\sin y) + ie^x (x\sin y + y\cos y)$

Equating real & imaginary part,

$\therefore u = e^x (x\cos y - y\sin y)$,

$u_x = e^x (\cos y) + (x\cos y - y\sin y) e^x$

~~$u_y = e^x (x(-\sin y) - \cos y)) + (x\cos y - y\sin y) e^x$~~

$u_y = e^x \{-x\sin y - (1\sin y + y\cos y)\}$

~~$\therefore v = e^x (x\sin y + y\cos y)$,~~

$u_y = e^x (-y\cos y - x\sin y - \sin y)$

and

$v = e^x (x\sin y + y\cos y)$

$v_x = e^x (\sin y + 0) + e^x (x\sin y + y\cos y)$

$v_x = e^x (\sin y + x\sin y + y\cos y)$

$v_y = e^x (x\cos y + \cos y - y\sin y)$

$\boxed{\therefore u_x = v_y}$

$u_y = e^x (-y\cos y - x\sin y - \sin y)$

$= -e^x (y\cos y + x\sin y + \sin y)$

$\boxed{\therefore u_y = -v_x}$

$$\frac{\partial}{\partial x} f(x^2) = f'(x^2) \times 2x$$

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Proof Imp ✓

C-R eq^n in cartesian form

① $u_x = v_y$

② $u_y = -v_x$

Proof not necessary

C-R eq^n in polar form

We have,

$$f(z) = u + iv \quad \text{—— ①}$$

putting $z = re^{i\theta}$, then

$$f(re^{i\theta}) = u(r,\theta) + iv(r,\theta)$$

diff p.w.r to r & θ

$$f'(re^{i\theta}) * e^{i\theta} = \frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} \quad \text{—— ②}$$

and

$$f'(re^{i\theta}) * ire^{i\theta} = \frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta}$$

$$f'(re^{i\theta}) * e^{i\theta} = \frac{1}{ir}\frac{\partial u}{\partial \theta} + \frac{\cancel{i}}{\cancel{i}r}\frac{\partial v}{\partial \theta}$$

$$f'(re^{i\theta}) * e^{i\theta} = \frac{1}{r}\frac{\partial v}{\partial \theta} - \frac{1}{r} i \frac{\partial u}{\partial \theta} \quad \text{—— ③}$$

$$\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} - \frac{1}{r} i \frac{\partial u}{\partial \theta}$$

Equating real & img. part,

$$\boxed{\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}}$$

$$\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

$$\boxed{\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}}$$

Q. Show that $f(z) = z^6$ is analytic function.

Soln

Let $f(z) = u + iv$

$u + iv = z^6$

or, $u+iv = (re^{i\theta})^6$

or, $u+iv = r^6 e^{i6\theta}$

or, $u+iv = r^6 (\cos 6\theta + i \sin 6\theta)$

or, $u+iv = r^6 \cos 6\theta + i r^6 \sin 6\theta$

$u = r^6 \cos 6\theta$, $u_r = 6r^5 \cos 6\theta$, $u_\theta = -6r^6 \sin 6\theta$

$v = r^6 \sin 6\theta$, $v_r = 6r^5 \sin 6\theta$

$v_\theta = 6r^6 \cos 6\theta$

Now,

$$\frac{1}{r}\frac{\partial v}{\partial \theta} = \frac{1}{r}\{6r^6 \cos 6\theta\}$$

$$= 6r^5 \cos 6\theta$$

$$= u_r$$

$$-r\frac{\partial v}{\partial r} = -r\{6r^5 \sin 6\theta\}$$

$$= -6r^6 \sin 6\theta$$

$$= u_\theta$$

$$\boxed{\therefore \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}}$$

$$\boxed{\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}}$$

By C-R eqn, $f(z) = z^6$ is analytic.

Q. Show that $f(z) = z^n$ is analytic function.

<u>Soln</u>

Let $f(z) = u + iv$

$u + iv = z^n$

or, $u + iv = (re^{i\theta})^n$

or, $u + iv = r^n e^{in\theta}$

or, $u + iv = r^n(\cos n\theta + i\sin n\theta)$

or, $u + iv = r^n \cos n\theta + r^n \sin n\theta$

$u = r^n \cos n\theta$, $u_r = nr^{n-1}\cos n\theta$, $u_\theta = -nr^n \sin n\theta$

$v = r^n \sin n\theta$, $v_r = nr^{n-1}\sin n\theta$

$v_\theta = nr^n \cos n\theta$

Now,

$$\frac{1}{r}\frac{\partial v}{\partial \theta} = \frac{1}{r}\{nr^n \cos n\theta\}$$

$$= nr^{n-1}\cos n\theta$$

$$= u_r$$

$$-r\frac{\partial v}{\partial r} = -r\{nr^{n-1}\sin n\theta\}$$

$$= -nr^n \sin n\theta$$

$$= u_\theta$$

$$\boxed{\therefore \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}}$$

$$\boxed{\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}}$$

Hence proved

Q. Show that $f(z) = z + \frac{1}{z}$ is an analytic function.

Sol^n.

Let $f(z) = u + iv$

or, $u + iv = z + \frac{1}{z}$

$$= re^{i\theta} + \frac{1}{re^{i\theta}}$$

$$= re^{i\theta} + \frac{1}{r} e^{-i\theta}$$

$$= r(\cos\theta + i\sin\theta) + \frac{1}{r}(\cos\theta - i\sin\theta)$$

$$= \left(r + \frac{1}{r}\right)\cos\theta + i\left(r - \frac{1}{r}\right)\sin\theta$$

Equating real and imaginary part,

$u = \left(r + \frac{1}{r}\right)\cos\theta$, $u_r = \left(1 - \frac{1}{r^2}\right)\cos\theta$, $u_\theta = -\left(r + \frac{1}{r}\right)\sin\theta$

$v = \left(r - \frac{1}{r}\right)\sin\theta$

$v_r = \left(1 + \frac{1}{r^2}\right)\sin\theta$, $v_\theta = \left(r - \frac{1}{r}\right)\cos\theta$

By C-R eq^n,

$$u_r = \frac{1}{r} v_\theta$$

$$u_\theta = -r v_r$$

Now,

$$\frac{1}{r} v_\theta = \frac{1}{r}\left(r - \frac{1}{r}\right)\cos\theta$$

$$= \left(1 - \frac{1}{r^2}\right)\cos\theta = u_r$$

$$-rV_r = -r\left(1+\frac{1}{r^2}\right)\sin\theta$$

$$= -\left(r+\frac{1}{r}\right)\sin\theta$$

$$= u_\theta$$

$\therefore$ By C-R eq^n, $z+\frac{1}{z}$ is an analytic function.

Q. Show that $f(z) = \frac{1}{z^4}$, except $z=0$ is an analytic function.

Sol^n:

Let $f(z) = u+iv$

or, $u+iv = \frac{1}{z^4}$

$= z^{-4}$

$= (re^{i\theta})^{-4}$

$= \frac{1}{r^4} e^{-i4\theta}$

or, $u+iv = \frac{1}{r^4}(\cos 4\theta - i\sin 4\theta)$

or, $u+iv = \frac{1}{r^4}\cos 4\theta - i\frac{1}{r^4}\sin 4\theta$

Equating real & imaginary part,

$u = \frac{1}{r^4}\cos 4\theta$

$V = -\frac{1}{r^4}\sin 4\theta$

$u_r = -4\cdot\frac{1}{r^5}\cos 4\theta$, $u_\theta = -\frac{1}{r^4}\,4\sin 4\theta$

$V_r = +4\cdot\frac{1}{r^5}\sin 4\theta$, $V_\theta = -\frac{1}{r^4}\,4\cos 4\theta$

Q $f(z) = \frac{1}{z^n}$

By C-R eq^n,

$$u_r = \frac{1}{r} v_\theta \, , \quad u_\theta = -r v_r$$

$$\frac{1}{r} v_\theta = \frac{1}{r}\left(-\frac{1}{r^4} 4\cos 4\theta\right) = -\frac{4}{r^5}\cos 4\theta = u_r$$

$$-r v_r = -r\left(4 \cdot \frac{1}{r^5}\sin 4\theta\right) = \frac{-1}{r^4} 4\sin 4\theta = u_\theta$$

$\therefore$ By C-R eq^n, $\frac{1}{z^4}$ is an analytic function.

Q. Show that $f(z) = z|z|$ is not analytic function.

Sol^n:

Let $f(z) = u + iv$

or, $u + iv = z|z|$

$$= (x+iy)\sqrt{x^2+y^2}$$

$$= x \cdot \sqrt{x^2+y^2} + iy\sqrt{x^2+y^2}$$

Equating real & imaginary part,

$$\therefore u = x \cdot \sqrt{x^2+y^2}$$

$$\therefore v = y \cdot \sqrt{x^2+y^2}$$

$$u_x = x \cdot \frac{d}{dx}(x^2+y^2)^{1/2} + \frac{dx}{dx} \cdot \sqrt{x^2+y^2}$$

$$= x \cdot \frac{1}{\cancel{2}}(x^2+y^2)^{-1/2} \cdot \cancel{2}x + \sqrt{x^2+y^2}$$

$$= \frac{x^2}{\sqrt{x^2+y^2}} + \sqrt{x^2+y^2}$$

$$u_y = x \cdot \frac{d}{dy}(x^2+y^2)^{1/2}$$

$$= x \cdot \frac{1}{\cancel{2}}(x^2+y^2)^{-1/2} \cdot \cancel{2}y$$

$$= \frac{xy}{\sqrt{x^2+y^2}}$$

$$\cancel{V_x}\; V = y \cdot \sqrt{x^2+y^2}$$

$$V_y = y \cdot \frac{d}{dy}(x^2+y^2)^{1/2} + \frac{dy}{dy} \cdot \sqrt{x^2+y^2}$$

$$= y \cdot \frac{1}{\cancel{2}}(x^2+y^2)^{-1/2} \cdot \cancel{2}y + \sqrt{x^2+y^2}$$

$$= \frac{y^2}{\sqrt{x^2+y^2}} + \sqrt{x^2+y^2}$$

$$V_x = y \frac{d}{dx} \cancel{d} (x^2+y^2)^{1/2}$$

$$= y \cdot \frac{1}{\cancel{2}} (x^2+y^2)^{-1/2} \cdot \cancel{2}x$$

$$= \frac{xy}{\sqrt{x^2+y^2}}$$

Since,

$u_x \neq v_y$

Therefore, $z|z|$ is not analytic function.

Q. Show that $f(z) = \frac{1}{z^n}$, except $z=0$ is an analytic function.

Sol^n:

Let $f(z) = u+iv$

or, $u+iv = \frac{1}{z^n}$

$= z^{-n}$

$= (re^{i\theta})^{-n}$

$= \frac{1}{r^n} e^{-in\theta}$

or, $u+iv = \frac{1}{r^n}(\cos n\theta - i\sin n\theta)$

or, $u+iv = \frac{1}{r^n}\cos n\theta - i\frac{1}{r^n}\sin n\theta$

Equating real & imaginary part,

$u = \frac{1}{r^n}\cos n\theta$, $v = -\frac{1}{r^n}\sin n\theta$

$u_r = -n\cdot\frac{1}{r^{n+1}}\cos n\theta$, $u_\theta = -\frac{1}{r^n} n\sin n\theta$

$v_r = n\cdot\frac{1}{r^{n+1}}\sin n\theta$, $v_\theta = -\frac{1}{r^n} n\cos n\theta$

By C-R eq^n,

$u_r = \frac{1}{r}v_\theta$, $u_\theta = -r v_r$

$\frac{1}{r}v_\theta = \frac{1}{r}\left(-\frac{1}{r^n} n\cos n\theta\right) = -n\cdot\frac{1}{r^{n+1}}\cos n\theta = u_r$

$-rv_r = -r\left(n\cdot\frac{1}{r^{n+1}}\sin n\theta\right) = -n\cdot\frac{1}{r^{n+\cancel{1}-\cancel{1}}}\sin n\theta$

$= -\frac{1}{r^n} n\sin n\theta = u_\theta$

$\therefore$ By C-R eq^n, $\frac{1}{z^n}$ is an analytic function.

Note: $f(z) = u(x,y) + i\,v(x,y)$

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, u is harmonic function

v is said to be harmonic conjugate of u.

Imp

Harmonic function:

A real valued function $h = \phi(x,y)$ of two variable is said to be harmonic function in a certain domain of xy-plane if it has continuous partial derivative of the first & second order and satisfies the Laplace's equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

Q. Show that the function $u(x,y) = 3x^2y + x^2 - y^3 - y^2$ is a harmonic function. find corresponding harmonic conjugate function $v(x,y)$ and find analytic function.

Soln:

Here $u(x,y) = 3x^2y + x^2 - y^3 - y^2$.

$$\frac{\partial u}{\partial x} = 6xy + 2x$$

$$\frac{\partial^2 u}{\partial x^2} = 6y + 2$$

$$\frac{\partial u}{\partial y} = 3x^2 - 3y^2 - 2y$$

$$\frac{\partial^2 u}{\partial y^2} = -6y - 2$$

Now,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 2 - 6y - 2 = 0$$

$\therefore u(x,y)$ is a harmonic function.

For harmonic conjugate function,

Since $f(z)$ is analytic by C-R eqn,

$$u_x = v_y$$

$$u_y = -v_x$$

Now,

$V_y = U_x$

$\frac{\partial v}{\partial y} = 6xy + 2x$

$\partial v = (6xy + 2x)\, dy$

Integrating w.r to y.

अर्को variable को ~~दुबैको~~ respect मा partially derivative गर्दा

$v = 6x\frac{y^2}{2} + 2xy + \phi(x)$ ~~— ①~~

$= 3xy^2 + 2xy + \phi(x)$ — ①

Diff. ① p.w.r. to "x"

$V_x = 3y^2 + 2y + \phi'(x)$, by C-R $u_y = -v_x$, $v_x = -u_y$

or, $-u_y = 3y^2 + 2y + \phi'(x)$

using data of u_y

$-(3x^2 - 3y^2 - 2y) = 3y^2 + 2y + \phi'(x)$

or, $-3x^2 + \cancel{3y^2} + \cancel{2y} = \cancel{3y^2} + \cancel{2y} + \phi'(x)$

or, $\phi'(x) = -3x^2$

integrating w.r. to 'x',

$\phi(x) = \frac{-3x^3}{3} + c$

$\therefore \phi(x) = -x^3 + c$

from ①,

$v = 3xy^2 + 2xy - x^3 + c$.

$\therefore$ The analytic function is $f(z) = u + iv$

$= (3x^2y + x^2 - y^3 - y^2) + i(3xy^2 + 2xy - x^3 + c)$.

which is required solution.

Q. $u(x,y) = \sin x \cosh y$.

Soln.

Here, $u(x,y) = \sin x \cosh y$

$$\frac{\partial u}{\partial x} = \cos x \cosh y$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\frac{\partial^2 u}{\partial y^2} = \sin x \cosh y$$

Now,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

$\therefore$ $u(x,y)$ is a harmonic function.

For harmonic conjugate function,

Since $f(z)$ is analytic by C-R eqn

$$u_x = v_y$$

$$u_y = -v_x$$

Now,

$$v_y = u_x$$

$$\frac{\partial v}{\partial y} = \cos x \cosh y$$

or, $dv = (\cos x \cosh y)\, dy$

Integrating w.r. to y.

$$v = \cos x \cdot \sinh y + \phi(x) \quad \text{--- (1)}$$

Diff. (1) p.w.r to "x"

$$v_x = -\sin x \cdot \sinh y + \phi'(x)$$

or, $-u_y = -\sin x \cdot \sinh y + \phi'(x)$

$\rightarrow d\left(\frac{1}{x}\right) = -\frac{1}{x^2}$ $\rightarrow d\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

$\rightarrow d\left(\frac{1}{x^2+a^2}\right) = \frac{-1 \times 2x}{(x^2+a^2)^2}$

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or, $-\sin x \sinh y = -\sin x \sinh y + \phi'(x)$

or, $\phi'(x) = 0$

integrating w.r.to 'x'

$\phi(x) = c.$

from ①,

$v = \cos x \cdot \sinh y + c$

$\therefore$ The analytic function is $f(z) = u + iv$

$= \sin x \cosh y + i(\cos x \cdot \sinh y + c)$

which is required solution.

Que. Show that $v = 2xy - \frac{y}{x^2+y^2}$ is a harmonic function.

Find the harmonic conjugate u of v.

Soln:

Here, $v = 2xy - \frac{y}{x^2+y^2}$ —— ①

$V_x = 2y + y \cdot \frac{2x}{(x^2+y^2)^2}$

$= 2y + \frac{2xy}{(x^2+y^2)^2}$

and

$V_{xx} = 0 + \left\{ \frac{(x^2+y^2)^2 \cdot 2y - 2xy \cdot 2(x^2+y^2) \cdot 2x}{(x^2+y^2)^4} \right.$

$\left[\begin{array}{ll} u = 2xy & , u' = 2y \\ v = (x^2+y^2)^2 & , v' = 2(x^2+y^2)^{2-1} * (2x+0) \end{array} \right]$

or, $V_{xx} = \frac{\cancel{(x^2+y^2)}\{2y(x^2+y^2) - 8x^2y\}}{(x^2+y^2)^{\cancel{4}\,3}}$

or, $V_{xx} = \frac{2x^2y + 2y^3 - 8x^2y}{(x^2+y^2)^3}$

$$\text{or, } V_{xx} = \frac{2y^3 - 6x^2y}{(x^2+y^2)^3}$$

Diff. ① p.w.r to y.

$$V_y = 2x - \left\{ \frac{(x^2+y^2)\cdot 1 - y\cdot 2y}{(x^2+y^2)^2} \right\}$$

$$\left[\begin{array}{ll} u = y, & u' = 1 \\ v = x^2+y^2, & v' = 2y \end{array} \right]$$

$$V_y = 2x + \frac{(y^2 - x^2)}{(x^2+y^2)^2}$$

$$\text{or, } V_{yy} = 0 + \left\{ \frac{(x^2+y^2)^2 \cdot 2y - (y^2-x^2)\cdot 2(x^2+y^2)\cdot 2y}{(x^2+y^2)^4} \right\}$$

$$\left[\begin{array}{ll} u = y^2 - x^2, & u' = 2y \\ v = (x^2+y^2)^2, & v' = 2(x^2+y^2)^{2-1}\cdot 2y \end{array} \right]$$

$$\cancel{\text{or, } V_{yy} = \{(x^2+y^2)^2 \cdot 2y + (x^2+y^2)\}}$$

$$\text{or, } V_{yy} = (x^2+y^2)\left\{ \frac{2y(x^2+y^2) - (y^2-x^2)\cdot 4y}{(x^2+y^2)^4} \right\}$$

$$\text{or, } V_{yy} = \frac{2x^2y + 2y^3 - 4y^3 + 4yx^2}{(x^2+y^2)^3}$$

$$\text{or, } V_{yy} = \frac{6x^2y - 2y^3}{(x^2+y^2)^3}$$

Now,

$$V_{xx} + V_{yy} = \frac{2y^3 - 6x^2y}{(x^2+y^2)^3} + \frac{6x^2y - 2y^3}{(x^2+y^2)^3}$$

$$= 0$$

Hence, v is a harmonic function.

For harmonic conjugate.

Since f(z) is analytic, by C-R eqⁿ

$$u_y = -V_x$$

$\rightarrow \int \frac{1}{x} dx = \log x$

$\rightarrow \int \frac{2x}{x^2+a^2} dx = \log(x^2+a^2)$

$\rightarrow \int \frac{1}{x^2} dx = \frac{x^{-2+1}}{(-2+1)}$

$\rightarrow \int \frac{2x}{(x^2+a^2)^2} dx = \frac{(x^2+a^2)^{-2+1}}{(-2+1)}$

$$u_y = -\left\{ 2y + \frac{2xy}{(x^2+y^2)^2} \right\}$$

Integrating both sides,

$$u = -\left\{ \frac{2y^2}{2} + x \cdot \frac{(x^2+y^2)^{-2+1}}{-2+1} \right\} + \phi(x)$$

$$u = -y^2 + \frac{x}{x^2+y^2} + \phi(x) \quad \text{--- (2)}$$

Diff p.w.r. to x.

$$u_x = 0 + \frac{(x^2+y^2)\cdot 1 - x\cdot(2x+0)}{(x^2+y^2)^2} + \phi'(x)$$

$$u_x = \frac{y^2-x^2}{(x^2+y^2)^2}$$ by C-R eq^n, $u_x = v_y$

$$v_y = \frac{y^2-x^2}{(x^2+y^2)^2} + \phi'(x) \quad [\text{Using } v_y)$$

or, $2x + \cancel{\frac{y^2-x^2}{(x^2+y^2)^2}} = \cancel{\frac{y^2-x^2}{(x^2+y^2)^2}} + \phi'(x)$

or, $\phi'(x) = 2x$

Integrating.

or, $\phi(x) = x^2 + c$

from (2)

$$u = x^2 - y^2 + \frac{x}{x^2+y^2} + c$$

which is required solution.

10. a) Determine whether $u = \frac{x}{x^2+y^2}$ is harmonic. Find corresponding analytic function $f(z) = u + iv$.

Soln,

Here, $u(x,y) = \frac{x}{x^2+y^2}$ —— ①

$$\frac{\partial u}{\partial x} = \frac{(x^2+y^2)\cdot 1 - x\cdot 2x}{(x^2+y^2)^2} \quad \begin{bmatrix} u = x & , u' = 1 \\ v = x^2+y^2 & , v' = 2x \end{bmatrix}$$

$$= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$$

$$= \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \left\{ \frac{(x^2+y^2)^2\cdot(-2x) - (y^2-x^2)\cdot 2(x^2+y^2)\cdot 2x}{(x^2+y^2)^4} \right\}$$

$$\begin{bmatrix} u = y^2-x^2 & , u' = -2x \\ v = (x^2+y^2)^2 & , v' = 2(x^2+y^2)^{2-1}\cdot 2x \end{bmatrix}$$

or, $\frac{\partial^2 u}{\partial x^2} = (x^2+y^2) \left\{ \frac{(x^2+y^2)\cdot(-2x) - (y^2-x^2)\cdot 4x}{(x^2+y^2)^4} \right\}$

or, $\frac{\partial^2 u}{\partial x^2} = \frac{-2x^3-2xy^2-4xy^2+4x^3}{(x^2+y^2)^3}$

or, $\frac{\partial^2 u}{\partial x^2} = \frac{2x^3-6xy^2}{(x^2+y^2)^3}$

$$\frac{\partial u}{\partial y} = \frac{(x^2+y^2)\times 0 - x\cdot 2y}{(x^2+y^2)^2} \quad \begin{bmatrix} u = x & , u' = 0 \\ v = x^2+y^2 & , v' = 2y \end{bmatrix}$$

$$= \frac{-2xy}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \left\{ \frac{(x^2+y^2)^2\cdot(-2x) - (-2xy)\cdot 2(x^2+y^2)\cdot 2y}{(x^2+y^2)^4} \right\}$$

$$\begin{bmatrix} u = -2xy & , u' = -2x \\ v = (x^2+y^2)^2 & , v' = 2(x^2+y^2)^{2-1}\cdot 2y \end{bmatrix}$$

or, $\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2)\{(x^2+y^2)\cdot(-2x) + 8xy^2\}}{(x^2+y^2)^4}$

or, $\frac{\partial^2 u}{\partial y^2} = \frac{-2x^3 - 2xy^2 + 8xy^2}{(x^2+y^2)^3}$

or, $\frac{\partial^2 u}{\partial y^2} = \frac{-2x^3 + 6xy^2}{(x^2+y^2)^3}$

Now,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2x^3 - 6xy^2}{(x^2+y^2)^3} + \frac{(-2x^3+6xy^2)}{(x^2+y^2)^3}$$

$$= 0$$

$\therefore$ u is a harmonic function.

For harmonic conjugate function,

Since, $f(z)$ is analytic by C-R eq^n

$$u_x = v_y$$
$$u_y = -v_x$$

Now,

$$u_y = -v_x$$

~~or,~~ $\cancel{\frac{\partial u}{\partial y} = \frac{2xy}{(x^2+y^2)^2}}$

~~or, ∂~~

or, $v_x = -u_y$

or, $v_x = \frac{2xy}{(x^2+y^2)^2}$

or, $\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}$

or, $\partial v = \frac{2xy}{(x^2+y^2)^2}\,\partial x$

Integrating w.r.to x,

$$v = y \cdot \frac{(x^2+y^2)^{-2+1}}{(-2+1)} + \phi(y)$$

$$v = \frac{-y}{(x^2+y^2)} + \phi(y) \quad \text{--- } ②$$

Diff ② p.w.r. to y.

$$\begin{bmatrix} u = y & , u' = 1 \\ v = (x^2+y^2) & , v' = 2y \end{bmatrix}$$

$$\cancel{v_y = \frac{(x^2+y^2)\cdot(-y) - (-y)\cdot(2y)}{(x^2+y^2)^2} + \phi'(y)}$$

$$\cancel{\text{or, } v_y = \frac{-x^2y - y^3 + 2y^2}{(x^2+y^2)^2} + \phi'(y)}$$

$$\cancel{\text{or, } u_x = \frac{-x^2y - y^3 + 2y^2}{(x^2+y^2)^2} + \phi'(y)}$$

$$\text{or, } v_y = -\left[\frac{(x^2+y^2)1 - y(2y+0)}{(x^2+y^2)^2}\right] + \phi'(y)$$

$$\text{or, } v_y = \frac{-x^2 - y^2 + 2y^2}{(x^2+y^2)^2} + \phi'(y)$$

$$\therefore v_y = \frac{y^2 - x^2}{(x^2+y^2)^2} + \phi'(y)$$

By C-R eq^n,

$$u_x = v_y$$

$$\text{or, } \frac{y^2 - x^2}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2} + \phi'(y)$$

$$\therefore \phi'(y) = 0$$

Integrating,

$$\phi(y) = c$$

From ②,

$$v = \frac{-y}{(x^2+y^2)} + c$$

For analytic function,

$$f(z) = u + iv$$

$$= \frac{x}{x^2+y^2} + i\left\{\frac{-y}{x^2+y^2} + c\right\}$$

$$= \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2} + ic$$

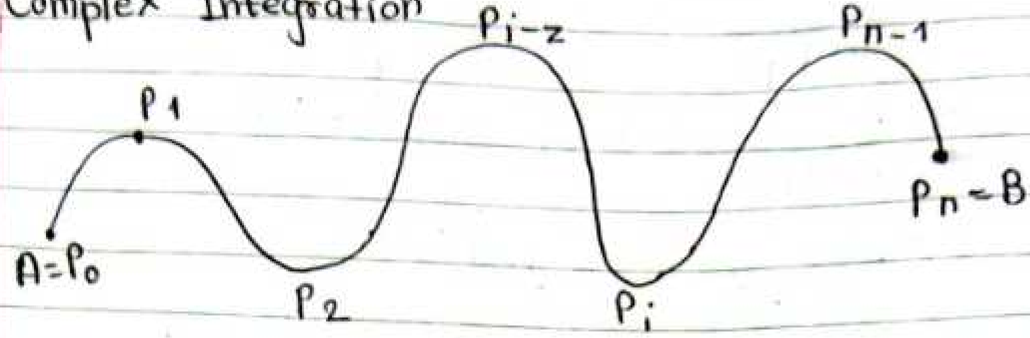
$$= \frac{1}{x^2+y^2}(x-iy) + ic$$

$$= \frac{1}{x^2+y^2}(x-iy) \times \frac{(x+iy)}{x+iy} + ic$$

$$= \frac{1}{\cancel{x^2+y^2}} \times \frac{\cancel{(x^2+y^2)}}{x+iy} + ic$$

$$f(z) = \frac{1}{z} + ic$$

Complex Integration



Let $f(z)$ be a continuous function of complex variable $z = x+iy$ defined at all points of a curve c having end point A & B. We subdivide the curve C into n parts $A_0 = P_0(z_0)$, $P_1(z_1)$, $P_2(z_2)$, ... $P_{i-1}(z_{i-1})$, $P_i(z_i)$, ..., $P_{n-1}(z_{n-1})$, $P_n(z_n) = B$

Let $\delta z_i = z_i - z_{i-1}$ and ξ_i be any point on the arc $P_{i-1} - P_i$. if the limit of a sum $\sum_{n=i}^{\infty} f(\xi_i)\delta z_i$

exists n tends to infinity, i.e. the chord δz_i tends to zero, then we can write $\int_c f(z)dz = \lim_{n\to\infty} \sum_{i=1}^{n} f(\xi_i)\delta z_i$

Writing $f(z) = u+iv$ and $z = x+iy$

$$dz = dx + i\,dy$$

so,

$$\int_c f(z)\,dz = \int_c (u+iv)(dx+i\,dy)$$

$$= \int_c (u\,dx - v\,dy) + i\int (v\,dx + u\,dy)$$

Q) Integrate $\int_c z^2 dz$, where c is the curve passing through the point $1+i$ and $2(1+2i)$ and specified as

i) the arc $y = x^2$

ii) the straight line joining the points $1+i$ & $2(1+2i)$

iii) the arc of $z = t + it^2$

<u>Soln.</u>

Here,

$$\int_c z^2 dz = \int_c (x+iy)^2 (dx + idy)$$

$$= \int_c (x^2 - y^2 + i2xy)(dx + idy)$$

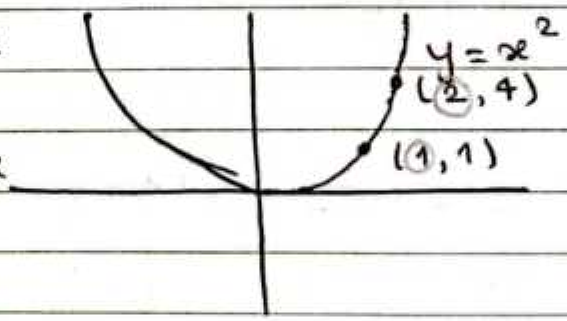
$$= \int_c \{(x^2-y^2)dx - 2xy\,dy\}$$

$$+ i\int_c \{(x^2-y^2)dy + 2xy\,dx\} \quad \text{--- (1)}$$

i) the arc $y = x^2$

$$\frac{dy}{dx} = 2x$$

$$dy = 2x\,dx$$



$$= \int_1^2 \{(x^2 - x^4)dx - 2x \cdot x^2\, 2x\,dx\}$$

$$+ i\int_1^2 \{(x^2 - x^4)\,2x\,dx + 2x \cdot x^2\,dx\}$$

$$= \int_1^2 (x^2 - 5x^4)dx + i\int_1^2 (4x^3 - 2x^5)dx$$

$$= \left[\frac{x^3}{3} - \frac{5}{5}x^5\right]_1^2 + i\left[\frac{4}{4}x^4 - \frac{2}{6}x^6\right]_1^2$$

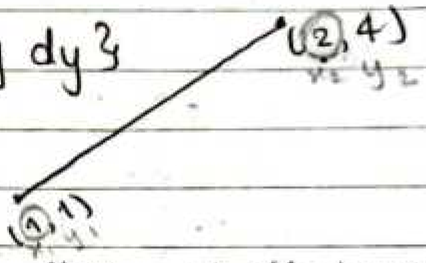
$$= \left[\frac{8}{3} - \frac{1}{3} - 2^5 + 1\right] + i\left[2^4 - \frac{2^6}{3} - 1^4 + \frac{1}{3}\right]$$

$$= -\frac{86}{3} - 6i$$

$$= -\left[\frac{86}{3} + 6i\right]$$

ii) from (1)

$$\int_c z^2\,dz = \int_c \{(x^2-y^2)\,dx - 2xy\,dy\} + i\int_c \{(x^2-y^2)\,dy + 2xy\,dx\}$$



$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

or, $y - 1 = \frac{3}{1}(x - 1)$

or, $y - 1 = 3x - 3$

or, $y = 3x - 2$

or, $\frac{dy}{dx} = 3$

or, $dy = 3dx$

$$= \int_1^2 \left[\{x^2 - (3x-2)^2\}dx - 2x\cdot(3x-2)\cdot 3dx\right] + i\int_1^2 \left[\{x^2 - (3x-2)^2\}\cdot 3dx + 2x\cdot(3x-2)\,dx\right]$$

$$= \int_1^2 \{x^2 - (9x^2 - 12x + 4)\,dx - 6x^2 - 4x\cdot 3dx\} + i\int_1^2 \{x^2 - (9x^2 - 12x + 4)\cdot 3dx + 6x^2 - 4x\}\,dx$$

$$= \int_1^2 (x^2 - 9x^2 + 12x - 4 - 6x^2 - 12x)\,dx + i\int_1^2 (3x^2 - 27x^2 + 36x + 12 + 6x^2 - 4x)\,dx$$

$$= \frac{-182}{3} + 36 - 4 + i\{-72+48-12\}$$

$$= \frac{-86}{3} - 6i$$

$$= -\left[\frac{86}{3} + 6i\right]$$

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iii) $z = t + it^2$

$x + iy = t + it^2$

$x = t \quad , \frac{dx}{dt} = 1 \quad , dx = dt$

$y = t^2 \quad , \frac{dy}{dt} = 2t \quad , dy = 2t\,dt$

from $1+i$ to $2+i4$

$(1,1)$ to $(2,4)$

when $x=1$ as $t=1$

$x=2$ as $t=2$

$$= \int_1^2 \{(t^2 - t^4)\,dt - 2t\cdot t^2 \cdot 2t\,dt\}$$

$$+ i\int_1^2 \{(t^2 - t^4)\cdot 2t\,dt + 2t\cdot t^2\,dt\}$$

$$= \int_1^2 (t^2 - 5t^4)\,dt + i\int_1^2 (4t^3 - 2t^5)\,dt$$

$$= \left[\frac{t^3}{3} - \frac{5}{5}t^5\right]_1^2 + i\left[\frac{4}{4}t^4 - \frac{2}{6}t^6\right]_1^2$$

$$= \left[\frac{8}{3} - \frac{1}{3} - 2^5 + 1\right] + i\left[2^4 - \frac{2^6}{3} - 1^4 + \frac{1}{3}\right]$$

$$= -\frac{86}{3} - 6i$$

$$= -\left[\frac{86}{3} + 6i\right]$$

$$= \int_1^2 (-14x^2 - 4)\,dx + i\int_1^2 (-18x^2 + 32x + 12)\,dx$$

$$= \int_1^2 \{(x^2 - 9x^2 + 12x - 4)\,dx - 18x^2\,dx + 12x\,dx +$$

$$i\int_1^2 \{(3x^2 - 27x^2 + 36x - 12)\,dx + (6x^2 - 4x)\,dx\}$$

$$= \int_1^2 (-26x^2 + 24x - 4)\,dx + i\int_1^2 -18x^2 + 32x - 12\,dx$$

$$= -26\left[\frac{x^3}{3}\right]_1^2 + 24\left[\frac{x^2}{2}\right]_1^2 - 4[x]_1^2 + i\left\{\left[-18\frac{x^3}{3}\right]_1^2\right.$$

$$\left. + 32\left[\frac{x^2}{2}\right]_1^2 - 12[x]_1^2\right\}$$

Ex 3.1

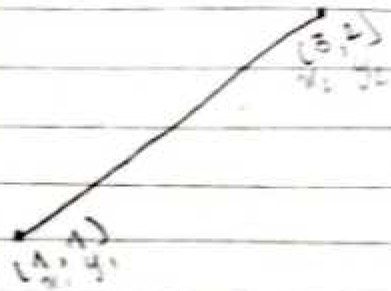
① $\int_c \text{Re}\, z \, dz$, where c is the shortest path from $1+i$ to $3+2i$

<u>Sol</u>

Solⁿ:

We have $z = x+iy$, $\text{Re}(z) = x$

$\text{Im}(z) = y$



Here, $\int_c (x+0i)(dx+idy)$

$= \int_c x\,dx + i\int_c x\,dy$ where c;

$y - y_1 = \frac{y_2-y_1}{x_2-x_1}(x-x_1)$

$y - 1 = \frac{1}{2}(x-1)$

or, $2y - 2 = x - 1$

or, $y = \frac{x+1}{2}$

or, $\frac{dy}{dx} = \frac{1}{2}$

or, $dy = \frac{1}{2}dx$

$= \int_1^3 x\,dx + i\int_1^3 x\,\frac{1}{2}\,dx$

$= \left[\frac{x^2}{2}\right]_1^3 + \frac{1}{2}i\left[\frac{x^2}{2}\right]_1^3$

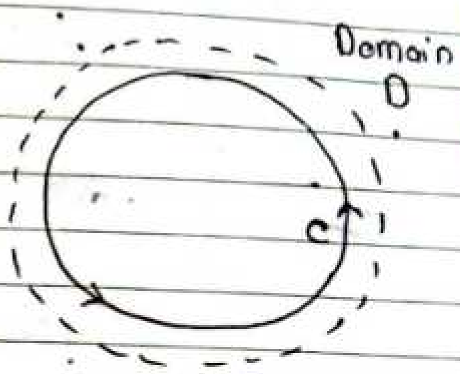
$= 4 + i2$

Green's Theorem:

$$\oint_C (F_1 dx + F_2 dy) = \iint_R \left(\frac{dF_2}{dx} - \frac{dF_1}{dy} \right) dx\, dy$$

Imp # Cauchy Integral Theorem

Statement: If a function $f(z)$ is analytic and its derivative $f'(z)$ is continuous at all points inside and on a simple closed curve c, then $\oint_C f(z)\, dz = 0$



Proof: Let $f(z) = u + iv$ and

$z = x + iy$ implies

$dz = dx + i dy$

Thus,

$$\oint_C f(z)\, dz = \oint_C (u + iv)(dx + i dy)$$

$$= \oint_C (u\,dx - v\,dy) + i \oint_C (v\,dx + u\,dy)$$

Since $f(z)$ is a analytic function in the region R and Its derivative $f'(z)$ exists in R and continuous, therefore $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ are also continuous in the region R enclosed by c, By Green's Theorem,

$$= \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx\, dy + i \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx\, dy$$

By C-R

$$= \iint_R \left(\frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} \right) dx\, dy + i \iint_R \left(\frac{\partial v}{\partial y} - \frac{\partial v}{\partial y} \right) dx\, dy$$

$= 0 + i0$

$= 0$

$\therefore \oint_C f(z)\, dz = 0$

Hence proved

Q) Evaluate the intergral $\oint_c \frac{1}{z} dz$ where c is the unit circle ie $|z| = 1$

Soln,

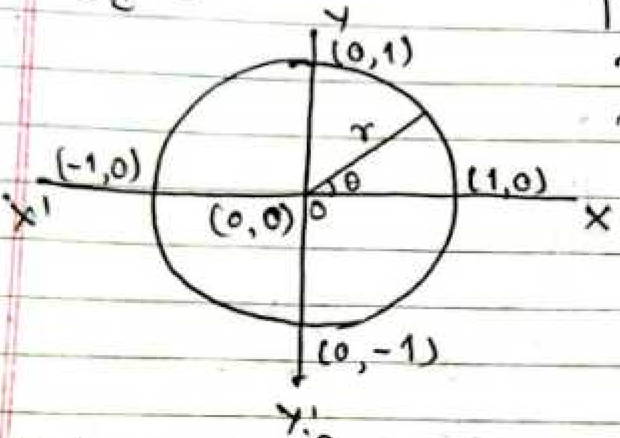
Here,

$\oint_c \frac{1}{z} dz$ where $c ; |z| = 1$

$|x+iy| = 1$

$x^2+y^2 = 1^2$

$x^2+y^2 = 1$



put $z = re^{i\theta}$

$z = 1 \cdot e^{i\theta}$

$z = e^{i\theta}$

$\frac{dz}{d\theta} = ie^{i\theta}$

$dz = ie^{i\theta} d\theta$ when θ varies from $\theta = 0$ to $\theta = 2\pi$

Now,

$$\int_c \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{\cancel{e^{i\theta}}} \cdot i\cancel{e^{i\theta}} d\theta$$

$$= i\int_0^{2\pi} d\theta$$

$$= i\left[\theta\right]_0^{2\pi}$$

$$= [2\pi - 0]\, i$$

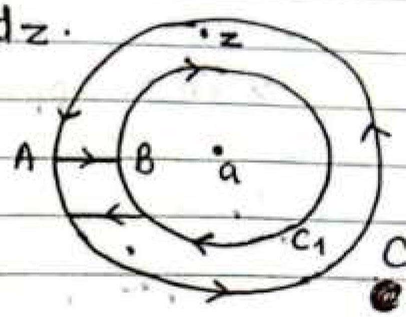
$$= 2\pi i$$

$$\oint \frac{e^{3z}}{z-2}\,dz = 2\pi i f(2)$$

Extension of Cauchy's integral theorem

Statement: If a function $f(z)$ is analytic in the region R between two simple closed curve C and C_1 then

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz.$$



V.V.V. Imp

Cauchy Integral Formula

Statement:

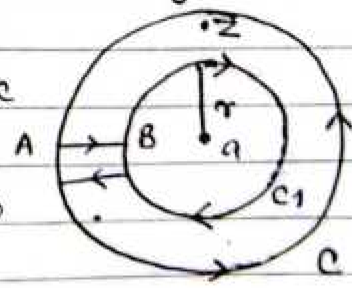
If $f(z)$ is analytic within and on a closed curve C and if "a" is any point within c then

$$\oint_C \frac{1}{z-a} f(z)\,dz = 2\pi i f(a)$$

Proof:

Take the function $\frac{f(z)}{z-a}$ which is analytic at all points except $z=a$. Draw a small circle C_1 with "a" as centre and radius r lying entirely within C.

Thus the function $\frac{f(z)}{z-a}$ is analytic in the region between C & C_1 so by cauchy integral theorem.



$$\oint_c \frac{f(z)}{z-a}dz = \oint_{c_1} \frac{f(z)}{z-a}dz \,, \quad \text{put } z-a = re^{i\theta}$$

$$\frac{d}{d\theta}(z-a) = ire^{i\theta}$$

$$dz = ire^{i\theta}d\theta$$

$$= \oint_{c_1} \frac{f(a+re^{i\theta})}{re^{i\theta}} \, ire^{i\theta}d\theta$$

$$= i\oint_{c_1} f(a+re^{i\theta})\,d\theta$$

when z approaches to a then radius tends to zero within circle C_1 when θ varies from $\theta = 0$ to $\theta = 2\pi$

$$= i\int_0^{2\pi} f(a+0)\,d\theta$$

$$= i\,f(a)\int_0^{2\pi} d\theta$$

$$= i\,f(a)\,[\theta]_0^{2\pi}$$

$$= 2\pi i f(a)$$

Q. Evaluate the following integral by using cauchy integral formula.

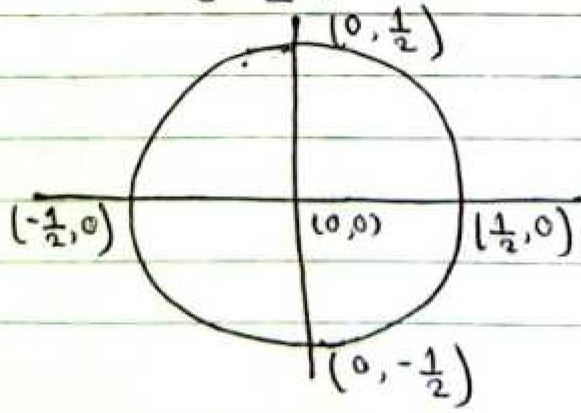
1) $\oint_c \frac{z^2+1}{z^2+z}dz$ where c is the circle $|z| = \frac{1}{2}$

Soln :

Here, $\oint_c \frac{z^2+1}{z^2+z}dz$, c ; $|z| = \frac{1}{2}$

$|x+iy| = \frac{1}{2}$

$x^2+y^2 = \frac{1}{4}$



$$= \oint_c \frac{z^2+1}{z(z+1)}\,dz$$

The pole of integrand are given by putting

$z(z+1) = 0$

$z = 0, -1$

$\therefore z = 0$ lies within circle c; $|z| = \frac{1}{2}$

and $z = -1$ doesnot lie within circle.

By cauchy integral formula,

$$\oint_c \frac{\frac{z^2+1}{z+1}}{z}\,dz = 2\pi i f(0)$$
$$= 2\pi i$$

where, $f(z) = \frac{z^2+1}{z+1}$

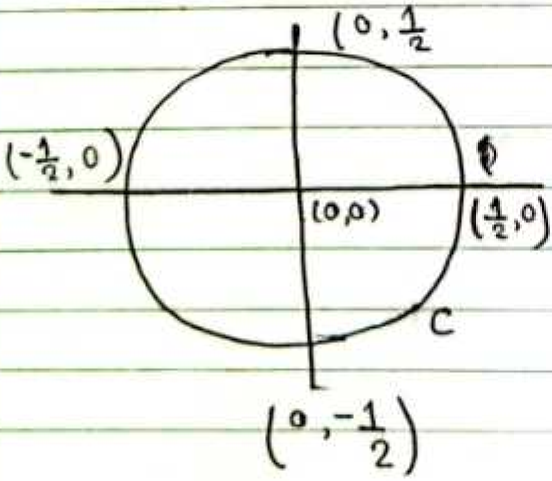
$$f(0) = \frac{0+1}{0+1} = 1$$

2) $\oint_c \frac{z^2+1}{z^2-1}\,dz$ where c is the circle $|z| = \frac{1}{2}$

Soln:

Here,

$$\oint_c \frac{z^2+1}{z^2-1}\,dz, \quad c: |z| = \frac{1}{2}$$



$|x+iy| = \frac{1}{2}$

$x^2+y^2 = \frac{1}{4}$

$$= \oint_c \frac{z^2+1}{(z-1)(z+1)} dz$$

The pole of integrand are given by putting

$(z-1)(z+1) = 0$

$z = 1, z = -1$

$\therefore z = 1$ & -1 doesn't lie in circle $c; |z| = \frac{1}{2}$

by cauchy integral formula

$$\oint_c \frac{z^2+1}{(z-1)(z+1)} dz = 0.$$

3) $\oint_c \frac{e^{4z}}{(z-2)(z-1)} dz$, where c is the circle $|z| = 3$

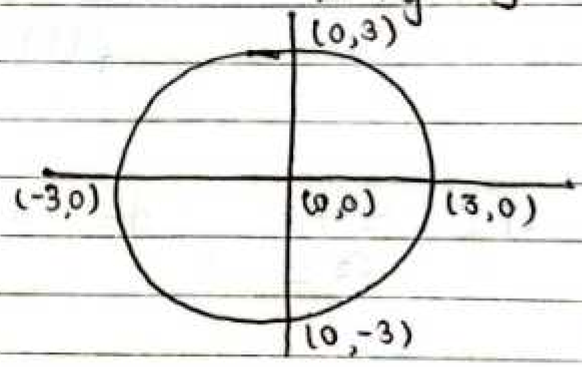
$|x+iy| = 3$

$x^2+y^2 = 9$

Solⁿ:

Here,

$$\oint_c \frac{e^{4z}}{(z-2)(z-1)} dz$$



The pole of integrand are given by putting

$(z-1)(z-2) = 0$

$z = 1, 2$

$\therefore z = 1$ & 2 lie in circle $c; |z| = 3$

by cauchy integral formula, pole at $z = 2$

$$\oint_c \frac{\frac{e^{4z}}{z-1}}{z-2} dz = 2\pi i f(2), \text{ where } f(z) = \frac{e^{4z}}{z-1}$$

$$= 2\pi i e^8$$

$$f(2) = \frac{e^8}{2-1} = e^8$$

anticlockwise: counter clockwise
clockwise : $(-)$ * Ans

, pole at $z=1$

$$\oint_c \frac{\frac{e^{4z}}{z-2}}{z-1}\,dz = 2\pi i f(1), \text{ where } f(z)=\frac{e^{4z}}{z-2}$$
$$= -2\pi i e^4$$

$$= \frac{e^4}{1-2}$$
$$= -e^4$$

$$\therefore \oint_c \frac{e^{4z}}{(z-2)(z-1)}\,dz = \text{sum of these integral value}$$
$$= 2\pi i e^8 - 2\pi i e^4$$
$$= 2\pi i(e^8 - e^4)$$

imp

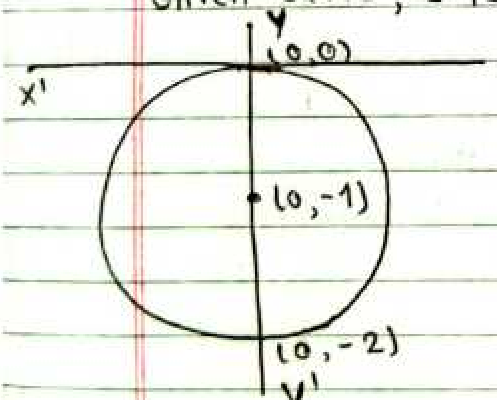
Q. Integrate : $\oint_c \frac{z^2}{z^4-1}\,dz$ where c is the circle $|z+i|=1$ in the counter clockwise.

Soln:

Here, $\oint_c \frac{z^2}{z^4-1}\,dz$

$$= \oint_c \frac{z^2}{(z^2-1)(z^2+1)}\,dz$$
$$= \oint_c \frac{z^2}{(z-1)(z+1)(z^2-i^2)}\,dz$$
$$= \oint_c \frac{z^2}{(z-1)(z+1)(z+i)(z-i)}\,dz$$

Given curve, c is the circle $|z+i|=1$



or, $|x+iy+i|=1$
or, $|x+i(y+1)|=1$
or, $x^2+(y+1)^2=1$ $[(x-h)^2+(y-k)^2=r^2]$
Centre $(0,-1)$, major axis y-axis i.e. $x=0$
$(y+1)^2=1$
$y+1=\pm 1$
+ve , $y=0$
-ve , $y=-2$

The pole of integrand are given by putting

$$(z-1)(z+1)(z-i)(z+i)=0$$

$$z=1,-1,i,-i$$

$\therefore z=-i$ lies within given circle and $z=1$, -1 & i doesnot line within circle

By Cauchy integral formula

$$\oint_c \frac{\frac{z^2}{(z-1)(z+1)(z-i)}}{z+i}\,dz = 2\pi i f(-i)$$

$$= 2\pi i \times -\frac{1}{4i}$$

$$= -\frac{\pi}{2}$$

where $f(z) = \frac{z^2}{(z-1)(z+1)(z-i)}$

$$f(-i) = \frac{(-i)^2}{((-i)^2-1)(-i-i)}$$

$$= \frac{i^2}{(i^2-1)(-2i)}$$

$$= \frac{-1}{(-1-1)(-2i)}$$

$$= -\frac{1}{4i}$$

Cauchy's integral formula for derivative of an analytic function. (Only statement)

Let $f(z)$ be an analytic function inside a simple closed curve c in the region R and if "a" is a point in the interior of c, then

$$\oint \frac{1}{(z-a)^{n+1}} f(z)\,dz = \frac{1}{n!}\, 2\pi i f^n(a)$$

~~Integrate:~~ $\cancel{\oint_C \frac{z^8}{(2z-1)^6}\,dz}$

Imp

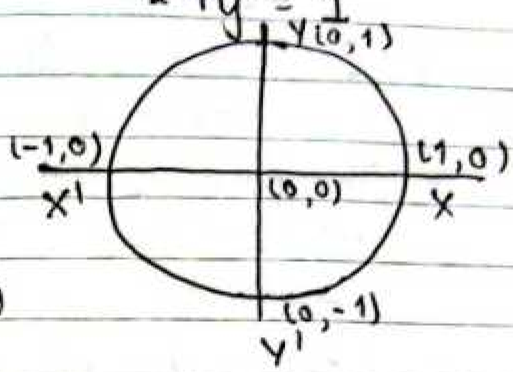
Que: Integrate : $\oint_c \frac{z^6}{(2z-1)^6} dz$ where c is the circle $|z| = 1$.

Soln :

Here, $\oint_c \frac{z^6}{2^6\left(z-\frac{1}{2}\right)^6} dz$, c ; $|z| = 1$

$x^2+y^2 = 1$

$\oint_c \frac{\frac{1}{2^6} z^6}{\left(z-\frac{1}{2}\right)^6} dz$



The pole of integrand are given by putting $\left(z-\frac{1}{2}\right)^6 = 0$

$z = \frac{1}{2}$ has order 6

$\therefore z = \frac{1}{2}$ lies within circle $|z| = 1$.

By cauchy integral formula

$$\oint_c \frac{\frac{1}{2^6} z^6}{\left(z-\frac{1}{2}\right)^6} dz = \frac{1}{5!} 2\pi i \; f^5\left(\frac{1}{2}\right)$$

$$= \frac{1}{\cancel{5}\cdot\cancel{4}\cdot\cancel{3}\cdot\cancel{2}\cdot 1} \cancel{2}\pi i \cdot \frac{1}{2^6} \frac{6\cdot\cancel{5}\cdot}{\cancel{4}\cdot\cancel{3}}$$

$$= \frac{6\pi i}{64}$$

$$= \frac{3\pi i}{32}$$

where $f(z) = \frac{1}{2^6} z^6$

$f'(z) = \frac{1}{2^6} 6z^5$

$f^5(z) = \frac{1}{2^6} 6\cdot5\cdot4\cdot3\cdot2\cdot1 z^1$

$f^5\left(\frac{1}{2}\right) = \frac{1}{2^6} 6\cdot5\cdot4\cdot3\cdot\cancel{2}\cdot\frac{1}{\cancel{2}}$

$$d\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

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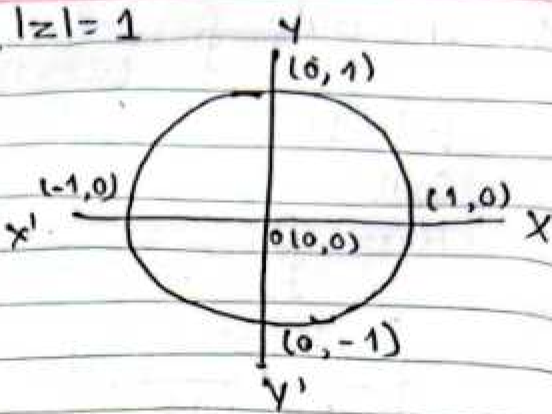
Q) $\oint_c \frac{z+1}{z^3-2z^2} dz \quad c: |z| = 1$



<u>Soln</u>

Here,

$$\oint_c \frac{z+1}{z^3-2z^2} dz$$

$$= \oint_c \frac{z+1}{z^2(z-2)} dz$$

The pole of integrand are given by putting $z^2(z-2) = 0$

$z = 0$ has order 2 lies within circle $|z| = 1$

& $z = 2$ doesn't lie within circle $|z| = 1$.

By cauchy integral formula,

$$\oint_c \frac{\frac{z+1}{z+2}}{z^2} dz = \frac{\cancel{z}1\, 2\pi i}{1!} f'(0)$$

$$= 2\pi i \times \left(-\frac{3}{4}\right)$$

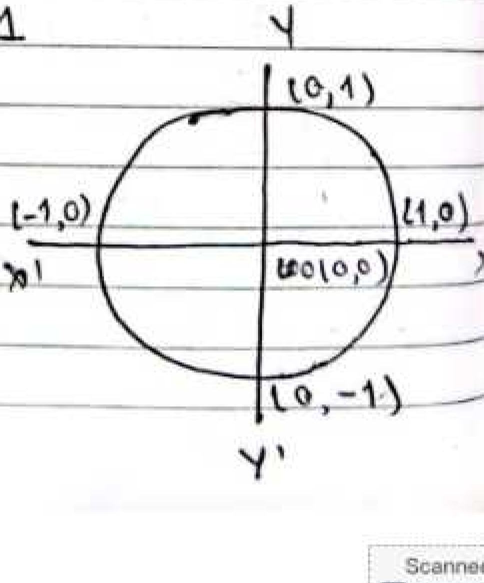
$$= -\frac{3}{2}\pi i$$

where, $f(z) = \frac{z+1}{z-2}$

$$= \frac{(z-2)\cdot 1 - (z+1)\cdot 1}{(z-2)^2}$$

$$f(0) = -\frac{3}{4}$$

Q) $\oint_c \frac{1}{z^8(z+\cancel{1}2)} dz \quad c: |z| = 1$



<u>Soln</u>

Here,

$$\oint_c \frac{1}{z^8(z+2)} dz$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -x^{-1-1} = -1x^{-2}$$

$$\frac{d^n}{dx^n}\left(\frac{1}{x+a}\right) = (-1)^n n! \frac{1}{(x+a)^{n+1}}$$

The pole of integrand are given by putting $z^8(z+2)=0$

$z=0$ has order 8 lies within circle $|z|=1$

$z=-2$ doesn't lie within circle $|z|=1$.

By cauchy integral formula,

$$\oint_c \frac{\frac{1}{z+2}}{z^8}\,dz = \frac{1}{7!} 2\pi i f^7(0)$$

where $f(z) = \frac{1}{z+2}$

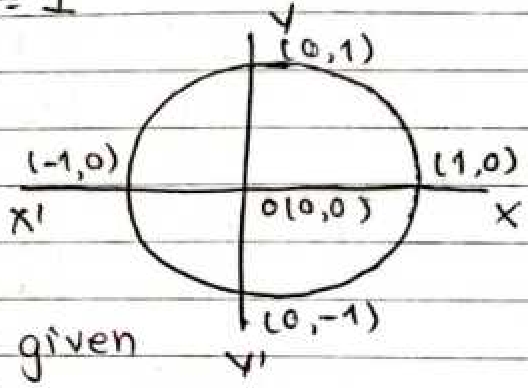
$$= \frac{-1}{\cancel{7!}} 2\pi i \times \cancel{7!} \cdot \frac{1}{2^8}$$

$$f^7(z) = (-1)^7\, 7! \frac{1}{(z+2)^8}$$

$$= \frac{-\pi i}{2^7}$$

$$f^7(0) = -7! \frac{1}{2^8}$$

g) $\oint_c \frac{1}{z^8(z+4)}\,dz \quad c ; |z| = 1$



Soln

Here,

$$\oint_c \frac{1}{z^8(z+4)}\,dz$$

The pole of Integrand are given by potting $z^8(z+4)=0$

$z=0$ has order 8 lies within circle $|z|=1$

$z=-4$ doesn't lie within circle $|z|=1$.

By cauchy integral formula,

$$\oint_c \frac{\frac{1}{z+4}}{z^8}\,dz = \frac{1}{7!} 2\pi i f^7(0)$$

where $f(z) = \frac{1}{z+4}$

$$= -\frac{1}{\cancel{7!}} 2\pi i \times \cancel{7!} \cdot \frac{1}{4^8}$$

$$f^7(z) = (-1)^7\, 7! \frac{1}{(z+4)^8}$$

$$= \frac{-2\pi i}{4^8} = \frac{-\pi i}{2^{15}}$$

$$f^7(0) = -7! \frac{1}{4^8}$$

Q. $\oint_c \frac{\cot z}{(z-\pi/2)^2}\, dz$ where c is the ellipse

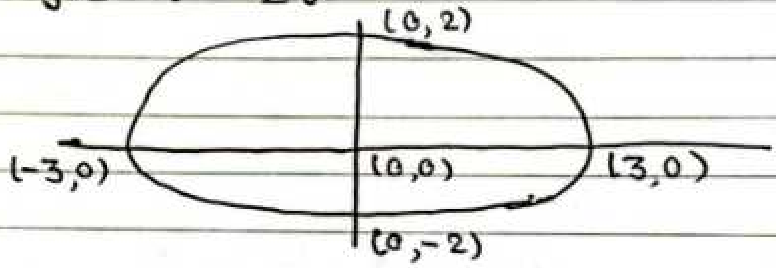
$4x^2+9y^2=36$

$\frac{x^2}{9}+\frac{y^2}{4}=1$

$\frac{x^2}{3^2}+\frac{y^2}{2^2}=1$

Solⁿ,

Here, $\oint_c \frac{\cot z}{\left(z-\frac{\pi}{2}\right)^2}\, dz$



$$= \oint_c \frac{\cot z}{\left(z-\frac{\pi}{2}\right)^2}\, dz$$

The pole of integrand are given by putting :

$$\left(z-\frac{\pi}{2}\right)^2=0$$

$z=\frac{\pi}{2}$ has order 2 lies within ellipse $4x^2+9y^2 = 36$

By cauchy integral formula.

$$\oint_c \frac{\cot z}{\left(z-\frac{\pi}{2}\right)^2}\, dz = \frac{1}{1!}\, 2\pi i\, f'(\pi/2)$$

where, $f(z)=\cot z$

$f'(z) = -\operatorname{cosec}^2 z$

$f'\left(\frac{\pi}{2}\right) = -\operatorname{cosec}^2\frac{\pi}{2}$

$= -1$

$$= \frac{1}{1!}\, 2\pi i \times -1$$

$$= -2\pi i$$

Assignment: Old Que Bank no. 1(a), (b), 2(b)
↳ deadline friday

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Sequence and Series

Definition:

An infinite sequence is obtained by assigning to each positive integer n a number z_n, called a term of sequence and written as $\{z_1, z_2, z_3 \ldots z_n \ldots\}$ or $\{z_n\}$ which is an infinite sequence.

The infinite sequence are connected by algebraic sign, is called infinite series

That is $z_1 + z_2 + z_3 + \ldots + z_n + \ldots$

i.e. $\sum_{n=1}^{\infty} z_n$.

Series of complex Variables

Let $f_1(z) + f_2(z) + f_3(z) + \ldots$ be an infinite series of function of complex variable z.

Taylor series and Maclaurin series (Only statement)

→ Taylor series:-

Statement: If $f(z)$ be analytic function in C with centre at "a" and radius r_0 then at each point z inside C, the series $f(a) + \frac{(z-a)^1}{1!} f'(a)$

$$+ \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \ldots + \frac{(z-a)^n}{n!} f^n(a) +$$

$\ldots$ converges to $f(z)$

i.e. $f(z) = f(a) + \frac{(z-a)}{1!} f'(a) + \frac{(z-a)^2}{2!} f''(a) + \ldots$

i.e. $f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n \quad \ldots + \frac{(z-a)^n}{n!} f^n(a) + \ldots$

where $a_n = \frac{f^n(a)}{n!}$

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→ Maclaurin series

The special core of Taylor's series, put $a=0$.
That means centre is zero.

$$f(z) = f(0) + \frac{z}{1!} f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \cdots\cdots$$

is called maclaurin series expansion.

Taylor's series of some special function (maclaurin series)

1) $\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots\cdots$ for $|z| < 1$

$$\boxed{\text{i.e. } \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n}$$

2) $\frac{1}{1+z} = 1 - z + z^2 - z^3 + \cdots\cdots$

$$\boxed{\text{i.e. } \frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n}$$ for $|z| < 1$.

3) $\frac{1}{(1-z)^2} = 1 + 2z + 3z^2 + 4z^3 + \cdots\cdots$ for $|z| < 1$.

$$\boxed{\text{i.e. } \frac{1}{(1-z)^2} = \sum_{n=0}^{\infty} (n+1) z^n}$$

$$\boxed{4)\ \frac{1}{(1+z)^2} = \sum_{n=0}^{\infty} (-1)^n (n+1) z^n}$$

5) $e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots\cdots$

$$\boxed{\text{i.e. } e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}}$$

Eg:

$$e^{2z} = 1 + \frac{(2z)}{1!} + \frac{(2z)^2}{2!} + \frac{(2z)^3}{3!} + \ldots\ldots$$

$$\boxed{\text{i.e. } e^{2z} = \sum_{n=0}^{\infty} \frac{(2z)^n}{n!}}$$

Eg: $$\boxed{e^{-z} = \sum_{n=0}^{\infty} \frac{(-z)^n}{n!}}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{n!}$$

i.e. $$e^{-z} = 1 - \frac{z}{1!} + \frac{z^2}{2!} - \frac{z^3}{3!} + \ldots\ldots$$

6) $$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \ldots\ldots$$

$$\boxed{\text{i.e. } \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}}$$

7) $$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \ldots\ldots$$

$$\boxed{\text{i.e. } \cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}}$$

8) $$\sinh z = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \ldots\ldots$$

$$\boxed{\text{i.e. } \sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!}}$$

9) $$\cosh z = 1 + \frac{z}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \ldots\ldots$$

$$\boxed{\text{i.e. } \cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}}$$

10) $\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} + \ldots\ldots, \quad |z| < 1$

11) $\log\left(\frac{1}{1+z}\right) = z + \frac{z^2}{2} + \frac{z^3}{3} + \ldots\ldots, \quad |z| < 1.$

Que: Expand the following functions of complex variable in a taylorsseries about $z = 0$. (maclaurin series)

① $f(z) = e^{-z}$

We have, Maclaurin series expansion of $f(z)$ is

$$f(z) = f(0) + \frac{z}{1!}f'(0) + \frac{z^2}{2!}f''(0) + \frac{z^3}{3!}f'''(0) + \ldots \quad \text{eq}^n \text{ ①}$$

Here,

$f(z) = e^{-z}$, $\quad f(0) = e^{-0} = 1$

$f'(z) = -e^{-z}$ $\quad f'(0) = -e^{-0} = -1$

$f''(z) = e^{-z}$ $\quad f''(0) = e^{-0} = 1$

$f'''(z) = -e^{-z}$ $\quad f'''(0) = -e^{-0} = -1$

from eq^n ①

$$e^{-z} = 1 - \frac{z}{1!} + \frac{z^2}{2!} - \frac{z^3}{3!} + \ldots\ldots$$

② $f(z) = \frac{1}{1-z^4}$

Sol^n

Here, $f(z) = \frac{1}{1-z^4}$

Since, $\frac{1}{1-z} = 1 + z + z^2 + z^3 + \ldots\ldots$

$$\frac{1}{1-(z^4)} = 1+z^4+(z^4)^2+(z^4)^3+\ldots\ldots$$

$$= 1+z^4+z^8+z^{12}+\ldots\ldots$$

③ $\dfrac{2-z}{(1-z)^2}$

<u>Soln</u>

Here, $f(z) = \dfrac{2-z}{(1-z)^2}$ —— ①

By partial function,

$$\frac{2-z}{(1-z)^2} = \frac{A}{1-z} + \frac{B}{(1-z)^2} \quad —— ②$$

or, $2-z = A(1-z)+B$

or, $2-z = A - Az + B$ or, $2-z = (A+B) - Az$

Equating the corresponding terms,

or, $A = 1$, $A+B = 2$

$\therefore A = 1$ $\quad 1+B = 2$

$\therefore B = 1$

from ②

$$f(z) = \frac{1}{1-z} + \frac{1}{(1-z)^2}$$

$$= \sum_{n=0}^{\infty} z^n + \sum_{n=0}^{\infty} \frac{(n+1)!\,z^n}{n!}$$

$$= \sum_{n=0}^{\infty} z^n + \sum_{n=0}^{\infty} (n+1)\, z^n$$

$$\therefore f(z) = \sum_{n=0}^{\infty} (n+2)\, z^n$$

$$= 2+3z+4z^2+5z^3+6z^4+\ldots\ldots$$

Q) Find the first four terms of the Taylor's series expansions of the complex function $f(z) = \frac{z+1}{(z-3)(z-4)}$ about the centre $z=2$ i.e. $z-2=0$.

Soln.

Here,

$$f(z) = \frac{z+1}{(z-3)(z-4)}$$

Let, $\frac{z+1}{(z-3)(z-4)} = \frac{A}{z-4} + \frac{B}{z-3}$ —— (1)

or, $z+1 = A(z-3) + B(z-4)$

or, $z+1 = Az - 3A + Bz - 4B$

or, $z+1 = z(A+B) + (-3A-4B)$

when $z=3$,

or, $B = -4$

when $z=4$,

or, $A = 5$

$$f(z) = \frac{5}{z-4} - \frac{4}{z-3}$$

$$= \frac{5}{-2+(z-2)} - \frac{4}{-1+(z-2)}$$

$$= \frac{5}{-2\left\{1-\frac{(z-2)}{2}\right\}} + \frac{4}{1-(z-2)}$$

$$= -\frac{5}{2}\left\{1+\left(\frac{z-2}{2}\right)+\left(\frac{z-2}{2}\right)^2+\left(\frac{z-2}{2}\right)^3 \cdots\cdots\right\} + 4\left\{1+(z-2)+(z-2)^2+(z-2)^3+\cdots\cdots\right\}$$

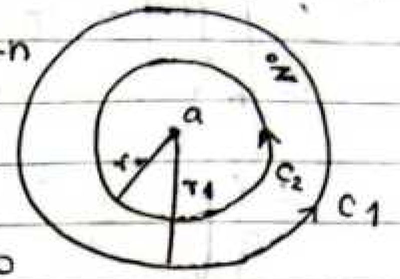
$$= \frac{3}{2} + \frac{11}{4}(z-2) + \frac{27}{8}(z-2)^2 + \cdots\cdots$$

which convergence for $|z-2| < 1$.

Laurent's series theorem (Statement)

If $f(z)$ is analytic in the ring shaped region R bounded by two concentric circles C_1 & C_2 with centre a and radius r_1 & r_2 $(r_1 > r_2)$ then for all z in R

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$



where

$$a_n = \frac{1}{2\pi i} \int_c \frac{1}{(w-a)^{n+1}} f(w)\, dw$$

and $$b_n = \frac{1}{2\pi i} \int_c \frac{1}{(w-a)^{-n+1}} f(w)\, dw$$

Q. Expand $f(z) = \dfrac{\sin z}{\left(z-\frac{\pi}{4}\right)^2}$ by using Laurent's series which converges for $0 < \left|z-\frac{\pi}{4}\right| < R$.

Soln:

Here, $f(z) = \dfrac{\sin z}{\left(z-\frac{\pi}{4}\right)^2}$

$$= \frac{\sin\left(z-\frac{\pi}{4}+\frac{\pi}{4}\right)}{\left(z-\frac{\pi}{4}\right)^2}$$

$$= \frac{\frac{1}{\sqrt{2}}\left(\sin\left(z-\frac{\pi}{4}\right) + \cos\left(z-\frac{\pi}{4}\right)\right)}{\left(z-\frac{\pi}{4}\right)^2}$$

$$= \frac{1}{\sqrt{2}\left(z-\frac{\pi}{4}\right)^2}\left[\left\{\left(z-\frac{\pi}{4}\right) - \frac{\left(z-\frac{\pi}{4}\right)^3}{3!} + \frac{\left(z-\frac{\pi}{4}\right)^5}{5!} \cdots\cdots\right\} + \left\{1 - \frac{\left(z-\frac{\pi}{4}\right)^2}{2!} + \frac{\left(z-\frac{\pi}{4}\right)^4}{4!} \cdots\cdots\right\}\right]$$

$$= \frac{1}{\sqrt{2}}\left[\left\{\left(z-\frac{\pi}{4}\right)^{-1} - \frac{\left(z-\frac{\pi}{4}\right)}{3!} + \frac{\left(z-\frac{\pi}{4}\right)^3}{5!} \cdots\cdots\right\} + \left\{\left(z-\frac{\pi}{4}\right)^{-2} - \frac{1}{2!} + \frac{\left(3-\frac{\pi}{4}\right)^2}{4!} \cdots\cdots\right\}\right]$$

which is Laurent's series expansion of $f(z)$ that convergence for $0 < \left|z-\frac{\pi}{4}\right| < R$.

Que: Find the Taylor's or Laurent's series which represents the function

$$f(z) = \frac{1}{(z+2)(z+3)}$$

i) when $|z| < 2$

ii) when $2 < |z| < 3$

iii) when $|z| > 3$

Soln:

Here, $f(z) = \frac{1}{(z+2)(z+3)}$

$$f(z) = \frac{A}{z+2} + \frac{B}{z+3}$$

or, $\frac{1}{(z+2)(z+3)} = \frac{A(z+3) + B(z+2)}{(z+2)(z+3)}$

→ z कम है भने constant common
माथि
" ठूलो " " z common

when $z=-3$, when $z=-2$

$1=-B$ $1=A$

$\therefore B=-1$ $\therefore A=1$

$$f(z)=\frac{1}{z+2}-\frac{1}{z+3} \quad \text{—— (1)}$$

i) when $|z|<2$

or, $\frac{|z|}{2}<1$

from (1)

$$f(z)=\frac{1}{2(1+z/2)}-\frac{1}{3(1+z/3)}$$

$$=\frac{1}{2}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{2}\right)^n-\frac{1}{3}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{3}\right)^n$$

iii) when $|z|>3$

$1>\frac{3}{|z|}$

i.e. $\frac{3}{|z|}<1$

from (1)

$$f(z)=\frac{1}{z\left(1+\frac{2}{z}\right)}-\frac{1}{z\left(1+\frac{3}{z}\right)}$$

$$=\frac{1}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{2}{z}\right)^n-\frac{1}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{3}{z}\right)^n$$

ii) $2<|z|<3$

$2<|z|$ or $|z|<3$

$\frac{2}{|z|}<1$ or $\frac{|z|}{3}<1$

from ①

$$f(z) = \frac{1}{z+2} + \frac{1}{z+3}$$

$$= \frac{1}{z\left(1+\frac{2}{z}\right)} - \frac{1}{3\left[1+\frac{z}{3}\right]}$$

$$= \frac{1}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{2}{z}\right)^n - \frac{1}{3}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{3}\right)^n$$

Que: $f(z) = \dfrac{z^2-1}{(z+2)(z+3)}$

i. when $|z| < 2$

ii. when $2 < |z| < 3$

iii. when $|z| > 3$

Soln

$$f(z) = \frac{z^2-1}{(z+2)(z+3)}$$

$$f(z) = 1 - \frac{(5z+7)}{(z+2)(z+3)}$$

$$= 1 - \left\{\frac{A}{z+2} + \frac{B}{z+3}\right\}$$

$$f(z) = A(z+3) + B(z+2)$$

$$\therefore A = 3, \quad B = 8$$

$$f(z) = 1 - \left\{\frac{-3}{z+2} + \frac{8}{z+3}\right\}$$

$$f(z) = 1 + \frac{9}{z+2} - \frac{8}{z+3}$$

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i. when $|z| < 2$.

$\frac{|z|}{2} < 1$

from ①, $f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$

$= 1 + \frac{3}{2}\left(1+\frac{z}{2}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$

$= 1 + \frac{3}{2}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{2}\right)^n - \frac{8}{3}\sum(-1)^n\left(\frac{z}{3}\right)^n$

ii. when $2 < |z| < 3$.

Soln.

$\left|\frac{2}{z}\right| < 1$ & $\left|\frac{z}{3}\right| < 1$

$f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$

$= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$

$= 1 + \frac{3}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{2}{z}\right)^n - \frac{8}{3}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{3}\right)^n$

iii. $|z| > 3$

$\left|\frac{z}{3}\right| > 3$

$f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$

$= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{z}\left(1+\frac{3}{z}\right)^{-1}$

$= 1 + \frac{3}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{2}{z}\right)^n - \frac{8}{z}\sum_{n=0}^{\infty}(-1)^n\left(\frac{3}{z}\right)^n$

$\frac{1}{i} = -i$

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Q. Let $f(z) = \frac{2z-3i}{z^2-3iz-2}$, expand $f(z)$ in the region

i) $1 < |z| < 2$

Soln,

Here, $f(z) = \frac{2z-3i}{(z-2i)(z-i)}$

$f(z) = \frac{1}{z-2i} + \frac{1}{z-i}$ when $1 < |z| < 2$

$1 < |z|$ & $|z| < 2$

$\frac{1}{|z|} < 1$ & $\frac{|z|}{2} < 1$

$= \frac{1}{-2i\left(1-\frac{z}{2i}\right)} + \frac{1}{z\left(1-\frac{i}{z}\right)}$

$\therefore f(z) = \frac{i}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2i}\right)^n + \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{i}{z}\right)^n$

ii) $0 < |z+i| < 2$

Soln

Here, $f(z) = \frac{2z-3i}{(z-2i)(z-i)}$

when $0 < |z+i| < 2$

put $z+i = u$

$0 < |u| < 2$

$\therefore \frac{|u|}{2} < 1$

~~Here~~ So that, $f(z) = \frac{1}{u-i-2i} + \frac{1}{u-i-i}$

$= \frac{1}{u-3i} + \frac{1}{u-2i}$

$$= \frac{1}{-3i} \frac{1}{\left(1-\frac{u}{3i}\right)} - \frac{1}{2i} \frac{1}{\left(1-\frac{y}{2i}\right)}$$

$$= \frac{i}{3} \sum_{n=0}^{\infty} \left(\frac{u}{3i}\right)^n + \frac{i}{2} \sum_{n=0}^{\infty} \left(\frac{y}{2i}\right)^n$$

$$= \frac{i}{3} \sum_{n=0}^{\infty} \left(\frac{z+i}{3i}\right)^n + \frac{i}{2} \sum_{n=0}^{\infty} \left(\frac{z+i}{2i}\right)^n$$

Q) Year: 2015
Fall

2a) $f(z) = \frac{1}{(z^2-z^3)}$ in the region $0<|z|<1$.

$$= \frac{1}{z^2(1-z)}$$

$$= \frac{1}{z^2}\{1+z+z^2+z^3+\cdots\}$$

$$= \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \cdots$$

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8. Find the expansion for $\frac{7z-2}{(z+1)z(z-2)}$ in the regions given by

a) $0<|z+1|<1$
b) $1<|z+1|<3$
c) $|z+1|>3$

Soln:

Here, $f(z) = \frac{7z-2}{z(z+1)(z-2)}$

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Let $\frac{7z-2}{z(z+1)(z-2)} = \frac{A}{z} + \frac{B}{z+1} + \frac{C}{z-2}$

or, $7z-2 = A(z+1)(z-2) + B\,z(z-2) + C\,z(z+1)$

when $z=-1$, when $z=2$

or, $-9 = 3B$ or, $12 = C6$

$\therefore B = -3$ $\therefore C = 2$

when $z=0$

or, $\cancel{+2} = \cancel{+}A\cancel{2}$

$\therefore A = 1$

So that $f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$ ——— (1)

~~when~~ put $u = z+1$

a) $0 < |u| < 1$

or, $|u| < 1$

from (1)

$$f(z) = \frac{1}{u-1} - \frac{3}{u} + \frac{2}{u-3}$$

$$= -\frac{3}{u} - \frac{1}{1-u} - \frac{2}{3}\frac{1}{(1-u/3)}$$

$$= -\frac{3}{u} - \sum_{n=0}^{\infty} u^n - \frac{2}{3}\sum_{n=0}^{\infty}\left(\frac{u}{3}\right)^n$$

$$= \frac{-3}{z+1} - \sum_{n=0}^{\infty}(z+1)^n - \frac{2}{3}\sum_{n=0}^{\infty}\left(\frac{z+1}{3}\right)^n$$

b) $1 < |z+1| < 3$

put $u = z+1$

$1 < |u| < 3$

$1 < |u|$, $|u| < 3$

or, $\frac{1}{|u|} < 1$ $\frac{|u|}{3} < 1$

from (1),

$$f(z) = \frac{1}{u-1} - \frac{3}{u} + \frac{2}{u-3}$$

$$= -\frac{3}{u} + \frac{1}{u\left(1-\frac{1}{u}\right)} - \frac{2}{3}\left(\frac{1}{1-u/3}\right)$$

$$= -\frac{3}{u} + \frac{1}{u}\sum_{n=0}^{\infty}\left(\frac{1}{u}\right)^n - \frac{2}{3}\sum_{n=0}^{\infty}\left(\frac{u}{3}\right)^n$$

$$= -\frac{3}{z+1} + \frac{1}{z+1}\sum_{n=0}^{\infty}\left(\frac{1}{z+1}\right)^n - \frac{2}{3}\sum_{n=0}^{\infty}\left(\frac{z+1}{3}\right)^n$$

c) $|z+1| > 3$

put $u = z+1$

$|u| > 3$

$\therefore \frac{3}{|u|} < 1$

from (1),

$$f(z) = \frac{1}{u-1} - \frac{3}{u} + \frac{2}{u-3}$$

$$= -\frac{3}{u} + \frac{1}{u\left(1-\frac{1}{u}\right)} + \frac{2}{u}\left(\frac{1}{1-3/u}\right)$$

$$= -\frac{3}{u} + \frac{1}{u}\sum_{n=0}^{\infty}\left(\frac{1}{u}\right)^n + \frac{2}{u}\sum_{n=0}^{\infty}\left(\frac{3}{u}\right)^n$$

$$= -\frac{3}{u} + \frac{1}{z+1}\sum_{n=0}^{\infty}\left(\frac{1}{z+1}\right)^n + \frac{2}{z+1}\sum_{n=0}^{\infty}\left(\frac{3}{z+1}\right)^n$$

Singularities, Zeros & Poles

Imp. Definition.

A complex function $f(z)$ has a singularity or is singular at a point $z=a$ if $f(z)$ is not analytic at $z=a$ but neighbourhood of $z=$"a" contain at a point at which $f(z)$ is analytic the point $z=a$ is known as a singular point of the function $f(z)$. if $f(z)=\frac{1}{z}$, then we have, $f'(z)=-\frac{1}{z^2}$, so that

the function $f(z)$ is analytic at all points except $z=0$. The point $z=0$ is a singular point of $f(z)=\frac{1}{z}$.

Types of singularities

① Isolated Singular point:

The singular point $z=a$ of the function $f(z)$ is called an isolated singular point if the point $z=a$ has a neighbourhood without further singularities of $f(z)$.

e.g. $f(z)=\tan z$

$$=\frac{\sin z}{\cos z}$$

$$, z=\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2} \text{ etc.}$$

Isolated singularities of $f(z)$ at $z=a$ can be expanded by the laurent's series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

$$= \sum_{n=0}^{\infty} a_n (z-a)^n + \frac{b_1}{z-a} + \frac{b_2}{(z-a)^2} + \frac{b_3}{(z-a)^3} + \dots + \frac{b_n}{(z-a)^n} \dots$$

The principal part of laurent's series expansions only finite term

$$\frac{b_1}{z-a} + \frac{b_2}{(z-a)^2} + \frac{b_3}{(z-a)^3} + \dots + \frac{b_n}{(z-a)^n}$$, then $z=a$

is called pole of order n of the function $f(z)$.

② Essential Singularity:

The principal part of laurent's series expansion has infinitely many terms then $z=a$ is called an essential singular point.

eg ① $e^{\frac{1}{z}} = 1 + \frac{\frac{1}{z}}{1!} + \frac{\frac{1}{z^2}}{2!} + \frac{\frac{1}{z^3}}{3!} + \dots$

$\therefore z=0$ is an essential singular point.

③ Removable Singularity

If $b_n = 0$ for all n then the singularity $z=z_0$ is called a removable ~~sig~~ singularity.

e.g ① $f(z) = \frac{1}{\left(z-\frac{\pi}{4}\right)} \sin\left(z-\frac{\pi}{4}\right)$

$$= \frac{1}{\left(z-\frac{\pi}{4}\right)} \left\{ \left(z-\frac{\pi}{4}\right) - \frac{\left(z-\frac{\pi}{4}\right)^3}{3!} + \frac{\left(z-\frac{\pi}{4}\right)^5}{5!} \dots \right\}$$

$$= 1 - \frac{\left(z-\frac{\pi}{4}\right)^2}{3!} + \frac{\left(z-\frac{\pi}{4}\right)^4}{5!}$$

$\therefore z = \frac{\pi}{4}$ is called removable singularity.

Zeros of analytic function

A zero of an analytic function $f(z)$ in the region R and $z=a$ in R such that $f(a)=0$, which is a simple zeros.

eg: $f(z) = z^2+1$
$f(\pm i) = 0$

A zero has order n if $f(a)=0$, $f'(a)=0$, $f''(a)=0$ .... $f^{n-1}(a)=0$ but $f^n(a) \neq 0$

In the eg, the function $f(z)$ has a simple zero at $z=\pm i$.

Eg: $f(z) = (1-z^4)^2$ has a zeros of order second.

$f(\pm 1) = 0$

$f'(z) = 2(1-z^4).(-4z^3)$

$f'(\pm 1) = 0$

but $f''(1) \neq 0$

Imp # Residues of Complex function

The coefficient of $\frac{1}{z-a}$ (i.e. b_1) in Laurent's series expansion is called residue of $f(z)$ at $z=a$ and we denote it by $b_1 = \frac{1}{2\pi i}\int_c f(z)\,dz$

$$\frac{1}{2\pi i}\int_c f(z)\,dz = \left\{ \operatorname*{Res}_{z=a} f(z) \right\}$$

In other word, $\oint_c f(z)\,dz = 2\pi i \left\{ \operatorname*{Res}_{z=a} f(z) \right\}$

Residue at a simple pole

1) If $f(z)$ has a pole at $z=a$ then

$$\left\{ \operatorname*{Res}_{z=a} f(z) \right\} = \lim_{z \to a} (z-a)\, f(z)$$

Proof:

We have, $f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$

$$= \sum_{n=0}^{\infty} a_n (z-a)^n + \frac{b_1}{z-a}$$

we have, $f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + \frac{b_1}{z-a}$

multiplying it by $(z-a)$ on both sides,

$(z-a) f(z) = a_0(z-a) + a_1(z-a)^2 + a_2(z-a)^3 + \dots + b_1$

$\lim_{z\to a} (z-a) f(z) = b_1$

$$\boxed{\therefore \left\{ \begin{matrix} \text{Res } f(z) \\ \text{at } z=a \end{matrix} \right\} = \lim_{z\to a} (z-a) f(z)}$$

Q. Find the residue of $f(z) = \frac{z^2+3}{(z-4)(z-5)}$

Soln:

The pole of the function are given by potting

$$(z-4)(z-5) = 0$$

$$z = 4, 5$$

$$\left\{ \begin{matrix} \text{Res } f(z) \\ \text{at } z=4 \end{matrix} \right\} = \lim_{z\to 4} (z-4) f(z)$$

$$= \lim_{z\to 4} \cancel{(z-4)} \frac{z^2+3}{\cancel{(z-4)}(z-5)}$$

$$= -19$$

$$\left\{ \begin{matrix} \text{Res } f(z) \\ \text{at } z=5 \end{matrix} \right\} = \lim_{z\to 5} (z-5) f(z)$$

$$= \lim_{z\to 5} \cancel{(z-5)} \frac{z^2+3}{(z-4)\cancel{(z-5)}}$$

$$= 28$$

$$f(z) = \frac{z^2+3}{(z-4)(z-5)}$$

Solⁿ

$$\frac{z^2+3}{(z-4)(z-5)} = \frac{A}{z-4} + \frac{B}{z-5}$$

$$z^2+3 = A(z-5) + B(z-4)$$

when $z = 5$, when $z = 4$

$28 = B$ $\quad\quad 19 = -A$

$\therefore B = 28$ $\quad\quad \therefore A = -19$

Residue at a pole of order n

If a function $f(z)$ has a pole of order n at $z=a$ then,

$$\left\{\begin{matrix} \text{Res } f(z) \\ z=a \end{matrix}\right\} = \frac{1}{(n-1)!} \lim_{z\to a} \frac{d^{n-1}}{dz^{n-1}}\{(z-a)^n f(z)\}$$

Que: Find the residues of $f(z) = \dfrac{z^2+3}{(z-4)(z-5)^2}$

Solⁿ:

The pole of the function are given by putting $(z-4)(z-5)^2 = 0$

$z=4$ is a simple pole

& $z=5$ has order 2.

$$\left\{\begin{matrix} \text{Res } f(z) \\ \text{at } z=4 \end{matrix}\right\} = \lim_{z\to 4} (z-4) f(z)$$

$$= \lim_{z\to 4} \cancel{(z-4)} \frac{(z^2+3)}{\cancel{(z-4)}(z-5)^2}$$

$$= 19$$

$$\left\{ \begin{array}{l} \text{Res } f(z) \\ \text{at } z=5 \end{array} \right\} = \frac{1}{1!} \lim_{z \to 5} \frac{d}{dz} (z-5)^2 f(z)$$

$$= \lim_{z \to 5} \frac{d}{dz} \cancel{(z-5)^2} \frac{z^2+3}{(z-4)\cancel{(z-5)^2}}$$

$$= \lim_{z \to 5} \frac{(z-4)2z - (z^2+3)}{(z-4)^2}$$

$$= \frac{10-28}{+1}$$

$$= -18$$

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Que: Find the singular points and residues of the function.

$$f(z) = \frac{z+2}{(z-2)(z^2+1)^2}$$

Soln

The pole of the function are given by putting.

$$(z-2)(z^2+1)^2 = 0$$

$z = 2$ is a simple pole.

& $z^2 = -1$

or, $z^2 = i^2$

$\therefore z = \pm i$ has order 2

$$\left\{ \begin{array}{l} \text{Res } f(z) \\ \text{at } z=2 \end{array} \right\} = \lim_{z \to 2} (z-2) f(z)$$

$$= \lim_{z \to 2} \cancel{(z-2)} \frac{(z+2)}{\cancel{(z-2)}(z^2+1)^2}$$

$$= \frac{4}{25}$$

$$\left\{\begin{matrix}\text{Res } f(z)\\ \text{at } z=\pm i\end{matrix}\right\} = \frac{1}{1!}\lim_{z\to i}\frac{d}{dz}\left\{(z-i)^2 f(z)\right\}$$

$$=\lim_{z\to i}\frac{d}{dz}(z-i)^2\frac{z+2}{(z-2)(z^2+1)^2}$$

$$\cancel{=\lim_{z\to i}\frac{d}{dz}(z-i)^2\frac{z+2}{(z^2-i^2)^2(z-2)}}$$

$$\cancel{=\lim_{z\to i}\frac{d}{dz}(z-i)^2\frac{z+2}{(z-2)\{(z+i)(z-i)\}^2}}$$

$$=\lim_{z\to i}\frac{d}{dz}(z-i)^2\frac{z+2}{(z-2)(z^2-i^2)^2}$$

$$=\lim_{z\to i}\frac{d}{dz}(z-i)^2\frac{z+2}{(z-2)\{(z+i)(z-i)\}^2}$$

$$=\lim_{z\to i}\frac{d}{dz}\cancel{(z-i)^2}\frac{z+2}{(z-2)(z+i)^2\cancel{(z-i)^2}}$$

$$=\lim_{z\to i}\left[\frac{(z+i)^2(z-2)\cdot 1-(z+2)\frac{d}{dz}\{(z-2)(z+i)^2\}}{\{(z-2)(z+i)^2\}^2}\right]$$

$$=\lim_{z\to i}\left[\frac{(z+i)^2(z-2)-(z+2)\{2(z+i)(z-2)+(z+i)^2\}}{(z+i)^4(z-2)^2}\right]$$

$$=\lim_{z\to i}\frac{(z+i)^2(z-2)-2(z^2-4)(z+i)-(z+2)(z+i)^2}{(z+i)^4(z-2)^2}$$

$$=\frac{(2i)^2(i-2)-2(i^2-4)(i+i)-(i+2)(i+i)^2}{(i+i)^4(i-2)^2}$$

$$=\frac{-4(i-2)-2(-5)(2i)-(i+2)(2i)^2}{(2i)^4(i-2)^2}$$

$$=\frac{-4(i-2)+20i-(i+2)(4i^2)}{(2i)^4(i-2)^2}$$

$$=\frac{-4i+8+20i+4i+8}{16(i^2-4i+4)}$$

$$= \frac{20i+16}{16(3-4i)}$$

$$= \frac{4(5i+4)(3+4i)}{16(3-4i)(3+4i)}$$

$$= \frac{15i-20+12+16i}{16(9+16)}$$

$$= \frac{31i-8}{100}$$

similarly residue at $z=-i = -\left(\frac{31i-8}{100}\right)$

Cauchy's Residues Theorem

Statement: If $f(z)$ is analytic inside a simple closed curve C except at a finite number of poles within c then

$$\oint_c f(z)\,dz = 2\pi i \sum_{j=1}^{n} \left\{ \underset{\text{at } z=a_j}{\text{Res}}\, f(z) \right\}.$$

Que: Integrate:

$$\oint_c \frac{z-23}{z^2-4z-5}\,dz \quad \text{where } c: |z-2| = 4.$$

Soln

The pole of integrand are given by putting

$z^2 - 4z - 5 = 0$

$\therefore z = 5, -1$

For figure,

$|z-2| = 4$

or, $|x+iy-2| = 4$

or, $|(x-2)+iy| = 4$

or, $(x-2)^2 + y^2 = 4^2$, centre $(2,0)$

put $y=0$, $(x-2)^2 = 4^2$

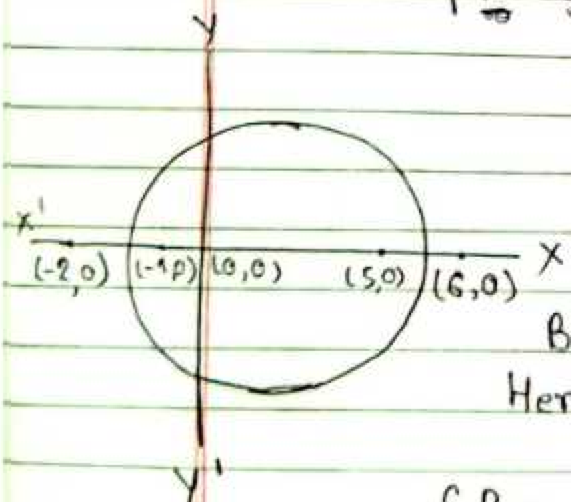
$x - 2 = \pm 4$

Taking +ve, $x - 2 = 4$

$\therefore x = 6$

Taking -ve, $x - 2 = -4$

$\therefore x = -2$



Both $z = 5, -1$ lie within given circle.

Here, $f(z) = \dfrac{z-23}{(z+1)(z-5)}$

$$\left\{ \underset{\text{at } z=5}{\text{Res}}\, f(z) \right\} = \lim_{z\to 5} (z-5)\, f(z)$$

$$= \lim_{z \to 5} \cancel{(z-5)} \frac{z-23}{(z+1)\cancel{(z-5)}}$$

$$= -3$$

$$\left\{ \begin{array}{l} \text{Res } f(z) \\ \text{at } z=-1 \end{array} \right\} = \lim_{z \to -1} (z+1) f(z)$$

$$= \lim_{z \to -1} \cancel{(z+1)} \frac{z-23}{\cancel{(z+1)}(z-5)}$$

$$= 4$$

By Cauchy's Residues Theorem,

$$\oint_c f(z)\, dz = 2\pi i \,[\text{sum of Residue}]$$
$$= 2\pi i\, [-3+4]$$
$$= 2\pi i$$

$$\therefore \oint_c \frac{z-23}{z^2-4z-5}\, dz = 2\pi i$$

Q. Integrate:

$$\oint_c \frac{1}{z^8(z+4)}\, dz \quad \text{where } c: |z| = 2.$$

or, $|z| = 2$

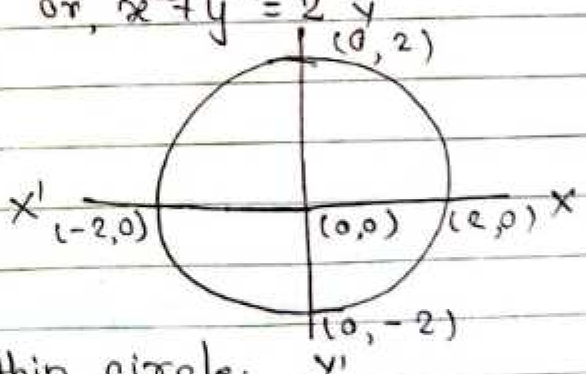
or, $|x+iy| = 2$

or, $x^2+y^2 = 2^2$

Soln,

Here,

$$f(z) = \frac{1}{z^8(z+4)}$$



The poles are $z^8(z+4) = 0$

$z = 0$ (has order 8) lies within circle

& $z = -4$ doesnot lies within circle.

$$\left\{ \begin{array}{l} \text{Res } f(z) \\ \text{at } z=0 \end{array} \right\} = \frac{1}{7!} \lim_{z \to 0} \frac{d^7}{dz^7} z^8 f(z)$$

$$\frac{d^n}{dx^n}\left(\frac{1}{x+a}\right) = (-1)^n n! \frac{1}{(x+a)^{n+1}}$$

$$= \frac{1}{7!} \lim_{z\to 0} \frac{d^7}{dz^7}\left\{ \cancel{z^8} \frac{1}{\cancel{z^8}(z+4)} \right\}$$

$$= \frac{1}{\cancel{7!}} \lim_{z\to 0} (-1)^7 \cancel{7!} \frac{1}{(z+4)^8}$$

$$= -\frac{1}{4^8}$$

By cauchy's Residues Theorem,

$$\oint_c f(z)\,dz = 2\pi i\left[-\frac{1}{4^8}\right]$$

$$= \frac{-\pi i}{2^{15}}$$

Q. Integrate: $\oint_c \frac{1}{z^8(z+4)}\,dz$ where $c : |z+2| = 3$

or, $|x+iy+2| = 3$

or, $(x+2)^2 + y^2 = 3^2$

it's centre $(-2,0)$

put $y=0$,

$(x+2)^2 = 3^2$

$x+2 = \pm 3$

Taking +ve, $x+2=3$

$\therefore x = 1$

Taking -ve, $x+2=-3$

$\therefore x = -5$

Soln,

Here, $f(z) = \oint_c \frac{1}{z^8(z+4)}$

The poles are $z^8(z+4)=0$

$z=0$ (has order 8) &

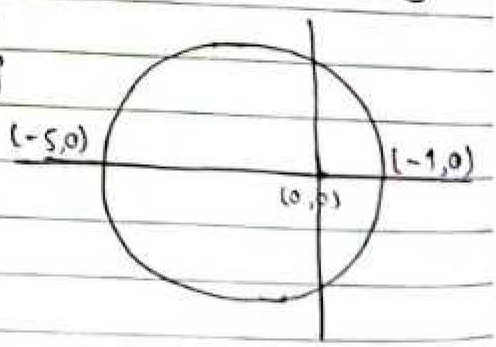
$z=-4$ both lie within circle,

$$\left\{\begin{matrix}\text{Res } f(z) \\ \text{at } z=0\end{matrix}\right\} = \frac{1}{7!} \lim_{z\to 0} \frac{d^7}{dz^7}\left\{z^8 f(z)\right\}$$

$$= \frac{1}{7!} \lim_{z\to 0} \frac{d^7}{dz^7}\left\{ \cancel{z^8} \frac{1}{\cancel{z^8}(z+4)} \right\}$$

$$= \frac{1}{\cancel{7!}} \lim_{z\to 0} (-1)^7 \cancel{7!} \frac{1}{(z+4)^8}$$

$$= -\frac{1}{4^8}$$



and $\left\{ \begin{matrix} \text{Res } f(z) \\ z=-4 \end{matrix} \right\} = \lim_{z \to -4} \cancel{(z+4)} \dfrac{1}{z^8 \cancel{(z+4)}}$

$$= \frac{1}{(-4)^8}$$

$$= \frac{1}{4^8}$$

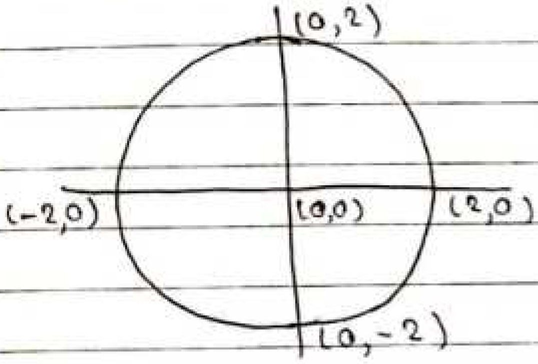
By Cauchy's Residues Theorem,

$$\oint_c f(z)\,dz = 2\pi i \left[-\frac{1}{4^8} + \frac{1}{4^8} \right]$$

$$= 0$$

Spring - 2013

2a) $\oint_c \left(\dfrac{z^2 - \sin z}{4z^2 - 1} \right) dz$ where $c : |z| = 2$, counter-clockwise (anticlockwise)



Soln

$$= \oint_c \frac{\frac{z^2 - \sin z}{4}}{z^2 - (1/2)^2}\, dz$$

$$= \oint_c \frac{\frac{z^2 - \sin z}{4}}{\left(z - \frac{1}{2}\right)\left(z + \frac{1}{2}\right)}\, dz$$

The poles are $z = \frac{1}{2}, -\frac{1}{2}$ are within circle;

$\left\{ \begin{matrix} \text{Res } f(z) \\ \text{at } z = 1/2 \end{matrix} \right\} = \lim_{z \to \frac{1}{2}} \left(z - \frac{1}{2}\right) f(z)$

$$= \lim_{z \to \frac{1}{2}} \cancel{\left(z - \frac{1}{2}\right)} \frac{\frac{z^2 - \sin z}{4}}{\cancel{\left(z - \frac{1}{2}\right)}\left(z + \frac{1}{2}\right)}$$

$$= \frac{\frac{\frac{1}{4}\sin\frac{1}{2}}{4}}{1}$$

$$= \frac{1}{16}\sin\frac{1}{2}$$

$$\left\{\begin{matrix} \text{Res } f(z) \\ \text{at } z=-1/2 \end{matrix}\right\} = \lim_{z\to -1/2} \left(z+\frac{1}{2}\right) f(z)$$

$$= \lim_{z\to -\frac{1}{2}} \cancel{\left(z+\frac{1}{2}\right)} \frac{\frac{z^2 - \sin z}{4}}{\left(z-\frac{1}{2}\right)\cancel{\left(z+\frac{1}{2}\right)}}$$

$$= \frac{\frac{\frac{1}{4}\sin\left(-\frac{1}{2}\right)}{4}}{-1} \qquad [\sin(-1) = -\sin 1]$$

$$= \frac{1}{16}\sin\left(\frac{1}{2}\right)$$

By Cauchy's Residue Theorem,

$$\oint_c f(z)\,dz = 2\pi i\left[\frac{1}{16}\sin\left(\frac{1}{2}\right) + \frac{1}{16}\sin\left(\frac{1}{2}\right)\right]$$

$$= \frac{\pi i}{4}\sin\left(\frac{1}{2}\right)$$

② $1+2r+3r^2+4r^3+\ldots = \frac{1}{(1-r)^2}$

① $a+ar+ar^2+ar^3+\ldots\ldots$ $S_\infty = \frac{a}{1-r}$ where $r = \frac{t_n}{t_{n-1}}$

Unit 2: 7+8+2.5+7

Z-Transform and It's Application

Z-Transform:

Let f(t) be a function of time period "t" where t is non negative defined at 0, T, 2T, 3T which are the discrete values of t. Then z-transform of f(t) is denoted by ~~f(t)~~ $Z[f(t)]$ and defined by

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$ which is function and converges.

The fixed positive number T is reffered to a sampling period.

i.e. $$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$= F(z)$$

or, $$Z^{-1}[F(z)] = f(t) \text{ or } f(nT)$$

Q. Find z-transform using def^n method.

i) $Z[1]$

We have,

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

where $f(t) = 1$

i.e. $$Z[1] = \sum_{n=0}^{\infty} 1\, z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{z^n}$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \ldots\ldots$$

$$= \frac{1}{1-\frac{1}{z}} \qquad \left(r = \frac{1}{z}\right)$$

$$\boxed{Z[1] = \frac{z}{z-1}}$$

$$\boxed{Z^{-1}\left(\frac{z}{z-1}\right) = 1}$$

ii) $Z[(-1)^n] = \sum_{n=0}^{\infty} (-1)^n z^{-n}$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{z^n}$$

$$= 1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \cdots\cdots \qquad \left(r = -\frac{1}{z}\right)$$

$$= \frac{1}{1-\left(\frac{-1}{z}\right)}$$

$$\boxed{\therefore Z[(-1)^n] = \frac{z}{z+1}}$$

$$\boxed{Z^{-1}\left(\frac{z}{z+1}\right) = (-1)}$$

iii) $Z[a^n]$

We have, $Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

where $f(t) = a^n$

i.e. $Z[a^n] = \sum_{n=0}^{\infty} a^n z^{-n}$

$$= \sum_{n=0}^{\infty} \frac{a^n}{z^n}$$

$$= 1 + \frac{a}{z} + \frac{a^2}{z^2} + \frac{a^3}{z^3} + \cdots\cdots$$

$$= \frac{1}{1 - \frac{a}{z}}$$

$$\boxed{z[a^n] = \frac{z}{z-a}}$$

$$\boxed{z^{-1}\left(\frac{z}{z-a}\right) = a^n}$$

eg: $z[2^n] = \frac{z}{z-2}$ eg: $z^{-1}\left[\frac{z}{z-2}\right] = 2^n$

iv) $z[(-1)^n a^n] = \sum_{n=0}^{\infty} (-1)^n a^n z^{-n}$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\frac{a}{z}\right)^n$$

$$= 1 - \frac{a}{z} + \frac{a^2}{z^2} - \frac{a^3}{z^3} \cdots\cdots$$

$$= \frac{1}{1 - \left(\frac{-a}{z}\right)}$$

$$\boxed{\therefore z[(-1)^n a^n] = \frac{z}{z+a}}$$

$$\boxed{z^{-1}\left(\frac{z}{z+a}\right) = (-1)^n a^n}$$

eg: $z^{-1}\left(\frac{z}{z+2}\right) = (-1)^n 2^n$

v) $Z[n] = \sum_{n=0}^{\infty} n\, z^{-n}$

$$= \sum_{n=0}^{\infty} \frac{n}{z^n}$$

$$= 0 + \frac{1}{z} + \frac{2}{z^2} + \frac{3}{z^3} + \frac{4}{z^4} + \ldots\ldots$$

$$= \frac{1}{z}\left[1 + 2\cdot\frac{1}{z} + 3\cdot\frac{1}{z^2} + 4\cdot\frac{1}{z^3} + \ldots\ldots\right]$$

$$= \frac{1}{z}\ \frac{1}{\left(1-\frac{1}{z}\right)^2}$$

$$\boxed{Z[n] = \frac{z}{(z-1)^2}}$$

$$\boxed{Z^{-1}\left[\frac{z}{(z-1)^2}\right] = n}$$

vi) $Z\left[(-1)^{n+1} n\right] = \sum_{n=0}^{\infty} (-1)^{n+1} n z^{n}$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{n+1} n}{z^n}$$

$$= 0 + \frac{1}{z} - \frac{2}{z^2} + \frac{3}{z^3} - \frac{4}{z^4} + \ldots\ldots$$

$$= \frac{1}{z}\left[1 - 2\cdot\frac{1}{z} + 3\cdot\frac{1}{z^2} - 4\cdot\frac{1}{z^3} + \ldots\right]$$

$$= \frac{1}{z}\ \frac{1}{\left[1-\left(\frac{-1}{z}\right)\right]^2}$$

$$\boxed{Z[n] = \frac{z}{(z+1)^2}}$$

$$\boxed{Z^{-1}\left[\frac{z}{(z+1)^2}\right] = (-1)^{n+1} n}$$

$$e^z = 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \cdots$$

$$\log(1+z) = \frac{z}{1} - \frac{z^2}{2} + \frac{z^3}{3} \cdots$$

$$\log(1-z) = \log(1+(-z))$$
$$= -z - \frac{z^2}{2} - \frac{z^3}{3} - \cdots$$
$$= -\left(z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots\right)$$
$$-\log(1-z) = z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots$$

vii) $$Z[na^n] = \sum_{n=0}^{\infty} n a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} n\left(\frac{a}{z}\right)^n$$

$$= 0 + \frac{a}{z} + \frac{2a^2}{z^2} + \frac{3a^3}{z^3} + \frac{4a^4}{z^4} + \cdots$$

$$= \frac{a}{z}\left[1 + 2\cdot\frac{a}{z} + 3\frac{a^2}{z^2} + 4\frac{a^3}{z^3} + \cdots\right]$$

$$= \frac{a}{z}\,\frac{1}{\left(1-\frac{a}{z}\right)^2}$$

$$\boxed{Z[na^n] = \frac{az}{(z-a)^2}}$$

$$\boxed{Z^{-1}\left[\frac{az}{(z-a)^2}\right] = na^n}$$

viii) $$Z\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$$

$$= 1 + \frac{1}{1!}\frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots$$

$$= e^{1/z}$$

ix) $$Z\left[\frac{1}{n+1}\right] = \sum_{n=0}^{\infty} \frac{1}{n+1} z^{-n}$$

$$= 1 + \frac{1}{2}\cdot\frac{1}{z} + \frac{1}{3}\frac{1}{z^2} + \frac{1}{4}\frac{1}{z^3} + \cdots$$

$$= z\left[\frac{1}{z} + \frac{1}{2}\left(\frac{1}{z}\right)^2 + \frac{1}{3}\left(\frac{1}{z}\right)^3 + \cdots\right]$$

$$= -z\log\left(1 - \frac{1}{z}\right)$$

Unit Step function:

A function defined by the relation

$$f(n) = \begin{cases} 1 & \text{for } n = 0, 1, 2, \ldots \\ 0 & \text{for } n < 0 \end{cases}$$

is said to be unit step function.

Unit Ramp function:

A function defined by $f(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$

Unit Impulse function:

A function is of the form $f(n) = \delta(n) = \begin{cases} 1 & \text{for } n = 0, \\ 0 & \text{for } n \neq 0 \end{cases}$

is said to be unit impulse function or Dirac-delta function.

Q. Find the z-transform for discrete time unit impulse $\delta(n)$

Soln.

We have,

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Now, $Z[\delta(n)] = \sum_{n=0}^{\infty} \delta(n) \frac{1}{z^n}$.

$$= \delta(0) + \delta(1)\frac{1}{z} + \delta(2)\frac{1}{z^2} + \ldots\ldots$$

$$= 1 + 0 + 0 \ldots\ldots$$

$$Z[\delta(n)] = 1$$

$$\boxed{Z^{-1}[1] = \delta(n)}$$

Q. Find the z-transform of discrete time unit step signal u(n)

Soln:

We have,

$u(n) = 1$ for $n = 0, 1, 2, 3, \ldots\ldots$

$= 0$ for $n < 0$

Now,

$$Z[u(n)] = \sum_{n=0}^{\infty} u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} u(n) \frac{1}{z^n}$$

$$= u(0) + u(1)\frac{1}{z} + u(2)\frac{1}{z^2} + \ldots\ldots$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \ldots\ldots$$

$$= \frac{1}{1 - \frac{1}{z}}$$

$$\boxed{Z[u(n)] = \frac{z}{z-1}}$$

Q. Find the z-transform of Unit Ramp function.

Soln:

We have, unit Ramp function $f(t) = t$ for $t \geq 0$

$= 0$ for $t < 0$

Now, $Z[f(t)] = \sum_{n=0}^{\infty} t \cancel{z}^{-n}$

$$= Z[t]$$

$$= \sum_{n=0}^{\infty} nT z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{nT}{z^n}$$

$$= 0 + \frac{T}{z} + \frac{2T}{z^2} + \frac{3T}{z^3} + \dots\dots$$

$$= \frac{T}{z}\left[1 + 2\cdot\frac{1}{z} + 3\cdot\left(\frac{1}{z}\right)^2 + \dots\right]$$

$$= \frac{T}{z}\,\frac{1}{\left(1-\frac{1}{z}\right)^2}$$

$$\boxed{\therefore z[f(t)] = \frac{Tz}{(z-1)^2}}$$

Theorem: (Linearity property of Z-transform)

Let $f(t)$ & $g(t)$ are two function of t in the discrete time period $0, T, 2T, \dots$ and a & b are constant then $Z[af(t) + bg(t)] = aZ[f(t)] + bZ[g(t)]$

Proof:

By def^n,

$$Z[af(t) + bg(t)] = \sum_{n=0}^{\infty}\{af(nT) + bg(nT)\}z^{-n}$$

$$= a\sum_{n=0}^{\infty} f(nT)\, z^{-n} + b\sum_{n=0}^{\infty} g(nT)\, z^{-n}$$

$$= aZ[f(t)] + bZ[g(t)]$$

Proved

Q) Find the z-transform of $e^{\frac{in\pi}{2}}$ and then find $z\left[\cos\frac{n\pi}{2}\right]$ & $z\left[\sin\frac{n\pi}{2}\right]$.

Soln,

We have,

$$z[a^n] = \frac{z}{z-a}$$

$$\text{Now, } z\left[\left(e^{\frac{i\pi}{2}}\right)^n\right] = \frac{z}{z-e^{i\pi/2}}$$

$$= \frac{z}{z-\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)}$$

$$= \frac{z}{z-i}$$

$$= \frac{z}{z-i} \times \frac{z+i}{z+i}$$

$$= \frac{z^2+iz}{z^2+1}$$

also,

$$z\left[\cos\frac{n\pi}{2}+i\sin\frac{n\pi}{2}\right] = \frac{z^2}{z^2+1} + i\frac{z}{z^2+1}$$

$$\text{or, } z\left[\cos\frac{n\pi}{2}\right] + iz\left[\sin\frac{n\pi}{2}\right] = \frac{z^2}{z^2+1} + i\frac{z}{z^2+1}$$

Equating real & imaginary part,

$$\therefore z\left[\cos\frac{n\pi}{2}\right] = \frac{z^2}{z^2+1} \quad \&$$

$$z\left[\sin\frac{n\pi}{2}\right] = \frac{z}{z^2+1}.$$

State & prove 1st shifting theorem

If $Z[f(t)] = F(z)$ then $Z[e^{-at} f(t)] = F(e^{aT} z)$

Proof:

We have,

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$= F(z) \quad \text{——— (1)}$$

Now,

$$Z[e^{-at} f(t)] = \sum_{n=0}^{\infty} f(nT) e^{-anT} z^{-n}$$

$$= \sum_{n=0}^{\infty} f(nT) (e^{aT} z)^{-n}$$

$$= F(e^{aT} z)$$

eg (1) find $Z[e^{-at}]$

Soln:

$$= Z[e^{-at} \cdot 1]$$

$$= Z[1]_{z \to e^{aT} z}$$

$$= \left[\frac{z}{z-1}\right]_{z \to e^{aT} z}$$

$$= \frac{e^{aT} z}{e^{aT} z - 1}$$

$$= \frac{\cancel{e^{aT}} z}{\cancel{e^{aT}} (z - e^{-aT})}$$

$$= \frac{z}{z - e^{-aT}}$$

$$\boxed{\therefore Z(e^{-at}) = \frac{z}{z - e^{-aT}}}$$

V.V.Imp

Q. find $z(e^{-iat})$ and hence deduce the value of $z(\cos at)$ & $z(\sin at)$

Soln:

Here, $z(e^{-iat}\cdot 1)$

By 1st shifting theorem

$$z(1)_{z\to e^{iaT}z}$$

$$= \left[\frac{z}{z-1}\right]_{z\to e^{iaT}z}$$

$$= \frac{e^{iaT}z}{e^{iaT}z-1}$$

$$z(e^{-iat}) = \frac{\cancel{e^{iaT}}z}{\cancel{e^{iaT}}(z-e^{-iaT})}$$

$$\therefore z(e^{-iat}) = \frac{z}{z-e^{-iaT}}$$

Also,

$$z(e^{-iat}) = \frac{z}{z-(\cos aT - i\sin aT)}$$

$$= \cancel{\frac{z}{(z-\cos aT)+i\sin aT} \times \frac{z-\cos aT}{}}$$

$$\cancel{= z\{(z-\cos aT - i\sin aT\}}$$

$$= \frac{z}{(z-\cos aT)+i\sin aT} \times \frac{\{(z-\cos aT)-i\sin aT\}}{(z-\cos aT)-i\sin aT}$$

$$= \frac{z(z-\cos aT)-iz\sin aT}{(z-\cos aT)^2+\sin^2 aT}$$

or, $z(\cos at - i\sin at) = \dfrac{z(z-\cos aT)-iz\sin aT}{z^2-2z\cos aT+1}$

or, $z(\cos at) - iz(\sin at) = \dfrac{z(z-\cos aT)}{z^2-2z\cos aT+1} - i\dfrac{z\sin aT}{z^2-2z\cos aT+1}$

Equating real & imaginary part

$$\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$$

$$\sinh ax = \frac{e^{ax}-e^{-ax}}{2}$$

$$\boxed{\begin{aligned} z(\cos at) &= \frac{z(z-\cos aT)}{z^2-2z\cos aT+1} \\ \& \; z(\sin at) &= \frac{z\sin aT}{z^2-2z\cos aT+1} \end{aligned}}$$

Q. find z- transform of cosh at sin bt

i.e. $z(\cosh at \sin bt)$

$$= z\left\{\left(\frac{e^{at}+e^{-at}}{2}\right)\sin bt\right\}$$

$$= \frac{1}{2} z(e^{at}\sin bt) + \frac{1}{2} z(e^{-at}\sin bt)$$

$$= \frac{1}{2} z(\sin bt)_{z\to e^{-aT}z} + \frac{1}{2} z(\sin bt)_{z\to e^{aT}z}$$

$$= \frac{1}{2}\left[\frac{z\sin bT}{z^2-2z\cos bT+1}\right]_{z\to e^{-aT}z} + \frac{1}{2}\left[\frac{z\sin bT}{z^2-2z\cos bT+1}\right]_{z\to e^{aT}z}$$

$$= \frac{1}{2}\left[\frac{e^{-aT}z\sin bT}{(e^{-aT}z)^2-2e^{-aT}z\cos bT+1}\right] + \frac{1}{2}\left[\frac{e^{aT}z\sin bT}{(e^{aT}z)^2-2e^{aT}\cos bT+1}\right]$$

Scaling theorem:

If $z[f(t)] = F(z)$ then,

$$z[a^n f(t)] = F\left(\frac{z}{a}\right)$$

Proof:

We have, $z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$= F(z) \quad \cdots\cdots (1)$

Now, $z[a^n f(t)] = \sum_{n=0}^{\infty} a^n f(nT) z^{-n}$

$= \sum_{n=0}^{\infty} f(nT) \frac{z^{-n}}{a^{-n}}$

$= \sum_{n=0}^{\infty} f(nT) \left(\frac{z}{a}\right)^{-n}$

$= F\left(\frac{z}{a}\right)$

Q. $z(a^n \cos bt)$

$= z(\cos bt)_{z \to \frac{z}{a}}$

$= \left[\frac{z(z - \cos bT)}{z^2 - 2z\cos bT + 1}\right]_{z \to \frac{z}{a}}$

$= \dfrac{\frac{z}{a}\left(\frac{z}{a} - \cos bT\right)}{\frac{z^2}{a^2} - 2 \cdot \frac{z}{a} \cos bT + 1}$

$= \dfrac{z(z - a\cos bT)}{z^2 - 2az\cos bT + a^2}$

Q. Show that:

$$z(t^k) = -Tz \frac{d}{dz} z(t^{k-1})$$

We have, $z(t^k) = \sum_{n=0}^{\infty} (nT)^k z^{-n}$ ...... (1)

$$z(t^{k-1}) = \sum_{n=0}^{\infty} (nT)^{k-1} z^{-n} \quad \text{.....(2)}$$

Diff. w. r to z

$$\frac{d}{dz}(t^{k-1}) = \sum_{n=0}^{\infty} (nT)^{k-1}(-n) z^{-n-1}$$

$$-\frac{d}{dz}(t^{k-1}) = \sum_{n=0}^{\infty} \frac{(nT)^k}{\cancel{n}T} \cdot \cancel{n} \frac{z^{-n}}{z}$$

$$= \sum_{n=0}^{\infty} (nT)^T z^{-n}$$

$$= -Tz \frac{d}{dz} z(t^{k-1})$$

$$\boxed{\text{or, } z(t^k) = -Tz \frac{d}{dz} z(t^{k-1})}$$

(1) put $k = 1$,

$$z(t) = -Tz \frac{d}{dz} z(t^0)$$

$$= -Tz \frac{d}{dz}\left(\frac{z}{z-1}\right)$$

$$= -Tz \frac{(z-1)\cdot 1 - z\cdot 1}{(z-1)^2}$$

$$= \frac{Tz}{(z-1)^2}$$

$$\boxed{\therefore z(t) = \frac{Tz}{(z-1)^2}}$$

② put $k=2$

$$z(t^2) = -Tz \frac{d}{dz} z(t)$$

$$= -Tz \frac{d}{dz} \frac{Tz}{(z-1)^2}$$

$$= -T^2 z \frac{d}{dz} \frac{z}{(z-1)^2}$$

$$= -T^2 z \left\{ \frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1)}{(z-1)^4} \right\}$$

Theorem:

If $z[f(t)] = F(z)$ then

$$z[nf(t)] = -z \frac{d}{dz} F(z) \text{ or } \frac{d}{dz} z[f(t)]$$

$$= -z \frac{d}{dz} F(z)$$

Proof:

By definition,

$$z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$\text{or, } F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Diff both sides w.r. to z

$$\frac{d}{dz} F(z) = \sum_{n=0}^{\infty} f(nT)(-n) z^{-n-1}$$

$$\text{or, } -\frac{d}{dz} F(z) = \sum_{n=0}^{\infty} n f(nT) \frac{z^{-n}}{z}$$

$$\boxed{\text{or, } z[nf(t)] = -z \frac{d}{dz} F(z)}$$

eg 1) $z(n)$

$= z(n \cdot 1)$

$= -z \frac{d}{dz} z(1)$

$= -z \frac{d}{dz}\left(\frac{z}{z-1}\right)$

$= -z \frac{(-1)}{(z-1)^2}$

$= \frac{z}{(z-1)^2}$

eg 2) $z(n^2)$

$= z(n \cdot n)$

$= -z \frac{d}{dz} z(n)$

$= -z \frac{d}{dz}\left\{\frac{z}{(z-1)^2}\right\}$

$= -z\left\{\frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1)}{(z-1)^4}\right\}$

$= -z\left\{\frac{z^2 - \cancel{2z} + 1 - 2z^2 + \cancel{2z}}{(z-1)^4}\right\}$

$= -z \frac{(-z^2+1)}{(z-1)^4}$

$= z \frac{\cancel{(z-1)}}{(z-1)^{\cancel{4}\,3}}$

$= \frac{z}{(z-1)^3}$

State and prove 2nd shifting theorem.

If $Z[f(t)] = F(z)$ then $Z[f(t+T)] = z[F(z) - f(0)]$

Proof:

By def^n $Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$$F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} \quad \text{—— (1)}$$

Now,

$$Z[f(t+T)] = \sum_{n=0}^{\infty} f(nT+T) z^{-n}$$

$$= \sum_{n=0}^{\infty} f\{(n+1)T\} z^{-n}$$

, put $N = n+1$

when $n=0$ as $N=1$

& $n \to \infty$ as $N \to \infty$

$$= \sum_{N=1}^{\infty} f(NT) z^{-(N-1)}$$

$$= z \sum_{N=1}^{\infty} f(NT) z^{-N}$$

$$= z \left\{ \sum_{N=0}^{\infty} f(NT) z^{-N} - f(0) \right\}$$

$$= z[F(z) - f(0)]$$

Initial Value theorem

If $z[f(t)] = F(z)$ then $\lim_{t\to 0} f(t) = \lim_{z\to\infty} F(z)$

Proof:

We have,

$$z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$F(z) = \sum_{n=0}^{\infty} f(nT) \frac{1}{z^n}$$

$$\text{or, } F(z) = f(0) + \frac{f(T)}{z} + \frac{f(2T)}{z^2} + \frac{f(3T)}{z^3} + \dots$$

Taking limit as $z \to \infty$ on both sides,

$$\lim_{z\to\infty} F(z) = f(0) + \frac{f(T)}{\infty} + \frac{f(2T)}{\infty^2} + \frac{f(3T)}{\infty^3}$$

$$= f(0)$$

$$\boxed{\lim_{z\to\infty} F(z) = \lim_{t\to 0} f(t)}$$

State and prove final value theorem:

If $z(f(t)) = F(z)$ then $\lim_{t\to\infty} f(t) = \lim_{z\to 1} (z-1) F(z)$

Proof:

By def^n $z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$$\text{or, } F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} \quad \text{---} (1)$$

By 2^{nd} shifting theorem $z[f(t+T)]$

$$= z[F(z) - f(0)]$$

or, $z[F(z) - f(0)] = \sum_{n=0}^{\infty} f(nT+T)\, z^{-n}$ —— ②

Subtracting ① from ②

$zF(z) - zf(0) - F(z) = \sum_{n=0}^{\infty} \left[f\{(n+1)T\} - f(nT) \right] z^{-n}$

or, $(z-1)F(z) - z f(0) = \sum_{n=0}^{\infty} \left[f\{(n+1)T\} - f(nT) \right] z^{-n}$

Taking limit as $z \to 1$ on both sides.

or, $\lim_{z\to 1} (z-1)F(z) - f(0) = \sum_{n=0}^{\infty} \left[f\{(n+1)T\} - f(nT) \right]$

or, $\lim_{z\to 1} (z-1)F(z) - f(0) = \cancel{f(T)} - f(0) + \cancel{f(2T)} - \cancel{f(T)} + f(3T) - \cancel{f(2T)} + \ldots\ldots + f(\infty)$

or, $\lim_{z\to 1} (z-1)F(z) - \cancel{f(0)} = f(\infty) - \cancel{f(0)}$

$$\boxed{\text{or, } \lim_{z\to 1} (z-1)F(z) = \lim_{t\to\infty} f(t)}$$

Hence proved.

Inverse Z-transform

1) Partial fraction method
2) Inversion integral method (Residues method)
3) Direct Division method

Q) Find $z^{-1}\left(\frac{z}{z^2-5z+6}\right)$ by partial fraction method.

Let $F(z) = \frac{z}{z^2-5z+6}$

or, $\frac{F(z)}{z} = \frac{1}{(z-2)(z-3)}$

$$= \frac{A}{z-3} + \frac{B}{z-2}$$

$$= \frac{1}{z-3} - \frac{1}{z-2}$$

$$\text{or, } F(z) = \frac{z}{z-3} - \frac{z}{z-2}$$

Taking inverse z-transform on both sides

$$z^{-1}[F(z)] = z^{-1}\left(\frac{z}{z-3}\right) - z^{-1}\left(\frac{z}{z-2}\right)$$

$$\therefore z^{-1}\left(\frac{z}{z^2-5z+6}\right) = 3^n - 2^n$$

Residues method

Let $F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$$F(z) = f(0) + f(T)\frac{1}{z} + f(2T)\frac{1}{z^2} + f(3T)\frac{1}{z^3} + \ldots$$

$$\ldots + f(nT)\frac{1}{z^n} \ldots\ldots$$

multiplying it by z^{n-1} on both sides,

$$z^{n-1}F(z) = z^{n-1}f(0) + z^{n-2}f(T) + z^{n-3}f(2T) + \ldots$$

$$\ldots + f(nT)z^{-1}$$

This is laurent series expansion of $z^{n-1}F(z)$ around the point $z = 0$

Thus the coefficient $f(nT) = \frac{1}{2\pi i}\oint_c z^{n-1}F(z)\,dz$

$$\text{or, } z^{-1}F(z) = \frac{1}{\cancel{2\pi i}} \cancel{2\pi i}[\text{Sum of residue of } z^{n-1}F(z)]$$

$$\boxed{\text{or, } z^{-1}[F(z)] = \text{Sum of residues of } z^{n-1}F(z).}$$

Q) Find $z^{-1}\left(\frac{z}{z^2-5z+6}\right)$ by Residue method.

or

Find inverse z-transform of $F(z) = \frac{z}{z^2-5z+6}$

Soln

Let $F(z) = \frac{z}{z^2-5z+6}$

$z^{n-1}F(z) = \frac{z^n}{(z-2)(z-3)}$

The pole of $z^{n-1}F(z)$ is $z = 2, 3$

$$\left\{\begin{array}{c}\text{Residue of } z^{n-1}F(z) \\ \text{at } z=2\end{array}\right\} = \lim_{z\to 2} (z-2)\, z^{n-1}F(z)$$

$$= \lim_{z\to 2} \cancel{(z-2)} \frac{z^n}{\cancel{(z-2)}(z-3)}$$

$$= -2^n$$

$$\left\{\begin{array}{c}\text{Res of } z^{n-1}F(z) \\ \text{at } z=3\end{array}\right\} = \lim_{z\to 3} (z-3)\, z^{n-1}F(z)$$

$$= \lim_{z\to 3} \cancel{(z-3)} \frac{z^n}{(z-2)\cancel{(z-3)}}$$

$$= 3^n$$

We have,

$z^{-1}[F(z)] = $ sum of residues of $z^{n-1}F(z)$

$= 3^n - 2^n$

$$z^{-1}\left(\frac{z}{z^2-5z+6}\right) = 3^n - 2^n$$

Q. Find $z^{-1}\left(\frac{z}{(z-2)(z+3)^2}\right)$

Sol^n

Let, $F(z) = \frac{z}{(z-2)(z+3)^2}$

or, $z^{n-1}F(z) = \frac{z^n}{(z-2)(z+3)^2}$

The pole of $z^{n-1}F(z)$ is $z=2$ & -3 has order 2.

$$\left\{\begin{matrix}\text{Res of } z^{n-1}F(z) \\ \text{at } z=2\end{matrix}\right\} = \lim_{z\to 2}(z-2)\, z^{n-1}F(z)$$

$$= \lim_{z\to 2} \cancel{(z-2)} \cdot \frac{z^n}{\cancel{(z-2)}(z+3)^2}$$

$$= \frac{2^n}{25}$$

$$\left\{\begin{matrix}\text{Res of } z^{n-1}F(z) \\ \text{at } z=-3\end{matrix}\right\} = \frac{1}{1!}\lim_{z\to -3}\frac{d}{d^2}\left\{\cancel{(z+3)^2}\cdot\frac{z^n}{(z-2)\cancel{(z+3)^2}}\right\}$$

$$= \lim_{z\to -3}\left\{\frac{(z-2)\,nz^{n-1} - z^n\cdot 1}{(z-2)^2}\right\}$$

$$\cancel{=} \frac{-\cancel{5}n + \cancel{3}^n}{\cancel{25}} \quad = \frac{-5n(-3)^{n-1} - (-3)^n}{25}$$

$$= \frac{(-3)^n}{25}\left\{-5n\frac{1}{(-3)} - 1\right\} = \frac{(-3)^n(5n-3)}{75}$$

We have,

$z^{-1}[F(z)]$ = Sum of residues of $z^{n-1}F(z)$

$$= \frac{2^n}{25} + \frac{(-3)^n(5n-3)}{75}$$

$$= \frac{\cancel{2^n - 5n + 3^n}}{\cancel{25}}$$

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Q. Find the inverse z-transform of $\frac{2z}{(z-1)(z^2+1)}$.

Sol^n.

Let $F(z) = \frac{2z}{(z-1)(z^2+1)}$

$z^{n-1}F(z) = \frac{2z^n}{(z-1)(z^2+1)}$

The pole of $z^{n-1}F(z)$ is $z = +1, \pm i$ $[z^2 = -1$ i.e. $z = \pm i]$

$\left\{\begin{array}{l}\text{Res of } z^{n-1}F(z) \\ \text{at } z = +1\end{array}\right\} = \lim_{z\to 1} \cancel{(z-1)} \frac{2z^n}{\cancel{(z-1)}(z^2+1)} = \frac{2}{2} = 1$

$\left\{\begin{array}{l}\text{Res of } z^{n-1}F(z) \\ \text{at } z = i\end{array}\right\} = \lim_{z\to i} (z-i) \frac{2z^n}{(z-1)(z^2+1)}$

$= \lim_{z\to i} (z-i) \frac{2z^n}{(z-1)(z^2-i^2)} = \lim_{z\to i} \cancel{(z-i)} \frac{2z^n}{(z-1)(z+i)\cancel{(z-i)}}$

$= \frac{2i^n}{(i-1)(i+i)} = \frac{\cancel{2} i^n}{(i-1)\cancel{2}i} = \frac{i^n}{(i^2-i)}$

$= \frac{-i^n}{(1+i)}$

$\left\{\begin{array}{l}\text{Res of } z^{n-1}F(z) \\ \text{at } z = -i\end{array}\right\} = \frac{(-i)^n}{1-i}$

We have, $z^{-1}[F(z)]$ = Sum of residue of $z^{n-1}F(z)$

$= 1 - \left[\frac{i^n}{1+i} + \frac{(-i)^n}{1-i}\right]$

We have $z + \bar{z} = 2x = 2$ Real part of z

Here if $\frac{i^n}{1+i} + \frac{(-i)^n}{1-i} = 2$ Real part of $\frac{i^n}{(1+i)}$

$= 2$ Real part of $\frac{(e^{i\pi/2})^n}{\sqrt{2}e^{i\pi/4}}$ $\left[\begin{array}{l} i = e^{i\pi/2} \\ 1+i = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) \\ = \sqrt{2}e^{i\pi/4}\end{array}\right]$

$= 2$ Real part of $\frac{1}{\sqrt{2}}(e^{in\pi/2 - i\pi/4})$

$= \frac{2}{\sqrt{2}}$ Real part of $e^{i(\frac{n\pi}{2} - \frac{\pi}{4})} = \sqrt{2}\cos\left(\frac{n\pi}{2} - \frac{\pi}{4}\right)$

$\therefore z^{-1}[F(z)] = 1 - \sqrt{2}\cos\left(\frac{n\pi}{2} - \frac{\pi}{4}\right)$

Application of z-transform

- Difference equation:

Theorem:

If $z[y_n] = \bar{y}$ then $z[y_{n+k}] = z^k\left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots \frac{y_{k-1}}{z^{k-1}}\right]$

Proof:

Given by $z[y_n] = \sum_{n=0}^{\infty} y_n z^n = \bar{y}$ —— (1)

By definition $z[y_{n+k}] = \sum_{n=0}^{\infty} y_{n+k} z^{-n}$

put $n+k = N$ when $n=0$ as $N=k$

$n \to \infty$ as $N = \infty$

so that $z[y_{n+k}] = \sum_{N=k}^{\infty} y_N z^{-(-k+N)}$

$$= \sum_{N=k}^{\infty} y_N z^{-N} z^k$$

$$= z^k \sum_{N=k}^{\infty} y_N z^{-N}$$

$$= z^k\left[\sum_{N=0}^{\infty} y_N z^{-N} - \sum_{N=0}^{k-1} y_N z^{-N}\right]$$

$$= z^k\left[z(y_n) - \sum_{N=0}^{k-1} \frac{y_N}{z^N}\right]$$

$$\boxed{z(y_{n+k}) = z^k\left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \frac{y_{k-1}}{z^{k-1}}\right]}$$

Hence proved.

put k = 1,

$$z[y_{n+1}] = z\left[\bar{y} - y_0\right]$$

$$\boxed{z[y_{n+1}] = z\left[z(y_n) - y_0\right]}$$ → formula

put k = 2,

$$z[y_{n+2}] = z^2\left[\bar{y} - y_0 - \frac{y_1}{z}\right]$$

$$\boxed{z[y_{n+2}] = z^2\left[z(y_n) - y_0 - \frac{y_1}{z}\right]}$$ → formula

V.V. Imp. 8 marks

Q. Using z-transform, solve the difference equation.

$$y_{n+2} + 6y_{n+1} + 9y_n = 2^n \text{ when } y_0 = 0,\ y_1 = 0$$

Soln:

Here, $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$

Taking z-transform on both sides,

$$z[y_{n+2}] + 6z[y_{n+1}] + 9z[y_n] = z[2^n]$$

or, $z^2\left[z(y_n) - y_0 - \frac{y_1}{z}\right] + 6z\left[z(y_n) - y_0\right] + 9z(y_n) = \frac{z}{z-2}$

when $y_0 = 0$, $y_1 = 0$

or, $z^2\left[z(y_n) - 0 - \frac{0}{z}\right] + 6z\left[z(y_n) - 0\right] + 9z(y_n) = \frac{z}{z-2}$

or, $z^2 z(y_n) + 6z\, z(y_n) + 9z(y_n) = \frac{z}{z-2}$

or, $z(y_n)(z^2 + 6z + 9) = \frac{z}{z-2}$

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or, $z(y_n) = \dfrac{z}{(z-2)(z^2+6z+9)}$

or, $z(y_n) = \dfrac{z}{(z-2)(z+3)^2}$

or, $y_n = z^{-1} \dfrac{z}{(z-2)(z+3)^2}$ ——— (1)

Let $F(z) = \dfrac{z}{(z-2)(z+3)^2}$

multiplying it by z^{n-1} on both sides,

$z^{n-1}F(z) = \dfrac{z^n}{(z-2)(z+3)^2}$

The pole of $z^{n-1}F(z)$ are given by putting $(z-2)(z+3)^2 = 0$

$z = 2$ has order 1 &

$z = -3$ has order 2

$$\left\{\begin{matrix}\text{Res } z^{n-1}F(z) \\ \text{at } z=2\end{matrix}\right\} = \lim_{z\to 2} \cancel{(z-2)} \frac{z^n}{\cancel{(z-2)}(z+3)^2}$$

$$= \frac{2^n}{25}$$

$$\left\{\begin{matrix}\text{Res } z^{n-1}F(z) \\ \text{at } z=-3\end{matrix}\right\} = \lim_{z\to -3} \cancel{(z+3)^2} \frac{z^n}{(z-2)\cancel{(z+3)^2}}$$

$$= \lim_{z\to -3} \frac{d}{dz}\left(\frac{z^n}{z-2}\right)$$

$$= \lim_{z\to -3} \frac{(z-2)\, nz^{n-1} - z^n(1-0)}{(z-2)^2}$$

$$= \frac{-5n(-3)^{n-1} - (-3)^n}{25}$$

$$= \frac{(-3)^n}{25}\left\{-5n(-3)^{-1} - 1\right\}$$

$$= \frac{(-3)^n}{25}\left\{-5n \cdot \frac{1}{(-3)^1} - 1\right\}$$

$$= \frac{(-3)^n}{25}\left\{\frac{5n}{3} - 1\right\}$$

$$= \frac{(-3)^n}{25}\left(\frac{5n-3}{3}\right)$$

$$= \frac{(-3)^n(5n-3)}{75}$$

We have, $z^{-1}F(z) = $ sum of residue of $z^{n-1}F(z)$

$$z^{-1}F(z) = \frac{2^n}{25} + \frac{(-3)^n}{75}(5n-3)$$

from ①

$$\therefore y_n = z^{-1}F(z)$$

$$= \frac{2^n}{25} + \frac{(-3)^n}{75}(5n-3)$$

Que: Solve, $y_{n+2} - 3y_{n+1} + 2y_n = 0$ where $y_0 = 0$, $y_1 = 1$

Soln.

Here, $y_{n+2} - 3y_{n+1} + 2y_n = 0$

Taking z-transform on both sides

$$z(y_{n+2}) - 3z(y_{n+1}) + 2z(y_n) = z(0)$$

or, $z^2\left[z(y_n) - y_0 - \frac{y_1}{z}\right] - 3z\left[z(y_n) - y_0\right] + 2z(y_n) = 0$

when $y_0 = 0$, $y_1 = 1$

or, $z^2\left[z(y_n) - 0 - \frac{1}{z}\right] - 3z\left[z(y_n) - 0\right] + 2z(y_n) = 0$

or, $z^2 z(y_n) - z - 3z\, z(y_n) + 2z(y_n) = 0$

or, $z(y_n)(z^2 - 3z + 2) = z$

or, $z(y_n) = \frac{z}{z^2 - 3z + 2}$

or, $z(y_n) = \dfrac{z}{(z-2)(z-1)}$

or, $y_n = z^{-1} \dfrac{z}{(z-2)(z-1)}$

Let $F(z) = \dfrac{z}{(z-2)(z-1)}$

multiplying it by z^{n-1} on both sides,

$$z^{n-1} F(z) = \frac{z^n}{(z-2)(z-1)} \quad \text{—— (1)}$$

The pole of $z^{n-1} F(z)$ are given by putting $(z-2)(z-1) = 0$

$z = 2, 1$

$$\left\{ \begin{array}{l} \text{Res } z^{n-1} F(z) \\ \text{at } z=1 \end{array} \right\} = \lim_{z \to 1} \cancel{(z-1)} \frac{z^n}{(z-2)\cancel{(z-1)}}$$

$$= \frac{(1)^n}{(-1)}$$

$$= -1$$

$$\left\{ \begin{array}{l} \text{Res } z^{n-1} F(z) \\ \text{at } z=2 \end{array} \right\} = \lim_{z \to 2} \cancel{(z-2)} \frac{z^n}{\cancel{(z-2)}(z-1)}$$

$$= 2^n$$

$\therefore z^{-1} F(z) =$ sum of residue of $z^{n-1} F(z)$

$z^{-1} F(z) = -1 + 2^n$

from (1) $= z^{-1} F(z)$

$= 2^n - 1$

Q. $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0$, $y_1 = 0$

Soln

Here:

$$y_{n+2} + 2y_{n+1} + y_n = n$$

Taking z-transform on both sides

$$z(y_{n+2}) + 2z(y_{n+1}) + z(y_n) = z(n)$$

or, $z^2\left[z(y_n) - y_0 - \frac{y_1}{z}\right] + 2z[z(y_n) - y_0] + z(y_n) = \frac{z}{(z-1)^2}$

when $y_0 = 0$, $y_1 = 0$

or, $z^2 z(y_n) + 2z\, z(y_n) + z(y_n) = \frac{z}{(z-1)^2}$

or, $z(y_n)(z^2 + 2z + 1) = \frac{z}{(z-1)^2}$

or, $z(y_n) = \frac{z}{(z-1)^2(z^2+2z+1)}$

or, $z(y_n) = \frac{z}{(z-1)^2(z+1)^2}$

or, $y_n = z^{-1}\frac{z}{(z-1)^2(z+1)^2}$ ——— ①

Let $F(z) = \frac{z}{(z-1)^2(z+1)^2}$

multiplying it by z^{n-1} on both sides,

$$z^{n-1}F(z) = \frac{z^n}{(z-1)^2(z+1)^2}$$

The pole of $z^{n-1}F(z)$ are given by putting

$$(z-1)^2(z+1)^2 = 0$$

$z = 1$ has order 2 &

$z = -1$ has order 2

$$\frac{d(z+1)^2}{dv} \quad d\frac{u}{v} = v\frac{du}{dx} - u\frac{dv}{dx}$$

$$= 2(z+1)^{2} \cdot 1$$

$$\left\{ \begin{array}{l} \text{Res } z^{n-1}F(z) \\ \text{at } z=1 \end{array} \right\} = \lim_{z \to 1} \cancel{(z-1)^2} \frac{z^n}{\cancel{(z-1)^2}(z+1)^2}$$

$$= \lim_{z\to 1} \frac{d}{dz}\left(\frac{z^n}{(z+1)^2}\right)^{u}_{v}$$

$$= \lim_{z\to 1} \frac{(z+1)^2 nz^{n-1} - z^n(2z+2)}{(z+1)^4}$$

$$= \frac{4n(1)^{n-1} - 1^n \cdot 4}{16}$$

$$= \frac{4n-4}{16}$$

$$= \frac{n-1}{4}$$

$$\left\{ \begin{array}{l} \text{Res } z^{n-1}F(z) \\ \text{at } z=-1 \end{array} \right\} = \lim_{z\to -1} \cancel{(z+1)^2} \frac{z^n}{(z-1)^2\cancel{(z+1)^2}}$$

$$= \lim_{z\to -1} \frac{d}{dz}\left(\frac{z^n}{(z-1)^2}\right)$$

$$= \lim_{z\to -1} \frac{(z-1)^2 nz^{n-1} - z^n 2(z-1)}{(z-1)^4}$$

$$= \frac{4n(-1)^{n-1} - (-1)^n(-4)}{16}$$

$$= \frac{(-1)^n \, 4n(-1)^{-1} + 4(-1)^n}{16}$$

$$= \frac{(-1)^n\{-4n+4\}}{16}$$

$$= (-1)^n \left\{\frac{1-n}{4}\right\}$$

$$= (-1)^n \left(\frac{n-1}{-4}\right)$$

$$\therefore z^{-1}F(z) = \frac{n-1}{4} + (-1)^n \frac{n-1}{-4}$$

$$y_n = \frac{n-1}{4}\left\{1-(-1)^n\right\}$$

Theorem:

If $z(y_n) = \bar{y}$ then $z[y_{n+k}] = z^k\left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \cdots\cdots \frac{y_{k-1}}{z^{k-1}}\right]$

Proof:

By def^n $z[y_n] = \sum_{n=0}^{\infty} y_n z^{-n}$

$$= \bar{y} \quad \cdots\cdots (1)$$

Now, $z[y_{n+k}] = \sum_{n=0}^{\infty} y_{n+k} z^{-n}$. put $N = n+k$

$n=0$ as $N=k$ &

$n \to \infty$ as $N \to \infty$

$$= \sum_{N=k}^{\infty} y_N z^{-(N-k)}$$

$$= \sum_{N=k}^{\infty} y_N z^{-N} \cdot z^k$$

$$= z^k \sum_{N=k}^{\infty} y_N z^{-N}$$

$$= z^k\left[\sum_{N=0}^{\infty} y_N z^{-N} - \sum_{N=0}^{k-1} \frac{y_N}{z^N}\right]$$

$$= z^k\left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} \cdots\cdots \frac{y_{k-1}}{z^{k-1}}\right]$$

Hence proved.

Que: Solve. $y_{k+2} + 2y_{k+1} + y_k = k$ where $y_0 = 0, y_1 = 0$

Sol^n

Here:

$$y_{k+2} + 2y_{k+1} + y_k = k$$

Taking z-transform on both sides,

$$z(y_{k+2}) + 2z(y_{k+1}) + z(y_k) = z(k)$$

or, $$z^2\left[z(y_k) - y_0 - \frac{y_1}{z}\right] + 2z\left[z(y_k) - y_0\right] + z(y_k) = \frac{z}{(z-1)^2}$$

Same as $y_{n+2} + 2y_{n+2} + y_n = n$

$e^{i\alpha} = \cos\alpha + i\sin\alpha$
$e^{-i\alpha} = \cos\alpha - i\sin\alpha$

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Que: Solve:
$$U_{x+2} - 2\cos\alpha\, U_{x+1} + U_x = 0,\ U = 0,\ U_1 = 1$$

Soln

Here,
$$U_{x+2} - 2\cos\alpha\, U_{x+1} + U_x = 0$$

Taking z-transform on both sides,

or, $z(U_{x+2}) - 2\cos\alpha\, z(U_{x+1}) + z(U_x) = z(0)$

or, $z^2\left\{\overline{U} - U_0 - \frac{U_1}{z}\right\} - 2\cos\alpha\, z\left\{\overline{U} - U_0\right\} + \overline{U} = 0$

where, $\overline{U} = z(U_x)$

or, $z^2\overline{U} - 0 - z - 2z\cos\alpha\,\overline{U} - 0 + \overline{U} = 0$ using $U_0 = 0, U_1 = 1$

or, $\overline{U}(z^2 - 2z\cos\alpha + 1) - z = 0$

or, $\overline{U}(z^2 - 2z\cos\alpha + 1) = z$

or, $\overline{U} = \frac{z}{z^2 - 2z\cos\alpha + 1}$

or, $z(U_x) = \frac{z}{z^2 - 2z\cos\alpha + 1}$

or, $U_x = z^{-1}\left(\frac{z}{z^2 - 2z\cos\alpha + 1}\right)$ ——— (1)

Let $F(z) = \frac{z}{z^2 - 2z\cos\alpha + 1}$

multiplying by z^{x-1} on both sides,
$$z^{x-1}F(z) = \frac{z^x}{(z - e^{i\alpha})(z - e^{-i\alpha})}$$

The pole of $z^{x-1}F(z)$ are given by putting
$(z - e^{i\alpha})(z - e^{-i\alpha}) = 0$
$z = e^{i\alpha}$ (has order 1) &
$z = e^{-i\alpha}$ (has order 1)

$$\left\{\begin{matrix}\text{Res } z^{n-1}F(z) \\ \text{at } z = e^{i\alpha}\end{matrix}\right\} = \lim_{z \to e^{i\alpha}} \cancel{(z-e^{i\alpha})} \frac{z^n}{\cancel{(z-e^{i\alpha})}(z-e^{-i\alpha})}$$

$$= \frac{e^{i\alpha n}}{(e^{i\alpha}-e^{-i\alpha})}$$

$$\boxed{\cancel{= \frac{e^{i\alpha}\cdot e^{n}}{e^{i\alpha}(1-e^{i\alpha})} = \frac{e^{n}}{1-e^{i\alpha}}}}$$

$$= \frac{e^{i\alpha n}}{\frac{e^{i\alpha}-e^{-i\alpha}}{2i}\,(2i)}$$

$$= \frac{e^{in\alpha}}{2i\sin\alpha}$$

$$\left\{\begin{matrix}\text{Res } z^{n-1}F(z) \\ \text{at } z = e^{-i\alpha}\end{matrix}\right\} = \lim_{z \to e^{-i\alpha}} \cancel{(z-e^{-i\alpha})} \frac{z^n}{(z-e^{i\alpha})\cancel{(z-e^{-i\alpha})}}$$

$$= \frac{e^{-i\alpha n}}{(e^{-i\alpha}-e^{i\alpha})}$$

$$= -\frac{e^{-in\alpha}}{\left(\frac{e^{i\alpha}-e^{-i\alpha}}{2i}\right)2i}$$

$$= -\frac{e^{-in\alpha}}{2i\sin\alpha}$$

from ①

$$u_n = z^{-1}F(z)$$

$$= \text{sum of residues of } z^{n-1}F(z)$$

$$= \frac{e^{in\alpha}}{2i\sin\alpha} - \frac{e^{-in\alpha}}{2i\sin\alpha}$$

$$= \frac{1}{\sin\alpha}\,\frac{e^{in\alpha}-e^{-in\alpha}}{2i}$$

$$= \frac{\sin n\alpha}{\sin\alpha}$$

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Fourier transform (7+8+2.5)

1) The Fourier integral:

If $f(x)$ is continuous and $\int_{-L}^{L} |f(x)| dx$ converges to zero as L approaches to infinity, then $f(x)$ can be represented by

$$f(x) = \int_0^\infty [A(\omega) \cos x\omega + B(\omega) \sin x\omega] \, d\omega \quad \text{--- (1)}$$

where $A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx$ &

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx$$

→ Sine and Cosine integral:

The fourier integral represented by $f(x)$ is given by

$$f(x) = \int_0^\infty [A(\omega) \cos x\omega + B(\omega) \sin x\omega) d\omega \quad \text{--- (1)}$$

Case I: when $f(x)$ is even

where $A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx$

$$= \frac{2}{\pi} \int_0^\infty f(x) \cos \omega x \, dx$$

& $B(\omega) = 0$

$$\boxed{\text{Then eq (1)}, \; f(x) = \int_0^\infty A(\omega) \cos x\omega \, d\omega \text{ where } A(\omega) = \frac{2}{\pi} \int_0^\infty f(x) \cos \omega x \, dx.}$$

is called cosine integral.

Case 2: When $f(x)$ is odd

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos\omega x\,dx$$

$$A(\omega) = 0 \text{ \& } B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin\omega x\,dx$$

$$= \frac{2}{\pi}\int_{0}^{\infty} f(x)\sin\omega x\,dx$$

Then ① $\boxed{\begin{aligned} f(x) &= \int_0^\infty B(\omega)\sin x\omega\,d\omega, \text{ where} \\ B(\omega) &= \frac{2}{\pi}\int_0^\infty f(x)\sin\omega x\,dx \end{aligned}}$

is called sine integral.

Integration by parts.

$$uv_1 - u'v_2 + u''v_3 - \ldots\ldots$$

1. $$\int x^2\cos ax\,dx = x^2\left(\frac{1}{a}\sin ax\right) - 2x\left(\frac{1}{a}\frac{\cos ax}{-a}\right) + 2\left(-\frac{1}{a^2}\frac{1}{a}\sin ax\right)$$

2. $$\int x^2 e^{ax}\,dx = x^2\frac{e^{ax}}{a} - 2x\left(\frac{1}{a}\frac{e^{ax}}{a}\right) + 2\left(\frac{1}{a^2}\frac{e^{ax}}{a}\right)$$

3. $$\left.\begin{aligned} &\int e^{ax}\cos bx\,dx \\ &\int e^{ax}\sin bx\,dx \end{aligned}\right\} \frac{u'v - uv'}{a^2+b^2}$$

3. $$\int e^{ax}\cos bx\,dx = \frac{ae^{ax}\cos bx - (-b)e^{ax}\sin bx}{a^2+b^2}$$

$$\int e^{ax}\sin bx\,dx = \frac{ae^{ax}\sin bx - be^{ax}\cos bx}{a^2+b^2}$$

$$\int_{a}^{a} f(x)\,dx = 0$$

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4. $\int_0^\infty e^{-st} \sin ax\,dx = \frac{a}{s^2+a^2}$

5. $\int_0^\infty e^{-st} \cos ax\,dx = \frac{s}{s^2+a^2}$

Exercise 6.1

1. Show that

V.V.I. a. $$\int_0^\infty \left[\frac{\cos x\omega + \omega \sin x\omega}{1+\omega^2}\right] d\omega = \begin{cases} 0 & \text{if } x<0 \\ \pi/2 & \text{if } x=0 \\ \pi e^{-x} & \text{if } x>0 \end{cases}$$

Sol^n.

By proof condition both $\sin x\omega$ & $\cos x\omega$ include so we use fourier integral.

The fourier integral of $f(x)$ is represented by

$$f(x) = \int_0^\infty [A(\omega)\cos x\omega + B(\omega)\sin x\omega]\,d\omega \quad \text{--- (1)}$$

where, $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos\omega x\,dx$

$$= \frac{1}{\pi}\left[\int_{-\infty}^{0} f(x)\cos\omega x\,dx + \int_0^0 f(x)\cos\omega x\,dx + \int_0^\infty f(x)\cos\omega x\,dx\right]$$

$$= \frac{1}{\pi}\left[0 + 0 + \int_0^\infty \pi e^{-x}\cos\omega x\,dx\right.$$

$$= \int_0^\infty e^{-1x}\cos\omega x\,dx$$

By Laplace integral, $\int_0^\infty e^{-st}\cos at\,dt = \frac{s}{s^2+a^2}$

$$= \frac{1}{1+\omega^2}$$

and $B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin\omega x\,dx$

$$= \frac{1}{\pi}\left[0+0+\int_0^\infty \pi e^{-1x} \sin \omega x \, dx\right]$$

$$= \int_0^\infty e^{-1x} \sin \omega x \, dx$$

By Laplace integral, $\int_0^\infty e^{-st} \sin at \, dt = \frac{a}{s^2+a^2}$

$$= \frac{\omega}{1+\omega^2}$$

Substitute the value of $A(\omega)$ & $B(\omega)$ in eq^n ① we get

$$f(x) = \int_0^\infty \left(\frac{1}{1+\omega^2} \cos x\omega + \frac{\omega}{1+\omega^2} \sin x\omega\right) d\omega$$

or, $$\int_0^\infty \left(\frac{\cos x\omega + \omega \sin x\omega}{1+\omega^2}\right) d\omega = \begin{cases} 0 & \text{if } x<0 \\ \pi/2 & \text{if } x=0 \\ \pi e^{-x} & \text{if } x>0 \end{cases}$$

Hence proved.

V.V.I b) $$\int_0^\infty \frac{(1-\cos \pi\omega)}{\omega} \sin x\omega \, dx = \frac{\pi}{2} \text{ if } 0<x<\pi$$

$$= 0 \text{ if } x > \pi$$

Soln,

The proof condition only $\sin x\omega$ include so we use sine integral.

The sine integral of $f(x)$ is represented by

$$f(x) = \int_0^\infty B(\omega) \sin x\omega \, d\omega \quad \text{—— ①}$$

where, $B(\omega) = \frac{2}{\pi}\int_0^\infty f(x) \sin \omega x \, dx$

$$= \frac{2}{\pi}\left[\int_0^\pi f(x) \sin \omega x \, dx + \int_\pi^\infty f(x) \sin \omega x \, dx\right]$$

$$= \frac{2}{\pi}\int_0^{\pi} \frac{\pi}{2} \sin w x \, dx + 0$$

$$= \int_0^{\pi} \sin w x \, dx$$

$$= \frac{1}{-w}\left[\cos w x\right]_0^{\pi}$$

$$= -\frac{1}{w}\left[\cos w\pi - \cos 0\right]$$

$$= \frac{[1-\cos \pi w]}{w}$$

substitute the value of $B(w)$ in eq^n ① we get,

$$f(x) = \int_0^{\infty}\left(\frac{1-\cos\pi w}{w}\right)\sin x w \, dw$$

or, $$\int_0^{\infty}\frac{(1-\cos\pi w)}{w}\sin x w \, dw = \begin{cases} \pi/2 \text{ if } 0<x<\pi \\ 0 \text{ if } x>\pi \end{cases}$$

Find sine & cosine integral of e^{-kx} for $k>0, x>0$.

Soln :

The sine integral of $f(x)$ is represented by :

$$f(x) = \int_0^{\infty} B(w)\sin x w \, dw \quad \text{— ①}$$

where $$B(w) = \frac{2}{\pi}\int_0^{\infty} f(x)\sin w x \, dx$$

$$= \frac{2}{\pi}\int_0^{\infty} e^{-kx}\sin w x \, dx$$

By Laplace Integral, $$= \frac{2}{\pi}\frac{w}{k^2+w^2}$$

from ①,

$$f(x) = \int_0^{\infty}\frac{2}{\pi}\frac{w}{k^2+w^2}\sin x w \, dw$$

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$$\text{or, } \frac{\pi}{2} e^{-kx} = \int_0^\infty \frac{\omega}{k^2+\omega^2} \sin x\omega \, d\omega \quad \text{, put } k = 1$$

$$\text{or, } \frac{\pi}{2} e^{-x} = \int_0^\infty \frac{\omega}{1+\omega^2} \sin x\omega \, d\omega \quad \text{replace } x \to \omega, \ \omega \to x$$

$$\boxed{\text{or, } \frac{\pi}{2} e^{-\omega} = \int_0^\infty \frac{x}{1+x^2} \sin \omega x \, dx}$$

The cosine integral of $f(x)$ is represented by

$$f(x) = \int_0^\infty A(\omega) \cos x\omega \, d\omega \quad \text{—— (2)}$$

where, $A(\omega) = \frac{2}{\pi} \int_0^\infty f(x) \cos \omega x \, dx$

$$= \frac{2}{\pi} \int_0^\infty e^{-kx} \cos \omega x \, dx$$

By Laplace ~~integ~~ integral,

$$= \frac{2}{\pi} \frac{k}{k^2+\omega^2}$$

from (2),

$$f(x) = \int_0^\infty \frac{2}{\pi} \frac{k}{k^2+\omega^2} \cos x\omega \, d\omega$$

$$\text{or, } \frac{\pi}{2} e^{-kx} = \int_0^\infty \frac{k}{k^2+\omega^2} \cos x\omega \, d\omega \text{, put } k = 1$$

$$\text{or, } \frac{\pi}{2} e^{-x} = \int_0^\infty \frac{1}{1+\omega^2} \cos x\omega \, d\omega \quad \text{, replace } x \to \omega, \ \omega \to x$$

$$\boxed{\text{or, } \frac{\pi}{2} e^{-\omega} = \int_0^\infty \frac{1}{1+x^2} \cos \omega x \, dx}$$

1. $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
2. $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
3. $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

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2c. Find cosine integral of $f(x) = \frac{1}{1+x^2}$ if $x > 0$.

Solⁿ:

The cosine integral of $f(x)$ is represented by

$$f(x) = \int_0^\infty A(\omega)\cos x\omega \, d\omega \text{ — } ①$$

where $A(\omega) = \frac{2}{\pi}\int_0^\infty f(x)\cos\omega x \, dx$

$$= \frac{2}{\pi}\int_0^\infty \frac{1}{1+x^2}\cos\omega x \, dx$$

$$= \frac{\cancel{2}}{\cancel{\pi}} \frac{\cancel{\pi}}{\cancel{2}} e^{-\omega}$$

$$= e^{-\omega}$$

from ①

$$f(x) = \int_0^\infty e^{-1\omega}\cos\omega x \, dx$$

Que. show that $\int_0^\infty \frac{\sin\pi\omega \cdot \sin x\omega}{1-\omega^2} d\omega = \begin{cases} \pi/2 \sin x & \text{if } 0 \le x \le \pi \\ 0 & \text{if } x > \pi \end{cases}$

Soln.

The sine integral of $f(x)$ is represented by.

$$f(x) = \int_0^\infty B(\omega)\sin x\omega \, d\omega \text{ — } ①$$

where $B(\omega) = \frac{2}{\pi}\int_0^\infty f(x)\sin\omega x \, dx$

$$= \frac{2}{\cancel{\pi}}\int_0^\pi \frac{\cancel{\pi}}{2}\sin x \sin\omega x \, dx$$

$$= \frac{1}{2}\int_0^\pi 2\sin x\sin\omega x \, dx$$

$\sin(\pi-\theta)=\sin\theta$

$\sin(\pi+\theta)=-\sin\theta$

$$=\frac{1}{2}\int_0^{\pi}\{\cos(1-\omega)x-\cos(1+\omega)x\}\,dx$$

$$=\frac{1}{2}\left[\frac{\sin(1-\omega)x}{(1-\omega)}\right]_0^{\pi}-\frac{1}{2}\left[\frac{\sin(1+\omega)x}{(1+\omega)}\right]_0^{\pi}$$

$$=\frac{1}{2}\,\frac{\sin(1-\omega)\pi}{1-\omega}-\frac{1}{2}\,\frac{\sin(1+\omega)\pi}{1+\omega}$$

$$=\frac{1}{2}\left[\frac{\sin\pi\omega}{1-\omega}+\frac{\sin\pi\omega}{1+\omega}\right]$$

$$=\frac{1}{2}\sin\pi\omega\left[\frac{1}{1-\omega}+\frac{1}{1+\omega}\right]$$

$$=\frac{1}{2}\sin\pi\omega\left[\frac{1+\cancel{\omega}+1-\cancel{\omega}}{1-\omega^2}\right]$$

$$=\frac{\sin\pi\omega}{1-\omega^2}$$

from ①,

$$f(x)=\int_0^{\infty}\frac{\sin\pi\omega}{1-\omega^2}\sin x\omega\,d\omega$$

or, $$\int_0^{\infty}\frac{\sin\pi\omega}{1-\omega^2}\sin x\omega\,d\omega=\begin{cases}\frac{\pi}{2}\sin x & \text{if } 0\le x\le\pi\\ 0 & \text{if } x>\pi\end{cases}$$

Proved.

Q. Show that $=\int_0^{\infty}\frac{\omega^3\sin x\omega}{\omega^4+4}\,d\omega=\frac{\pi}{2}e^{-x}\cos x$ if $x>0$.

Soln,

The proof condition only $\sin x\omega$ include so we use sine integral.

The sine integral of $f(x)$ is represented by:-

$$f(x)=\int_0^{\infty}[B(\omega)\sin x\omega]\,d\omega \quad \text{———} \quad ①$$

where,

$$B(\omega) = \frac{2}{\pi}\int_0^\infty f(x)\sin\omega x\, dx$$

$$= \frac{2}{\pi}\left[\int_0^\infty \left\{\frac{\pi}{2} e^{-x}\cos x \sin\omega x\right\} dx\right]$$

$$= \frac{1}{2}\int_0^\infty 2e^{-x}\cos x\sin\omega x\, dx$$

$$= \frac{1}{2}\int_0^\infty \left\{ e^{-x}\left[\sin(1+\omega)x + \sin(\omega-1)x\right]\right\} dx$$

$$= \frac{1}{2}\left[\int_0^\infty e^{-x}\sin(1+\omega)x\, dx + \int_0^\infty e^{-x}\sin(\omega-1)x\, dx\right]$$

$$= \frac{1}{2}\left[\frac{(1+\omega)}{1^2+(1+\omega)^2} + \frac{(\omega-1)}{1^2+(\omega-1)^2}\right]$$

$$= \frac{1}{2}\left[\frac{(1+\omega)(2-2\omega+\omega^2) + (\omega-1)(2+2\omega+\omega^2)}{\omega^4+4}\right]$$

$$= \frac{1}{2}\left(\frac{2\omega^3}{\omega^4+4}\right)$$

$$\therefore B(\omega) = \frac{\omega^3}{\omega^4+4}$$

from ①,

$$f(x) = \int_0^\infty \frac{\omega^3}{\omega^4+4}\sin x\omega\, d\omega$$

or, $$\int_0^\infty \frac{\omega^3}{\omega^4+4}\sin x\omega\, d\omega = \frac{\pi}{2} e^{-x}\cos \text{ if } x > 0$$

Proved

Fourier

Cosine and Sine transform

Def^n:

- Fourier cosine transform

We have, the Fourier cosine integral of $f(x)$ is represented by $f(x)=\int_0^\infty A(\omega)\cos x\omega \, d\omega$ — ①

where, $A(\omega)=\frac{2}{\pi}\int_0^\infty f(t)\cos\omega t \, dt$

Substituting the value of $A(\omega)$ in eq^n ① we get,

$$f(x)=\int_0^\infty \left(\frac{2}{\pi}\int_0^\infty f(t)\cos\omega t \, dt\right)\cos x\omega \, d\omega$$

$$=\sqrt{\frac{2}{\pi}}\int_0^\infty \left(\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos\omega t \, dt\right)\cos x\omega \, d\omega$$

$$f(x)=\sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_c(\omega)\cos x\omega \, d\omega \quad \text{— ②}$$

where,

$$\hat{f}_c(\omega)=\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos\omega t \, dt,$$

is called fourier cosine transform.

other notation of fourier cosine transform is $\mathcal{F}$

$$\mathcal{F}_c[f(t)]=\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos\omega t \, dt$$

and eq^n ② is said to be inverse fourier cosine transform.

- Fourier sine transform

We have, the Fourier sine integral of $f(x)$ is represented by $f(x)=\int_0^\infty B(\omega)\sin x\omega \, d\omega$ — ①

where, $B(\omega)=\frac{2}{\pi}\int_0^\infty f(t)\sin\omega t \, dt$

$$\int_0^\infty e^{-st}\cos at = \frac{s}{s^2+a^2}$$

$$\int_0^\infty e^{-st}\sin at\,dt = \frac{a}{s^2+a^2}$$

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Substituting the value of $B(\omega)$ in eq^n ① we get,

$$f(x) = \int_0^\infty \left(\frac{2}{\pi}\int_0^\infty f(t)\sin\omega t\,dt\right)\sin x\omega\,d\omega$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty \left(\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\sin\omega t\,dt\right)\sin x\omega\,d\omega$$

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_s(\omega)\sin x\omega\,d\omega \quad \text{——— ②}$$

where,

$$\hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\sin\omega t\,dt,$$

is called Fourier sine transform.

other notation of fourier sine transform is

$$\mathcal{F}_s[f(t)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\sin\omega t\,dt$$

and eq^n ② is said to be inverse fourier sine transform.

~~Exercise 7.2~~

Theorem: (Linearity of Fourier cosine transform)

statement: Let $\mathcal{F}_c[f(x)]$ & $\mathcal{F}_c[g(x)]$ are Fourier cosine transform and a and b are scalars then

$$\mathcal{F}_c(af+bg) = a\mathcal{F}_c(f) + b\mathcal{F}_c(g)$$

Proof:

We have, $\mathcal{F}_c(af+bg) = \sqrt{\frac{2}{\pi}}\int_0^\infty (af+bg)\cos\omega x\,dx$

$$= a\sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos\omega x\,dx + b\sqrt{\frac{2}{\pi}}\int_0^\infty g(x)\cos\omega x\,dx$$

$$= a\mathcal{F}_c[f(x)] + b\mathcal{F}_c[g(x)]$$

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Q. find Fourier ~~sine~~ ~~&~~ cosine transform of
$f(x) = \cancel{2e^{-5x} + 5e^{-2x}}\ e^{-ax}$

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cos wx \, dx$$

By Laplace integral.

$$= \sqrt{\frac{2}{\pi}} \frac{a}{a^2+w^2}$$

Q. Find Fourier sine & cosine transform of
$f(x) = 2e^{-5x} + 5e^{-2x}$

Soln:

The Fourier cosine transform of $f(x)$ is

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty (2e^{-5x} + 5e^{-2x}) \cos wx \, dx$$

$$= 2\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-5x} \cos wx \, dx + 5\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-2x} \cos wx \, dx$$

By Laplace integral.

$$= 2\sqrt{\frac{2}{\pi}} \frac{5}{5^2+w^2} + 5\sqrt{\frac{2}{\pi}} \frac{2}{2^2+w^2}$$

$$= 10\sqrt{\frac{2}{\pi}} \left[\frac{1}{25+w^2} + \frac{1}{4+w^2}\right]$$

$$\therefore f_c[f(x)] = 10\sqrt{\frac{2}{\pi}} \left[\frac{1}{25+w^2} + \frac{1}{4+w^2}\right]$$

and Fourier sine transform of $f(x)$ is

$$f_s[f(x)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\sin wx\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty (2e^{-5x} + 5e^{-2x})\sin wx\,dx$$

$$= 2\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-5x}\sin wx\,dx + 5\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-2x}\sin wx\,dx$$

By Laplace integral,

$$= 2\sqrt{\frac{2}{\pi}}\frac{w}{25+w^2} + 5\sqrt{\frac{2}{\pi}}\frac{w}{4+w^2}$$

Q. Find Fourier cosine transform of $f(x) = e^{-mx}$ for $m>0$, and then show that $\int_0^\infty \left(\frac{\cos kx}{1+x^2}\right)dx = \frac{\pi}{2}e^{-k}$

Soln

The Fourier cosine transform of $f(x)$ is

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos wx\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-mx}\cos wx\,dx$$

By Laplace transform $\int_0^\infty e^{-st}\cos at\,dt = \frac{s}{s^2+a^2}$

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}}\frac{m}{m^2+w^2}$$

By inverse fourier cosine transform

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_c(w)\cos xw\,dw$$

$$e^{-mx} = \sqrt{\frac{2}{\pi}}\int_0^\infty \sqrt{\frac{2}{\pi}}\frac{m}{m^2+w^2}\cos xw\,dw$$

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$$\text{or, } \frac{\pi}{2} e^{-mx} = \int_0^\infty \frac{m}{m^2+w^2} \cos xw \, dw$$

, put $m=1$

$$\text{or, } \frac{\pi}{2} e^{-x} = \int_0^\infty \frac{1}{w^2+1} \cos xw \, dw$$

replace $x \to k$
$w \to x$

$$\text{or, } \frac{\pi}{2} e^{-k} = \int_0^\infty \frac{1}{x^2+1} \cos kx \, dx$$

Hence proved.

Q. Find sine and cosine transform of $f(x) = \begin{cases} x^2 & \text{for } 0<x<1 \\ 0 & x>1 \end{cases}$

Soln

The Fourier cosine transform of $f(x)$ is

$$\mathcal{F}_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 x^2 \cos wx \, dx + 0$$

$$= \sqrt{\frac{2}{\pi}} \left[x^2 \left(\frac{1}{w} \sin wx \right) - 2x \left(\frac{1}{w} \left(\frac{1}{-w} \right) \cos wx \right) + 2 \left(\frac{1}{-w^2} \frac{1}{w} \sin wx \right) \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{x^2}{w} \sin wx + \frac{2x}{w^2} \cos wx - \frac{2}{w^3} \sin wx \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\left\{ \frac{1}{w} \sin w + \frac{2}{w^2} \cos w - \frac{2}{w^3} \sin w \right\} - \left\{ 0 \right\} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin w}{w} + \frac{2}{w^2} \cos w - \frac{2}{w^3} \sin w \right]$$

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$$\int_0^\infty e^{-ax}\cos wx\,dx = \frac{a}{a^2+w^2}$$

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and the Fourier sine transform of $f(x)$ is

$$f_s[f(x)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\sin wx\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^1 x^2 \sin wx\,dx + 0$$

$$= \sqrt{\frac{2}{\pi}}\left[x^2\left(\frac{1}{-w}\cos wx\right) - 2x\left(\frac{1}{-w}\frac{1}{w}\sin wx\right) + 2\left(\frac{1}{-w^2}\frac{1}{(-w)}\cos wx\right)\right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{-x^2}{w}\cos wx + \frac{2x}{w^2}\sin wx - \frac{2}{w^3}\cos wx\right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\left\{-\frac{1}{w}\cos w + \frac{2}{w^2}\sin w - \frac{2}{w^3}\cos w\right\} - \left\{0+0-\frac{2}{w^3}\right\}\right]$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{2}{w^2}\sin w - \frac{1}{w}\cos w - \frac{2}{w^3}\cos w + \frac{2}{w^3}\right]$$

Q. Find cosine transform of $f(x) = \frac{1}{1+x^2}$ for $x > 0$.

Soln

The Fourier cosine transform of $f(x)$ is

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos wx\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty \frac{1}{1+x^2}\cos wx\,dx$$

$$= \sqrt{\frac{2}{\pi}} * \frac{\pi}{2}e^{-w}$$

$$= \sqrt{\frac{\cancel{2}}{\cancel{\pi}}\times\frac{\cancel{\pi}}{\cancel{2}}\times\frac{\pi}{2}}\;e^{-w}$$

$$= \sqrt{\frac{\pi}{2}}\,e^{-w}$$

$\rightarrow \int (-2x) e^{-x^2} dx = e^{-x^2}$ $\rightarrow \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

$\rightarrow \int uv\,dx = u\int v\,dx - \int \left(\frac{du}{dx} \int v\,dx\right) dx$

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Q. Find Fourier cosine transform of $f(x) = e^{-x^2}$

<u>Soln</u>

The Fourier cosine transform of $f(x)$ is

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx\, dx$$

$$f_c[e^{-x^2}] = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos wx\, dx$$

$$\text{Let } I = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos wx\, dx \ldots\ldots (1)$$

Diff (1) w.r. to "w".

$$\frac{dI}{dw} = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} (-x) \sin wx\, dx$$

$$\left[\frac{d}{dw}(\cos xw) = -x \sin xw\right]$$

$$= \frac{1}{2}\sqrt{\frac{2}{\pi}} \int_0^\infty (-2x) e^{-x^2} \sin wx\, dx$$

Integrating w.r. to x

$$\frac{dI}{dw} = \frac{1}{2}\sqrt{\frac{2}{\pi}} \left[\left[\sin wx\, e^{-x^2}\right]_0^\infty - \int_0^\infty w \cos wx\, e^{-x^2} dx \right]$$

$$\text{or, } \frac{dI}{dw} = -\frac{w}{2}\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos wx\, dx$$

$$\text{or, } \frac{dI}{dw} = -\frac{w}{2} I$$

$$\text{or, } \frac{dI}{I} = -\frac{w}{2} dw$$

Integrating both sides,

$$\text{or, } \log I = -\frac{w^2}{4} + \log c$$

$$\text{or, } \log\left(\frac{I}{c}\right) = -\frac{w^2}{4}$$

$$\text{or, } \frac{I}{c} = e^{-w^2/4}$$

or, $I = ce^{-\omega^2/4}$ —— (2)

when $\omega = 0$ in eq^n (2)

or, $I = c$ ⟶

also when $\omega = 0$ in eq^n (1)

$$I = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos 0 \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} dx$$

$$= \sqrt{\frac{2}{\pi}} \frac{\sqrt{\pi}}{2}$$

$\therefore I = \frac{1}{\sqrt{2}}$, $\therefore c = \frac{1}{\sqrt{2}}$

from (2)

$$I = \frac{1}{\sqrt{2}} e^{-\omega^2/4}$$

$$\therefore f_c[e^{-x^2}] = \frac{1}{\sqrt{2}} e^{-\omega^2/4}$$

Note: $\int_{-\infty}^{\infty} \cos(x-t)\omega \, d\omega = 2\int_0^{\infty} \cos(x-t)\omega \, d\omega$

$= \frac{1}{2}\int_{-\infty}^{\infty} \cos(x-t)\omega \, d\omega$

$= \int_0^{\infty} \cos(x-t)\omega \, d\omega$

V.V.I

Fourier transform

We have, the Fourier integral of $f(x)$ in a real form is $f(x) = \int_0^{\infty} [A(\omega)\cos x\omega + B(\omega)\sin x\omega] \, d\omega$ —— ①

where $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos\omega t \, dt$ &

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin\omega t \, dt$$

Substituting the value of $A(\omega)$ & $B(\omega)$ in eqⁿ ① we get

$$f(x) = \frac{1}{\pi}\int_0^{\infty}\int_{-\infty}^{\infty} [\cos\omega x \cos\omega t + \sin\omega x \sin\omega t] f(t) \, d\omega dt.$$

$$= \frac{1}{\pi}\int_0^{\infty}\int_{-\infty}^{\infty} \cos(\omega x - \omega t) f(t) \, d\omega dt$$

$$= \frac{1}{2\pi}\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \cos(\omega x - \omega t) f(t) \, d\omega dt + i\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sin(\omega x - \omega t) f(t) \, d\omega dt\right]$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} [\cos(\omega x - \omega t) + i\sin(\omega x - \omega t)] f(t) \, d\omega dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{i(\omega x - \omega t)} f(t) \, d\omega dt \quad [\because e^{i\theta} = \cos\theta + i\sin\theta]$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega t} f(t) \, dt\right) e^{i\omega x} \, d\omega$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \hat{f}(\omega) e^{ix\omega} \, d\omega \quad —— ②$$

where, $\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega t} f(t) \, dt$,

is called fourier transform.

other notation, $\boxed{\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega x} f(x) \, dx}$

and eqⁿ ② is called inverse fourier transform.

Q. Find Fourier transform of $f(x) = \begin{cases} x^2 & \text{for } |x|<1, \text{ or, } -1<x<1 \\ 0 & \text{otherwise} \end{cases}$

Sol^n:

The Fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega x} f(x)\,dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-1}^{1} e^{-i\omega x} f(x)\,dx + 0$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-1}^{1} (\cos\omega x - i\sin\omega x)\,x^2\,dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-1}^{1} x^2\cos\omega x\,dx - i\cdot\frac{1}{\sqrt{2\pi}}\int_{-1}^{1} x^2\sin\omega x\,dx$$

$$= 2\cdot\frac{1}{\sqrt{2\pi}}\int_{0}^{1} x^2\cos\omega x\,dx - i\cdot 0$$

$$= \sqrt{\frac{2}{\pi}}\int_{0}^{1} x^2\cos\omega x\,dx$$

$$= \sqrt{\frac{2}{\pi}}\left[x^2\left(\frac{1}{\omega}\sin\omega x\right) - 2x\left(\frac{1}{-\omega^2}\cos\omega x\right) + 2\left(\frac{1}{-\omega^3}\sin\omega x\right)\right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\left\{\frac{1}{\omega}\sin\omega + \frac{2}{\omega^2}\cos\omega - \frac{2}{\omega^3}\sin\omega\right\} - \{0-0+0\}\right]$$

$$= \sqrt{\frac{2}{\pi}}\left(\frac{1}{\omega}\sin\omega + \frac{2}{\omega^2}\cos\omega - \frac{2}{\omega^3}\sin\omega\right)$$

Note :
$f(x) = |x|$ is even function
Defn : $|x| = x$ if $x > 0$ (+ve interval)
$= -x$ if $x < 0$ (-v interval)

Q. Find the fourier transform
$f(x) = |x|$ for $-1 < x < 1$
$= 0$ otherwise.

Soln :
The fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} f(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-i\omega x} |x|\, dx + 0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} (\cos\omega x - i\sin\omega x)\, |x|\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \Big[\underbrace{\int_{-1}^{1} |x| \cos\omega x\, dx}_{\text{even * even = even}} \Big] - i\frac{1}{\sqrt{2\pi}} \underbrace{\int_{-1}^{1} |x| \sin\omega x\, dx}_{\text{even * odd = odd}}$$

$$= \frac{2}{\sqrt{2\pi}} \int_{0}^{1} |x| \cos\omega x\, dx - 0$$

$$= \sqrt{\frac{2}{\pi}} \int_{0}^{1} x \cos\omega x\, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[x \cdot \frac{1}{\omega} \sin\omega x + 1 * \frac{1}{\omega} \times \frac{1}{\omega} \cos\omega x \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{x}{\omega} \sin\omega x + \frac{1}{\omega^2} \cos\omega x \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\left\{ \frac{\sin\omega}{\omega} + \frac{1}{\omega^2} \cos\omega \right\} - \left\{ 0 + \frac{1}{\omega^2} \right\} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{1}{\omega} \sin\omega + \frac{1}{\omega^2} \cos\omega - \frac{1}{\omega^2} \right]$$

Q. Find the Fourier transform

$$f(x) = 1 - x^2 \text{ for } 0 < x < 1$$
$$= 0 \text{ otherwise}$$

Soln:

The fourier transform of $f(x)$ is

$$F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} f(x)\,dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{0}^{1} e^{-i\omega x} (1 - x^2)\,dx + 0$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_0^1 e^{-i\omega x} dx - \int_0^1 x^2 e^{-i\omega x} dx\right]$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \left[\frac{e^{-i\omega x}}{-i\omega}\right]_0^1 - \left[x^2 \frac{ie^{-i\omega x}}{-i\omega} - 2x \times \frac{1}{-i\omega} \times \frac{1}{-i\omega} e^{-i\omega x} + 2 \times \frac{1}{i^2\omega^2} \times \frac{1}{-i\omega} \times e^{-i\omega x}\right]_0^1 \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \left[\frac{e^{-i\omega}}{-i\omega} - \frac{1}{i\omega}\right] - \left[-x^2 \frac{e^{-i\omega x}}{i\omega} + \frac{2x}{\omega^2} e^{-i\omega x} + \frac{2}{i\omega^3} e^{-i\omega x}\right]_0^1 \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \left[\frac{e^{-i\omega}}{-i\omega} - \frac{1}{i\omega}\right] \right.$$

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Sol^n

The fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} f(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{0}^{1} e^{-i\omega x} (1-x^2)\, dx + 0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{0}^{1} \underset{u}{(1-x^2)}\, \underset{v}{e^{-i\omega x}}\, dx$$

$$[uv_1 - u'v_2 - u''v_3 \ldots\ldots]$$

$$= \frac{1}{\sqrt{2\pi}} \left[(1-x^2)\frac{e^{-i\omega x}}{(-i\omega)} - (0-2x)\frac{e^{-i\omega x}}{(-i\omega)^2} + (-2)\frac{e^{-i\omega x}}{(-i\omega)^3} \right]_0^1$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{(1-x^2)e^{-i\omega x}}{-i\omega} + \frac{2x}{i^2\omega^2} e^{-i\omega x} + \frac{2e^{-i\omega x}}{i^3\omega^3} \right]_0^1$$

$$= \frac{1}{\sqrt{2\pi}} \left[\left\{ 0 + \frac{2}{-\omega^2} e^{-i\omega} + \frac{2}{i^3\omega^3} \right\} - \left\{ \frac{1}{-i\omega} + 0 + \frac{2}{i^3\omega^3} \right\} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{2}{-\omega^2} e^{-i\omega} - \cancel{\frac{2}{i\omega^3}} + \frac{1}{i\omega} + \cancel{\frac{2}{i\omega^3}} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{i\omega} - \frac{2}{\omega^2} e^{-i\omega} \right]$$

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10. Find the Fourier transform of $f(x)=\begin{cases}1-x^2 & \text{for } |x|<1\\ 0 & \text{for } |x|>1\end{cases}$

and then show that $\displaystyle\int_0^{\infty}\left[\left(\frac{x\cos x-\sin x}{x^3}\right)\right]\cos\frac{x}{2}\,dx = -\frac{3\pi}{16}$

Soln:

The fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-i\omega x}f(x)\,dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-1}^{1}\cancel{e^{-i\omega t}(1-x^2)\,dt}+\cancel{\emptyset}$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-1}^{1}(1-x^2)\,e^{-i\omega x}\,dx+0$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-1}^{1}(1-x^2)(\cos\omega x-i\sin\omega x)\,dx$$

$$=\frac{1}{\sqrt{2\pi}}\left[\int_{-1}^{1}\underbrace{(1-x^2)}_{\text{even}}\underbrace{\cos\omega x}_{\text{even}}\,dx-i\int_{-1}^{1}\underbrace{(1-x^2)}_{\text{even}}\underbrace{\sin\omega x}_{\text{odd}}\,dx\right]$$

$$=\frac{1}{\sqrt{2\pi}}\left[\int_{-1}^{1}(1-x^2)\cos\omega x\,dx+0\right]$$

Since $(1-x^2)$ is even

$$=2\frac{1}{\sqrt{2\pi}}\int_0^1(1-x^2)\cos\omega x\,dx$$

$$=\frac{2}{\sqrt{2\pi}}\left[(1-x^2)\left(\frac{1}{\omega}\sin\omega x\right)-(-2x)\left(\frac{1}{\omega}\times\frac{1}{-\omega}\cos\omega x\right)+(-2)\left(\frac{1}{-\omega^3}\sin\omega x\right)\right]_0^1$$

$$=\sqrt{\frac{2}{\pi}}\left[\frac{(1-x^2)\sin\omega x}{\omega}-\frac{2x\cos\omega x}{\omega^2}-\frac{2}{-\omega^3}\sin\omega x\right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\left\{0 - \frac{2\cos\omega}{\omega^2} + \frac{2\sin\omega}{\omega^3}\right\} - \{0-0-0\}\right]$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{-2\omega\cos\omega + 2\sin\omega}{\omega^3}\right]$$

$$\hat{f}(\omega) \text{ or } \mathcal{F}[f(x)] = -\frac{2}{\omega^3}\sqrt{\frac{2}{\pi}}\,(\omega\cos\omega - \sin\omega)$$

Now,

$$[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{ix\omega}\, d\omega$$

$$(1-x^2) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} -\frac{2}{\omega^3}\sqrt{\frac{2}{\pi}}\,(\omega\cos\omega - \sin\omega)\, e^{ix\omega}\, d\omega$$

or, $$(1-x^2) = -2 \times \sqrt{\frac{2}{\pi}} \times \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} -\frac{1}{\omega^3}(\omega\cos\omega - \sin\omega)\, e^{ix\omega}\, d\omega$$

or, $$(1-x^2) = \frac{2}{\pi}\int_{-\infty}^{\infty} \frac{\sin\omega - \omega\cos\omega}{\omega^3}\, e^{ix\omega}\, d\omega$$

put $x = \frac{1}{2}$

or, $$\frac{3}{8}\pi = \int_{-\infty}^{\infty} \frac{\sin\omega - \omega\cos\omega}{\omega^3}\, e^{i\omega/2}\, d\omega$$

or, $$\frac{3}{8}\pi = \int_{-\infty}^{\infty} \left(\frac{\sin\omega - \omega\cos\omega}{\omega^3}\right)\left(\cos\frac{\omega}{2} + i\sin\frac{\omega}{2}\right) d\omega$$

Equating real & imaginary part.

The real part is:

$$\int_{-\infty}^{\infty} \frac{\sin\omega - \omega\cos\omega}{\omega^3}\cos\frac{\omega}{2}\, d\omega = \frac{3}{8}\pi$$

or, $$\int_{-\infty}^{\infty} \left(\frac{\sin x - x\cos x}{x^3}\right)\cos\frac{x}{2}\, dx = \frac{3\pi}{8}$$

Hence proved.

or, $$-2\int_{0}^{\infty} \left(\frac{x\cos x - \sin x}{x^3}\right)\cos\frac{x}{2}\, dx = \frac{3\pi}{8}$$

$$\therefore \int_{0}^{\infty} \frac{x\cos x - \sin x}{x^3}\cos\frac{x}{2}\, dx = -\frac{3}{16}\pi$$ proved

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Eg:- Find the Fourier transform represented by

$$f(x) = \begin{cases} 1 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$$

and then evaluate $\int_0^\infty \frac{\sin x}{x}\, dx$

Solⁿ

The fourier transform of $f(x)$ is represented as:

$$\mathcal{f}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} f(x)\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-i\omega x} \cdot 1\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^{1}$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \times \frac{1}{i\omega} \left[e^{i\omega} - e^{-i\omega} \right]$$

$$= \frac{1}{\sqrt{2\pi}\, i\omega} \times \left(\frac{e^{i\omega} - e^{-i\omega}}{2i} \right) \times 2$$

$$\left[\because \sin i\omega = \frac{e^{i\omega} - e^{-i\omega}}{2i} \right]$$

$$= \frac{1}{i} \sqrt{\frac{2}{\pi}}\, \frac{\sin\omega}{\omega}$$

We have,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{ix\omega}\, d\omega$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{i} \sqrt{\frac{2}{\pi}}\, \frac{\sin\omega}{\omega}\, e^{ix\omega}\, d\omega$$

$$= \frac{1}{i\pi} \int_{-\infty}^{\infty} \frac{e^{i\omega x} \sin\omega}{\omega}\, d\omega$$

$\int_0^\infty e^{-x^2}dx = \sqrt{\pi}/2$

$\int_{-\infty}^\infty e^{-x^2}dx = \sqrt{\pi}$

$\int_{-\infty}^\infty e^{-ax^2}dx = \int_{-\infty}^\infty e^{-(\sqrt{a}x)^2}dx$

$= \frac{1}{\sqrt{a}}\sqrt{\pi}$

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or, $1 = \frac{1}{i\pi}\int_{-\infty}^\infty \frac{e^{i\omega x}\sin\omega}{\omega}\,d\omega$

put $x=0$

or, $1 = \frac{1}{i\pi}\int_{-\infty}^\infty \frac{\sin\omega}{\omega}\,d\omega$

or, $i\pi = \int_{-\infty}^\infty \frac{\sin\omega}{\omega}\,d\omega$

Now, $i\pi = \int_{-\infty}^\infty \frac{\sin x}{x}\,dx$

or, $i\pi = 2\int_0^\infty \frac{\sin x}{x}\,dx$

$\therefore \int_0^\infty \frac{\sin x}{x}\,dx = \frac{i\pi}{2}$

Imp Q. Find the Fourier transform of e^{-ax^2}

Soln :

The Fourier transform of $f(x)$ is

$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty f(x)\,e^{-i\omega x}\,dx$

Now, $\mathcal{F}[e^{-ax^2}] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-ax^2}\cdot e^{-i\omega x}\,dx$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-a\left\{x^2 + ix\frac{\omega}{a}\right\}}\,dx$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-a\left\{x^2 + 2x\frac{i\omega}{2a} + \left(\frac{i\omega}{2a}\right)^2 - \frac{i^2\omega^2}{4a^2}\right\}}\,dx$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{\left\{-a\left(x+\frac{i\omega}{2a}\right)^2 - \frac{\omega^2}{4a}\right\}}\,dx$

$= \frac{1}{\sqrt{2\pi}}\,e^{\frac{-\omega^2}{4a}}\int_{-\infty}^\infty e^{-a\left(x+\frac{i\omega}{2}\right)^2}\,dx$

$$= \frac{1}{\sqrt{2\pi}} e^{\frac{-\omega^2}{4a}} \cdot \frac{\sqrt{\pi}}{\sqrt{a}} \quad \left[\text{Since, } \int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{\sqrt{\pi}}{\sqrt{a}} \right]$$

$$= \frac{1}{\sqrt{2a}} e^{\frac{-\omega^2}{4a}}$$

Q. Find the Fourier transform of $e^{-\frac{x^2}{2}}$

Soln.

The Fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

Now, $$\mathcal{F}[e^{-x^2/2}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-x^2}{2}} \cdot e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\{x^2 + ix\frac{\omega}{2}\}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\{x^2 + 2x\cdot\frac{i\omega}{2\times2} + \left(\frac{i\omega}{2\times2}\right)^2 - \frac{i^2\omega^2}{16}\}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\{-\frac{1}{2}\left(x+\frac{i\omega}{4}\right)^2 - \frac{\omega^2}{8}\}} dx$$

$$= \frac{1}{\sqrt{2\pi}} e^{\frac{-\omega^2}{8}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(x+\frac{i\omega}{4}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{\omega^2}{8}} \cdot \frac{\sqrt{\pi}}{\sqrt{2}}$$

$$= \frac{1}{2} e^{-\frac{\omega^2}{8}}$$

$\int uv\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\int v\,dx\right)dx$

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Fourier transform of the derivative of f(x).

Statement:

Let $f(x)$ be continuous on the x-axis and $f(x)\to 0$ as $|x|\to\infty$ and $f'(x)$ absolutely integrable on the x-axis then,

i) $\mathcal{F}[f'(x)] = iw\mathcal{F}[f(x)]$

ii) $\mathcal{F}[f''(x)] = (iw)^2\mathcal{F}[f(x)]$

Proof:

By definition of Fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\,e^{-iwx}\,dx \quad \text{--- (1)}$$

So, $$\mathcal{F}[f'(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \underset{v}{f'(x)}\,\underset{u}{e^{-iwx}}\,dx$$

Using integration by parts,

$$\mathcal{F}[f'(x)] = \frac{1}{\sqrt{2\pi}}\left[\left[e^{-iwx} f(x)\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty}(-iw)\,f(x)\,e^{-iwx}\,dx\right]$$

$$= iw\cdot\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\,e^{-iwx}\,dx \quad \text{from (1)}$$

$$\therefore \mathcal{F}[f'(x)] = iw\,\mathcal{F}[f(x)]$$

Similarly,

$$\mathcal{F}[f''(x)] = (iw)^2\,\mathcal{F}[f(x)]$$

Proved.

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Imp

Q. Find the Fourier transform of xe^{-x^2}.

Solⁿ:

Let another function $f(x) = e^{-x^2}$

Diff it w.r. to 'x'

$$f'(x) = -2xe^{-x^2}$$

or, $-\frac{1}{2} f'(x) = xe^{-x^2}$

Taking fourier transform on both sides.

$$\mathcal{F}[xe^{-x^2}] = \frac{-1}{2}\mathcal{F}[f'(x)]$$

Using theorem of derivative of Fourier transform

$$= -\frac{1}{2} iw \mathcal{F}[f(x)]$$

$$= \frac{-iw}{2}\mathcal{F}[e^{-x^2}]$$

$$= \frac{-iw}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2} \cdot e^{-iwx} dx$$

$$\mathcal{F}[xe^{-x^2}] = \frac{-iw}{2\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(x^2+iwx)} dx$$

$$= \frac{-iw}{2\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\{x^2+2\cdot x\cdot\frac{iw}{2}+\frac{i^2w^2}{4}-\frac{i^2w^2}{4}\}} dx$$

$$= \frac{-iw}{2\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(x+\frac{iw}{2})^2} \cdot e^{\frac{i^2w^2}{4}} dx$$

$$= \frac{-iw}{2\sqrt{2\cancel{\pi}}} e^{\frac{-w^2}{4}} \cancel{\sqrt{\pi}}$$

$$= \frac{-iw}{2\sqrt{2}} e^{\frac{-w^2}{4}}$$

Parseval's identity for Fourier transform

If $f(x)$ & $g(x)$ are piece wise continuous, bounded and absolutely integrable on x-axis with Fourier transform $F(\omega)$ & $G(\omega)$ respectively.

Then $\boxed{\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |f(x)|^2 dx}$

Parseval's identity for Fourier Cosine transform

If $f(x)$ & $g(x)$ are piece wise continuous, bounded and absolutely integrable on x-axis with Fourier casine transform $F_c(\omega)$ & $G_c(\omega)$ respectively.

Then $\boxed{\int_{0}^{\infty} |F_c(\omega)|^2 d\omega = \int_{0}^{\infty} |f(x)|^2 dx}$

Parseval's identity for Fourier Sine transform

If $f(x)$ & $g(x)$ are piece wise continuous, bounded and absolutely integrable on x-axis with Fourier sine transform $F_s(\omega)$ & $G_s(\omega)$ respectively.

Then $\boxed{\int_{0}^{\infty} |F_s(\omega)|^2 d\omega = \int_{0}^{\infty} |f(x)|^2 dx}$

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15. Find the fourier cosine transform of e^{-x} and then show that $\int_0^\infty \frac{dx}{(1+x^2)^2} = \pi/4$

<u>Soln</u>:

The fourier cosine transform of $f(x)$ is

$$f_c[f(x)] = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos wx\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_0^\infty \underset{u}{e^{-x}}\,\underset{v}{\cos wx}\,dx$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{-1e^{-x}\cos wx - e^{-x}(-w)\sin wx}{(-1)^2+w^2}\right]_0^\infty$$

$$= \sqrt{\frac{2}{\pi}}\left[0 - \frac{\{-1e^0\cos 0 - 0\}}{1+w^2}\right]$$

$$F_c(w) = \sqrt{\frac{2}{\pi}}\,\frac{1}{1+w^2}$$

We have, Parseval's identity of Fourier cosine transform.

$$\int_0^\infty |F_c(w)|^2\,dw = \int_0^\infty |f(x)|^2\,dx$$

$$\int_0^\infty \frac{2}{\pi}\,\frac{1}{(1+w^2)^2}\,dw = \int_0^\infty |e^{-x}|^2\,dx$$

$$= \int_0^\infty e^{-2x}\,dx$$

$$= \left[\frac{e^{-2x}}{-2}\right]_0^\infty$$

$$= -\frac{1}{2}(0-e^0)$$

$$= \frac{1}{2}$$

or, $\int_0^\infty \frac{1}{(1+\omega^2)^2} d\omega = \frac{\pi}{4}$, Replace $x \rightarrow \omega$

$\therefore \int_0^\infty \frac{1}{(1+x^2)^2} dx = \frac{\pi}{4}$ Proved

14. Find the Fourier sine transform of e^{-x} and by using parseval's identity, show that $\int_0^\infty \frac{x^2}{(1+x^2)^2} dx = \frac{\pi}{4}$

Soln:

The fourier sine transform of $f(x)$ is

$$f_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \sin \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{-1e^{-x} \sin \omega x - e^{-x} \omega \cos \omega x}{(-1)^2 + \omega^2} \right]_0^\infty$$

$$= \sqrt{\frac{2}{\pi}} \left[0 - \left\{ \frac{0 - \omega \cdot 1}{1+\omega^2} \right\} \right]$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{\omega}{1+\omega^2}$$

We have, parseval identity of Fourier sine transform

$$\int_0^\infty |F_s(\omega)|^2 d\omega = \int_0^\infty |f(x)|^2 dx$$

$$\int \frac{2}{\pi} \frac{\omega^2}{(1+\omega^2)^2} d\omega = \int_0^\infty |e^{-x}|^2 dx$$

$$= \int_0^\infty e^{-2x} dx$$

$$= \left[\frac{e^{-2x}}{2} \right]_0^\infty$$

$$= -\frac{1}{2}(0 - e^{0})$$

$$= \frac{1}{2}$$

$$\therefore \int_0^{\infty} \frac{\omega^2}{(1+\omega^2)^2} d\omega = \frac{\pi}{4}$$

Replace $x \to \omega$

$$\int_0^{\infty} \frac{x^2}{(1+x^2)^2} dx = \frac{\pi}{4}$$

Proved

Definition

Convolution:

The convolution of two function $f(x)$ & $g(x)$ is denoted by $f(x) * g(x)$ and ~~denoted~~ defined by

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp$$

$$= \int_{-\infty}^{\infty} f(x-p)\, g(p)\, dp$$

- Theorem (Convolution of Fourier transform)

Let $f(x)$ & $g(x)$ are piecewise continuous, bounded and absolutely integrable on the x-axis then

$$\mathcal{F}[f(x) * g(x)] = \cancel{\mathcal{F}[f(x)}\, \sqrt{2\pi}\, \mathcal{F}[f(x)] * \mathcal{F}[g(x)]$$

Proof:

By def^n of convolution, $f(x) * g(x) = \int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp$

$$f[g(q)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(q)\, e^{-i\omega q} dx$$

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Taking fourier transformation,

$$f[f(x) * g(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp \right) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)\, g(x-p)\, e^{-i\omega x}\, dp\, dx$$

Changing the order of integration.

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)\, g(x-p)\, e^{-i\omega x}\, dx\, dp$$

· put $x - p = q$

$$\frac{d}{dx}(x-p) = q \quad , \text{when } x \to \infty, \; q \to \infty$$

$$(1-0) = \frac{dq}{dx} \quad \& \; x \to -\infty, \; q \to -\infty$$

$$\therefore dx = dq$$

$$f[f(x) * g(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)\, g(q)\, e^{-i\omega(p+q)}\, dq\, dp$$

$$= \sqrt{2\pi}\; \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(p)\, e^{-i\omega p}\, dp \right) g(q)\, e^{-i\omega q}\, dq$$

$$= \sqrt{2\pi}\; f(f)\; f(g)$$

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eg: Verify the convolution theorem for the function
$$f(x) = g(x) = e^{-x^2}.$$

Soln:

We have convolution theorem is
$$\mathcal{F}(f*g) = \sqrt{2\pi}\,\mathcal{F}(f)\cdot\mathcal{F}(g).$$
Here $f(x) = g(x) = e^{-x^2}$.

And, $f*g = \int_{-\infty}^{\infty} f(p)\,g(x-p)\,dp$
$$= \int_{-\infty}^{\infty} e^{-p^2} e^{-(x-p)^2}\,dp$$
$$= \int_{-\infty}^{\infty} e^{-2\left[p - xp + \left(\frac{x}{2}\right)^2\right]} e^{-\frac{x^2}{2}}\,dp$$
$$= e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} e^{-\left(\sqrt{2}\left(p-\frac{x}{2}\right)\right)^2} dp$$
$$= e^{-\frac{x^2}{2}} \frac{1}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-t^2}\,dt$$
By putting $t = \sqrt{2}\left(p - \frac{x}{2}\right)^2$

$$\therefore f*g = e^{-\frac{x^2}{2}} \sqrt{\frac{\pi}{2}}$$

Taking fourier transform on both side,
$$f(f*g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\pi/2}\, e^{-i\omega t}\,.dt$$
$$= \frac{1}{2}\int_{-\infty}^{\infty} e^{-\left(\frac{t^2}{2} + i\omega t\right)}\,dt$$
$$= \frac{1}{2} e^{-\frac{\omega^2}{2}} \int_{-\infty}^{\infty} e^{-\left(\frac{t+i\omega}{\sqrt{2}}\right)^2} dt$$
$$= \frac{1}{2} e^{-\frac{\omega^2}{2}} \int_{-\infty}^{\infty} e^{-q^2}\,dp\,\sqrt{2}$$

by putting

$$q = \frac{t+i\omega}{\sqrt{2}} = \frac{1}{\sqrt{2}} e^{\frac{-\omega^2}{2}} 2 \frac{\sqrt{\pi}}{2}$$

$$f(f*g) = \sqrt{\frac{\pi}{2}} e^{\frac{-\omega^2}{2}} \quad \ldots\ldots (1)$$

Again, $f[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2} e^{-i\omega t} dt$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(t+\frac{i\omega}{2}\right)^2} e^{\frac{-\omega^2}{4}} dt$$

$$= \frac{e^{\frac{-\omega^2}{4}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-q^2} dq$$

$$= \frac{e^{\frac{-\omega^2}{4}} 2 \cdot \frac{\sqrt{\pi}}{2}}{\sqrt{2\pi}}$$

$$f(f) = \frac{1}{\sqrt{2}} e^{\frac{-\omega^2}{4}}$$

Similarly we get, $f(g) = \frac{1}{\sqrt{2}} e^{\frac{-\omega^2}{4}}$

Multiplying these, we get

$$f(f)\, f(g) = \frac{1}{2} e^{\frac{-\omega^2}{2}}$$

Multiplying both side by $\sqrt{2\pi}$ we get,

$$\sqrt{2\pi} \cdot [f(f) \cdot f(g)] = \sqrt{\frac{\pi}{2}} e^{\frac{-\omega^2}{2}} \quad \ldots\ldots (2)$$

From (1) and (2) we get

$$f(f*g) = \sqrt{2\pi}\, f(f) \cdot f(g).$$

Partial Differential Equations (7+8+8)

An equation involving one or more partial derivatives of an unknown function of two or more independent variables is called partial differential equation.

eg ① . $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ - one dimensional wave eq^n.

② $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ - one dimensional heat eq^n.

③ $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ - two dimensional laplace eq^n.

④ Three dimensional laplace eq^n:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

⑤ Two dimensional heat eq^n:

$$\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

2nd order homogeneous differential eq^n.

eg 1) $\left(\frac{d^2}{dx^2} - 5\frac{d}{dx} + 6\right) y = 0$

or, $y'' - 5y' + 6y = 0$

It's auxillary eq^n is:

$m^2 - 5m + 6 = 0$

$m = 3, 2$

$\boxed{\text{If } m = \alpha, \beta \text{ then } y = C_1 e^{\alpha x} + C_2 e^{\beta x}}$

It's solution is:

$$y = c_1 e^{3x} + C_2 e^{2x}$$

2) $y'' - 4y' + 4y = 0$

Its auxillary eq^n is

$m^2 - 4m + 4 = 0$

or, $(m-2)^2 = 0$

$\therefore m = 2, 2$

It's sol^n is,

$y = (c_1 + c_2 x)e^{2x}$

$\boxed{\text{If } m = \alpha, \alpha \text{ then } y = (c_1 + c_2 x)e^{\alpha x}}$

3) $y'' - 6y' + 13y = 0$

Its auxillary eq^n is

$m^2 - 6m + 13 = 0$

or, $m = 3 + i2, 3 - i2$

$\therefore m = 3 \pm i2$

It's solu~~s~~tion is,

$y = (c_1 \cos 2x + c_2 \sin 2x)e^{3x}$

$\boxed{\text{If } m = \alpha, n = \beta \text{ then } y = (c_1 \cos \beta x + c_2 \sin \beta x)e^{\alpha x}}$

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5b) Solve $u_{xx} + u_{yy} = 0$ by using separation method.

Sol^n:

Here,

the given ~~partia~~ partial differential equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \; \ldots\ldots (1)$$

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY \ldots\ldots (2)$ be the sol^n of (1)

from (1) $\frac{\partial^2}{\partial x^2}(XY) + \frac{\partial^2}{\partial y^2}(XY) = 0$

or, $X''Y + XY'' = 0$

or, $X''Y = -XY''$

or, $\frac{X''}{X} = -\frac{Y''}{Y} = (k \text{ say})$

It is true for same constant.

$\frac{X''}{X} = k$ $\qquad\qquad$ $\frac{-Y''}{Y} = k$

or, $X'' - kX = 0$ $\qquad\qquad$ or, $Y'' + kY = 0$

It's auxillary eqn is, $\qquad\qquad$ It's auxillary eqn is,

or, $m^2 - k = 0$ $\qquad\qquad$ or, $m^2 + k = 0$

or, $m^2 = k$ $\qquad\qquad$ or, $m^2 = -k$

$\therefore m = \sqrt{k}, -\sqrt{k}$ $\qquad\qquad$ or, $m^2 = i^2 k$

$\therefore X(x) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x})$ $\qquad\qquad$ $\therefore m = \pm i\sqrt{k}$

$\therefore Y(y) = (c_3 \cos\sqrt{k}y + c_4 \sin\sqrt{k}y)$

from ②,

$u(x,t) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x})(c_3 \cos\sqrt{k}y + c_4 \sin\sqrt{k}y)$

Que: $u_{xx} - u_{yy} = 0$

Soln

Here,

the given partial differential equation is

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0 \quad \ldots\ldots ①$$

Let $u(x,y) = X(x)Y(y)$

i.e. $u = XY \ldots\ldots ②$ be the soln of ①

from ① $\frac{\partial^2 (XY)}{\partial x^2} - \frac{\partial^2 (XY)}{\partial y^2} = 0$

or, $X''Y - XY'' = 0$

or, $X''Y = XY''$

or, $\frac{X''}{X} = \frac{Y''}{Y}$

It is true for same constant.

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$\frac{X''}{X} = k$ $\quad\quad\quad\quad$ $\frac{Y''}{Y} = k$

or, $X'' - kX = 0$ $\quad\quad$ or, $Y'' - kY = 0$

It's auxillary eq^n is $\quad\quad$ It's auxillary eq^n is

or, $m^2 - k = 0$ $\quad\quad$ or, $m^2 - k = 0$

or, $m^2 = k$ $\quad\quad$ or, $m^2 = k$

$\therefore m = \sqrt{k}, -\sqrt{k}$ $\quad\quad$ $\therefore m = \sqrt{k}, -\sqrt{k}$

$\therefore X(x) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x})$ $\quad\quad$ $\therefore Y(y) = (c_3 e^{\sqrt{k}y} + c_4 e^{-\sqrt{k}y})$

from ②,

$u(x,y) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x})(c_3 e^{\sqrt{k}y} + c_4 e^{-\sqrt{k}y})$

Exercise 8.2

7b) $u_x + u_y = (x+y)u$

Soln

Here,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = (x+y)u \cdots\cdots ①$$

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY \cdots\cdots ②$ be the sol^n of ①

from $\frac{\partial (XY)}{\partial x} + \frac{\partial (XY)}{\partial y} = (x+y)XY$

or, $X'Y + XY' = (x+y)XY$

or, $\frac{X'}{X} + \frac{Y'}{Y} = x + y$

or, $\frac{X'}{X} - x = -\frac{Y'}{Y} + y$

It is true for same constant.

$\frac{X'}{X} - x = -\frac{Y'}{Y} + y = k\,(say)$

$$\frac{X'}{X} - x = k$$

$$\text{or, } \frac{X'}{X} = x + k$$

Integrating both sides,

$$\text{or, } \log X = \frac{x^2}{2} + kx + \log c_1$$

$$\text{or, } \log\left(\frac{X}{c_1}\right) = \frac{x^2}{2} + kx$$

$$\text{or, } \frac{X}{c_1} = e^{\frac{x^2}{2} + kx}$$

$$\therefore X = c_1 e^{\frac{x^2}{2} + kx}$$

$$-\frac{Y'}{Y} + y = k$$

$$\text{or, } \frac{Y'}{Y} = y - k$$

Integrating both sides

$$\text{or, } \log Y \cancel{= kx - \log y \frac{2}{2} +}$$

$$= \frac{y^2}{2} - ky + \log c_2$$

$$\text{or, } \log\left(\frac{Y}{c_2}\right) = \frac{y^2}{2} - ky$$

$$\text{or, } \frac{Y}{c_2} = e^{\frac{y^2}{2} - ky}$$

$$\therefore Y = c_2 e^{y^2/2 - ky}$$

From ②,

$$u(x,y) = c_1 e^{\frac{x^2}{2} + kx} \cdot c_2 e^{\frac{y^2}{2} - ky}$$

$$\therefore u = C e^{\frac{x^2+y^2}{2} - k(x+y)}$$

Eg. Solve $u_x + u_y = 0$ by using separating variables.

Soln

Here,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0 \cdots\cdots ①$$

Let $u(x,y) = X(x)\, Y(y)$

i.e. $u = XY \cdots\cdots ②$ be the soln of ①

from $\frac{\partial (XY)}{\partial x} + \frac{\partial (XY)}{\partial y} = 0$

$$\text{or, } X'Y + XY' = 0$$

$$\text{or, } \frac{X'}{X} + \frac{Y'}{Y} = 0$$

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$$\text{or, } \frac{X'}{X} = -\frac{Y'}{Y} = k \text{ (say)}$$

It is true for same constant.

$\frac{X'}{X} = k$

~~or, $X' - kX = 0$~~

~~It's auxillary eqⁿ is,~~

Integrating both sides,

or, $\log X = kx + \log c_1$

or, $\log\left(\frac{X}{c_1}\right) = kx$

or, $\frac{X}{c_1} = e^{kx}$

$\therefore X = c_1 e^{kx}$

$-\frac{Y'}{Y} = k$

or, $\frac{Y'}{Y} = -k$

Integrating both sides,

or, $\log Y = -ky + \log c_2$

or, $\log\left(\frac{Y}{c_2}\right) = -ky$

or, $\frac{Y}{c_2} = e^{-ky}$

$\therefore Y = c_2 e^{-ky}$

From ②,

$u(x,y) = c_1 e^{kx} \cdot c_2 e^{-ky}$

$\therefore u = c\, e^{k(x-y)}$

Eg: Find the solution of the following differential equation $y^2 u_x - x^2 u_y = 0$ by using separating of variables.

Solⁿ

Here,

$$y^2 \frac{\partial u}{\partial x} - x^2 \frac{\partial u}{\partial y} = 0$$

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY$ ...... ② be the solⁿ of ①

from $y^2 \frac{\partial (XY)}{\partial x} - x^2 \frac{\partial (XY)}{\partial y} = 0$

or, $y^2 X'Y - x^2 XY' = 0$

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$$\text{or, } y^2 X' Y = x^2 X Y'$$

$$\text{or, } y^2 \frac{X'}{X} = x^2 \frac{Y'}{Y} \qquad \text{or, } \frac{X'}{X x^2} = \frac{Y'}{Y y^2}$$

It is true for same constant.

$$\frac{X'}{X x^2} = \frac{Y'}{Y y^2} = k \text{ (say)}$$

$$\text{or, } \frac{X'}{X x^2} = k \qquad\qquad \text{or, } \frac{Y'}{Y y^2} = k$$

$$\text{or, } \frac{X'}{X} = k x^2 \qquad\qquad \text{or, } \frac{Y'}{Y} = k y^2$$

Integrating both sides, $\qquad$ Integrating both sides,

$$\text{or, } \log X = \frac{k x^3}{3} + \log c_1 \qquad \text{or, } \log Y = \frac{k y^3}{3} + \log c_2$$

$$\text{or, } \log\left(\frac{X}{c_1}\right) = \frac{k x^3}{3} \qquad \text{or, } \log\left(\frac{Y}{c_2}\right) = \frac{k y^3}{3}$$

$$\text{or, } \frac{X}{c_1} = e^{\frac{k x^3}{3}} \qquad \text{or, } \frac{Y}{c_2} = e^{\frac{k y^3}{3}}$$

$$\therefore X = c_1 e^{k x^3/3} \qquad \therefore Y = c_2 e^{\frac{k y^3}{3}}$$

From ②,

$$u(x,y) = c_1 e^{\frac{k x^3}{3}} \cdot c_2 e^{\frac{k y^3}{3}}$$

$$\therefore u = C e^{k/3 (x^3 + y^3)}$$

Ex 8.2

7a) $u_x - u_y = 0$

Sol^n

Here,

$$\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = 0 \cdots ①$$

Let $u(x,y) = X(x)\, Y(y)$

i.e. $u = XY \cdots ②$ be the sol^n of ① from $\frac{\partial(XY)}{\partial x} - \frac{\partial(XY)}{\partial y} = 0$

or, $X'Y - XY' = 0$

or, $X'Y = XY'$

or, $\frac{X'}{X} = \frac{Y'}{Y} = k$ (say)

It is true for same constant

$\frac{X'}{X} = k$ $\qquad\qquad$ $\frac{Y'}{Y} = k$

Integrating both sides,

or, $\log X = kx + \log c_1$

or, $\log\left(\frac{X}{c_1}\right) = kx$

or, $\frac{X}{c_1} = e^{kx}$

$\therefore X = c_1 e^{kx}$

Integrating both sides,

or, $\log Y = k\cancel{x}y + \log c_2$

or, $\log\left(\frac{Y}{c_2}\right) = ky$

or, $\frac{Y}{c_2} = e^{ky}$

$\therefore Y = c_2 e^{ky}$

From ②,

$u(x,y) = c_1 e^{kx} \cdot c_2 e^{ky}$

$\therefore u = Ce^{k(x+y)}$

c) $u_{xy} - u = 0$

Soln

Here,

$\frac{\partial u}{\partial xy} - u = 0 \;\ldots\ldots$ ①

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY \ldots$ ② be the soln of ①.

From ①,

$\frac{\partial(XY)}{\partial xy} - XY = 0$

or, $X'Y' = XY$

or, $\frac{X'}{X} = \frac{Y}{Y'} = k$ (say)

or, $\frac{X'}{X} = k$

or, $\log X = kx + \log c_1$

or, $\log\left(\frac{X}{c_1}\right) = kx$

or, $\frac{X}{c_1} = e^{kx}$

$\therefore X = c_1 e^{kx}$

$\frac{Y}{Y'} = k$

or, $kY' = Y$

or, $\frac{Y'}{Y} = \frac{1}{k}$

Integrating,

or, $\log Y = \frac{y}{k} + \log c_2$

or, $\log\left(\frac{Y}{c_2}\right) = \frac{y}{k}$

or, $\frac{Y}{c_2} = e^{y/k}$

$\therefore Y = c_2 e^{y/k}$

from (2),

$u = c_1 e^{kx} \cdot c_2 e^{y/k}$
$= C e^{(kx + y/k)}$

e) $u_{xx} + 9u = 0$

Sol^n

Here,

$$\frac{\partial^2 u}{\partial x^2} + 9u = 0 \quad \dots\dots (1)$$

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY \dots\dots (2)$ be the sol^n of (1)

from (1) $\frac{\partial^2 (XY)}{\partial x^2} + 9u = 0$

or, $X''Y + 9XY = 0$

or, $X''Y = -9XY$

or, $\frac{X''}{X} = -9\frac{Y}{Y}$

or, $\frac{X''}{X} = -9$

or, $X'' + 9X = 0$

It's auxillary eq^n is:

or, $m^2 + 9 = 0$

or, $m^2 = -9$

or, $m^2 = i^2 3^2$

$\therefore m = \pm i3$

from (2),

$\therefore X = (c_1 \cos 3y + c_2 \sin 3y)$

d) $xu_{xy} + 2yu = 0$

Sol^n

or, $x\frac{\partial u}{\partial xy} + 2yu = 0 \dots\dots (1)$

Let $u(x,y) = X(x)\,Y(y)$

i.e. $u = XY \dots\dots (2)$ be the sol^n of (1), From ①,

$$x\frac{\partial(XY)}{\partial xy} + 2yXY = 0$$

or, $xX'Y' = -2yXY$

or, $x\frac{X'}{X} = -2y\frac{Y}{Y'} = k$ (say)

or, $\frac{X'}{X} = k$ | or, $-2y\frac{Y}{Y'} = k$

or, $\frac{X'}{X} = \frac{k}{x}$ | or, $\frac{Y'}{Y} = \frac{-\lambda}{2y}$

Integrating, | Integrating,

$\log X = k\log x + c_1$ | or, $\log Y = \frac{-\lambda}{2}\log y + c_2$

$= c_1 x^k$ | $\therefore Y = c_2 y^{-\lambda/2}$

From ②,

$$u = c_1 x^k c_2 y^{-\lambda/2}$$
$$= c x^k y^{-\lambda/2}$$

f) $u_x = y u_y$

Soln,

$$\frac{\partial u}{\partial x} = y\frac{\partial u}{\partial y} \text{.....①}$$

Let $u(x,y) = X(x)Y(y)$

i.e. $u = XY$.....② be the soln of ①,

From ①,

$X' = yY' = k$ (say)

$X' = k$ | $yY' = k$

Integrating, | or, $Y' = \frac{k}{y}$

or, $X = c_1 kx$ | or, $Y = k\log y$

$\therefore Y = c_2 y^k$

from ②,

$$u = c_1 kx \, c_2 y^k$$
$$= c y^k kx$$

Wave Equation - Vibration of String

1) Derivation of one dimensional wave equation
$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Imp 2023 fall 5b. 2) Solution of wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ with given boundary conditions $u(0,t)=0$, $u(l,t)=0$ and initial deflection $u(x,0)=f(x)$ & initial velocity $u_t(x,0)=g(x)$.

Soln:

Here, the wave eqn is $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ .....(1)

Let $u(x,t) = T(t)\,X(x)$ .....(2) be the soln of (1)

Then eqn (1) becomes $\frac{\partial^2 (XT)}{\partial t^2} = c^2 \frac{\partial^2}{\partial x^2}(XT)$

or, $XT'' = c^2 X'' T$

or, $\boxed{\frac{T''}{T} = c^2 \frac{X''}{X} = k}$ it is true for same constant say ($k = -p^2$)

$\frac{T''}{T} = k$ & $c^2 \frac{X''}{X} = k$

Case I : $k = -p^2$ (negative)

$\frac{T''}{c^2 T} = -p^2$ | $\frac{X''}{X} = -p^2$

or, $T'' + p^2c^2T = 0$

It's auxillary eqn is

$m^2 + p^2c^2 = 0$

$m = \pm icp$

$\therefore T(t) = (c_1 \cos cpt + c_2 \sin cpt)$

or, $X'' + p^2X = 0$

It's auxillary eqn is.

$m^2 + p^2 = 0$

$m = \pm ip$

$\therefore X(x) = (c_3 \cos px + c_4 \sin px)$

from ②,

$$u(x,t) = (c_1 \cos cpt + c_2 \sin cpt)(c_3 \cos px + c_4 \sin px) \quad \cdots\cdots ③$$

Case II : when $k = p^2$ (positive)

$\frac{T''}{c^2 T} = p^2$ $\qquad$ $\frac{X''}{X} = p^2$

$T'' - c^2p^2T = 0$ $\qquad$ $X'' - p^2X = 0$

It's auxillary eq^n is $\qquad$ It's auxillary eq^n is

$m^2 - c^2p^2 = 0$ $\qquad$ $m^2 - p^2 = 0$

$\therefore m = \pm cp$ $\qquad$ $\therefore m = \pm p$

$$\therefore T(t) = (c_5 e^{cpt} + c_6 e^{-cpt})(c_7 e^{px} + c_8 e^{-px}) \cdots\cdots ④$$

Case III : when $k = 0$

$\frac{T''}{c^2 T} = 0$ $\qquad$ $\frac{X''}{X} = 0$

or, $T'' = 0$ $\qquad$ or, $X'' = 0$

Integrating $\qquad$ Integrating

or, $T' = c_9$ $\qquad$ or, $X' = c_{11}$

or, $T = c_9 t + c_{10}$, $\qquad$ or, $X = c_{11}x + c_{12}$

$\therefore T = c_9 t + c_{10}$

from ②,

$$u(x,t) = (c_9 t + c_{10})(c_{11}x + c_{12}) \cdots\cdots ⑤$$

The three possible solutions of the wave equation are:

$$u(x,t) = (c_1 \cos cpt + c_2 \sin cpt)(c_3 \cos px + c_4 \sin px)$$

$$u(x,t) = (c_5 e^{cpt} + c_6 e^{-cpt})(c_7 e^{px} + c_8 e^{-px})$$

$$u(x,t) = (c_9 t + c_{10})(c_{11}x + c_{12})$$

In the wave equation, we will be dealing with problems on vibration. So the vibration of a string must be period function of x & t, this is true only for the solution must involve the trigonometric terms and the trigonometric term is periodic, the feasible sol^n is:

$$u(x,t) = (c_1 \cos cpt + c_2 \sin cpt)(c_3 \cos px + c_4 \sin px) \quad \text{......} (6)$$

Using given boundary $u(0,t) = 0$ in eq^n (6)

$$u(0,t) = (c_1 \cos cpt + c_2 \sin cpt)\, c_3$$

or, $(c_1 \cos cpt + c_2 \sin cpt)\, c_3 = 0$

$\therefore c_3 = 0$, $(c_1 \cos cpt + c_2 \sin cpt) \neq 0$

from (6)

$$u(x,t) = (c_1 \cos cpt + c_2 \sin cpt)(c_4 \sin px) \quad \text{....} (7)$$

again,

boundary condition $u(l,t) = 0$ in eq^n (7)

$u(l,t) = (c_1 \cos cpt + c_2 \sin cpt)\, c_4 \sin pl$

or, $(c_1 \cos cpt + c_2 \sin cpt)\, c_4 \sin pl = 0$

$\therefore \sin pl = 0$, $(c_1 \cos cpt + c_2 \sin cpt) \neq 0$, $c_4 \neq 0$.

or, $\sin pL = \sin n\pi$ for $n = 0, 1, 2, \ldots$

or, $pL = n\pi$

$\therefore p = \dfrac{n\pi}{L}$

from (7)

$$u(x,t) = \left(c_1 \cos \frac{cn\pi}{L} t + c_2 \sin \frac{cn\pi}{L} t\right) c_4 \sin \frac{n\pi}{L} x$$

The most general sol^n of wave eq^n is.

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos \frac{cn\pi}{L} t + B_n \sin \frac{cn\pi}{L} t \right] \sin \frac{n\pi x}{L}} \quad \text{.....} (8)$$

deflection

Using initial condition $u(x,0) = f(x)$ in eq^n (8)

$$\boxed{f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x \; \cdots\cdots (9)}$$

which half range sine series with interval $(0, l)$

where, $\boxed{A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx}$

Diff (8) p.w.r. to t we get,

$$u_t(x,t) = \sum_{n=1}^{\infty} \left[-A_n \frac{cn\pi}{l} \sin \frac{cn\pi}{l} t + B_n \frac{cn\pi}{l} \cos \frac{cn\pi}{l} t \right] \sin \frac{n\pi}{l} x$$

Again, using initial velocity $u_t(x,0) = g(x)$

$$g(x) = \sum_{n=1}^{\infty} \left[B_n \frac{cn\pi}{l} \sin \frac{n\pi}{l} x \right]$$

, which is also half range sine series.

where $B_n \frac{cn\pi}{\cancel{l}} = \frac{2}{\cancel{l}} \int_0^L g(x) \sin \frac{n\pi}{l} x \, dx$

$$\boxed{\therefore B_n = \frac{2}{cn\pi} \int_0^L g(x) \sin \frac{n\pi}{l} x \, dx}$$

Pg 231

Eg: Find the solution of one dimensional wave equation corresponding to the triangular initial deflection.

$$f(x) = \begin{cases} \frac{2k}{L} x & \text{if } 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & \text{if } \frac{L}{2} < x < L \end{cases}$$

and its initial velocity is zero.

Soln:

The most general solution of one dimensional wave eqn is

$$u(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos\frac{cn\pi}{L} t + B_n \sin\frac{cn\pi}{L} t \right] \sin\frac{n\pi x}{L} \quad \text{.....①}$$

where, $A_n = \frac{2}{L}\int_0^L f(x) \sin\frac{n\pi}{L} x \, dx$ &

$$B_n = \frac{2}{cn\pi}\int_0^L g(x) \sin\frac{n\pi}{L} x \, dx$$

$$= 0 \left[\text{since } g(x) = 0 \text{ (initial velocity is zero)}\right]$$

Now,

$$A_n = \frac{2}{L}\left[\int_0^{L/2} \frac{2k}{L} x \sin\frac{n\pi}{L} x \, dx + \int_{L/2}^{L} \frac{2k}{L}(L-x) \sin\frac{n\pi}{L} x \, dx\right]$$

$$= \frac{4k}{L^2}\left[x\left(\frac{\cos\frac{n\pi}{L}x}{\frac{n\pi}{L}}\right) - 1 \frac{\sin\frac{n\pi}{L}x}{\frac{n\pi}{L}\left(\frac{-n\pi}{L}\right)}\right]_0^{L/2}$$

$$+ \left[(L-x)\left(\frac{\cos\frac{n\pi x}{L}}{\frac{-n\pi}{L}}\right) - (0-1)\frac{\sin\frac{n\pi}{L}x}{-\frac{n^2\pi^2}{L^2}}\right]_{L/2}^{L}$$

or, $A_n = \frac{4k}{L^2}\left[\left\{\frac{-L^2}{2n\pi}\cos\frac{n\pi}{2} + \frac{L^2}{n^2\pi^2}\sin\frac{n\pi}{2}\right\} - \{0-0\} + \{0+0\} - \left\{\frac{-L^2}{2n\pi}\cos\frac{n\pi}{2} - \frac{L^2}{n^2\pi^2}\sin\frac{n\pi}{2}\right\}\right]$

or, $A_n = \frac{4k}{L^2} \times \frac{2L^2}{n^2\pi^2}\sin\frac{n\pi}{2}$

$\therefore A_n = \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}$

$$u(x,t) = \sum_{n=1}^{\infty}\left[\frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}\cos\frac{cn\pi}{L}t + 0\sin\frac{cn\pi}{L}t\right]\sin\frac{n\pi}{L}x$$

$$\therefore u(x,t) = \frac{8k}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\sin\frac{n\pi}{2}\cos\frac{cn\pi}{L} + \sin\frac{n\pi}{L}x$$

Ex 8.2

5. Find the deflection $u(x,t)$ of the vibrating string of length $L=\pi$, $c^2=1$ and its initial deflection is zero and initial velocity is

$$g(x) = \begin{cases} 0.01x & \text{if } 0 < x \frac{\pi}{2} \\ 0.01(\pi - x) & \text{if } \pi/2 < x < \pi \end{cases}$$

Soln:

The one dimensional wave eq^n is $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$

since $c^2 = 1$

Then $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ ..... (1)

The most general soln of (1) is $u(x,t)$

$$= \sum_{n=1}^{\infty}\left[A_n\cos\frac{n\pi}{L}t + B_n\sin\frac{n\pi t}{L}\right]\sin\frac{n\pi}{L}x$$

since $L=\pi$,

$$= \sum_{n=1}^{\infty}\left[A_n \cos nt + B_n \sin nt\right] \sin nx \quad \ldots\ldots (1)$$

where,

$$A_n = \frac{2}{\pi}\int_0^{\pi} f(x)\, \sin \frac{n\cancel{\pi}}{\cancel{\pi}} x \, dx$$

Since initial deflection is zero i.e. $f(x)=0$

Then $A_n = 0$

$$B_n = \frac{2}{cn\pi}\int_0^{\pi} g(x)\, \sin\frac{n\cancel{\pi}}{\cancel{\pi}} x\, dx$$

$$B_n = \frac{2}{cn\pi}\left\{\int_0^{\pi/2} \frac{1}{100} x \sin nx\, dx + \int_{\pi/2}^{\pi} \frac{1}{100}(\pi - x)\sin nx\, dx\right\}$$

$$= \frac{1}{50cn\pi}\left[\int_0^{\pi/2} \underset{u\ \cdot\ v}{x \sin nx}\, dx + \int_{\pi/2}^{\pi} \underset{u\ \cdot\ v}{(\pi - x)\sin nx}\, dx\right]$$

$$= \frac{1}{50cn\pi}\left[\left[x\left(\frac{\cos nx}{-n}\right) - 1 \cdot \frac{\sin nx}{-n^2}\right]_0^{\pi/2} + \left[(\pi - x)\left(\frac{\cos nx}{-n}\right) - (0-1)\frac{\sin nx}{-n^2}\right]_{\pi/2}^{\pi}\right]$$

$$= \frac{1}{50cn\pi}\left[\left\{\cancel{\frac{-\pi}{2n}\cos\frac{n\pi}{2}} + \frac{\sin\frac{n\pi}{2}}{n^2}\right\} - \{0+0\} + \{0-0\} - \left\{\cancel{\frac{-\pi}{2n}\cos\frac{n\pi}{2}} - \frac{1}{n^2}\sin\frac{n\pi}{2}\right\}\right]$$

$$= \frac{1}{50cn\pi}\,\frac{2}{n^2}\sin\frac{n\pi}{2}$$

$$\boxed{B_n = \frac{1}{25cn^3\pi}\sin\frac{n\pi}{2}}$$

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from ②,

$$u(x,t) = \sum_{n=1}^{\infty} \left[0 \cos nt + \frac{1}{25cn^3\pi} \sin\frac{n\pi}{2} \sin nt \right] \sin nx$$

$$= \frac{1}{25c\pi} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin\frac{n\pi}{2} \sin nt \sin nx.$$

V.V.I Pg. 232

Eg: Solve one dimensional wave equation with initial deflection is $0.01 \sin 3x$ and initial velocity is zero and $L=\pi$, $c^2 = 1$.

Solⁿ,

We have, one dimensional wave eqⁿ is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \text{ since } c^2 = 1.$$

then, $$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} \quad \ldots\ldots ①$$

The most general solⁿ of ① is $u(x,t)$

$$= \sum_{n=1}^{\infty} \left[A_n \cos\frac{n\pi}{L} t + B_n \sin\frac{n\pi}{L} t \right] \sin\frac{n\pi}{L} x,$$

since $\pi = L$

then, $$u(x,t) = \sum_{n=1}^{\infty} (A_n \cos nt + B_n \sin nt) \sin nx,$$

Since initial velocity is zero i.e. $g(x) = 0$ then $B_n = 0$

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos nt \sin nx$$

$$u(x,t) = A_1 \cos t \sin x + A_2 \cos 2t \sin 2x + A_3 \cos 3t \sin 3x + \ldots\ldots ②$$

We have, $$f(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi}{\pi} x$$

or, $0.01 \sin 3x = A_1 \sin x + A_2 \sin 2x + A_3 \sin 3x + \ldots$

or, $0\sin x + 0\sin 2x + 0.01\sin 3x = A_1 \sin x + A_2 \sin 2x + A_3 \sin 3x + \ldots\ldots$

Equating the coefficient of like terms.

$A_1 = 0, A_2 = 0, A_3 = 0.01, A_4 = 0, \ldots\ldots$

from ②,

$u(x,t) = 0.01 \cos 3t \sin 3x$

Pg 244 V.I

⑧ A tightly stretched string with fixed end points $x=0$ and $x=L$ is initially in a position given by $u = u_0 \sin^3\left(\frac{\pi x}{L}\right)$. If it is released from the rest from this position. Find the displacement.

Soln:

We have, one dimensional wave eqn is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \ldots\ldots ①$$

The most general solution of ① is $u(x,t)$

$$= \sum_{n=1}^{\infty} \left[A_n \cos \frac{cn\pi}{L} t + B_n \sin \frac{cn\pi}{L} t\right] \sin \frac{n\pi}{L} x,$$

since initial velocity is zero i.e. $g(x) = 0$, then, $B_n = 0$

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos \frac{cn\pi}{L} t \sin \frac{n\pi}{L} x$$

or, $u(x,t) = A_1 \cos \frac{c\pi}{L} t \sin \frac{\pi}{L} x + A_2 \cos \frac{2c\pi}{L} t \sin \frac{2\pi}{L} x$

$$+ A_3 \cos \frac{3c\pi}{L} t \sin \frac{3\pi}{L} x + \ldots\ldots ②$$

We have, $f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$

$\sin 3A = 3\sin A - 4\sin^3 A$
$\sin^3 A = \frac{3}{4}\sin A - \frac{1}{4}\sin 3A$

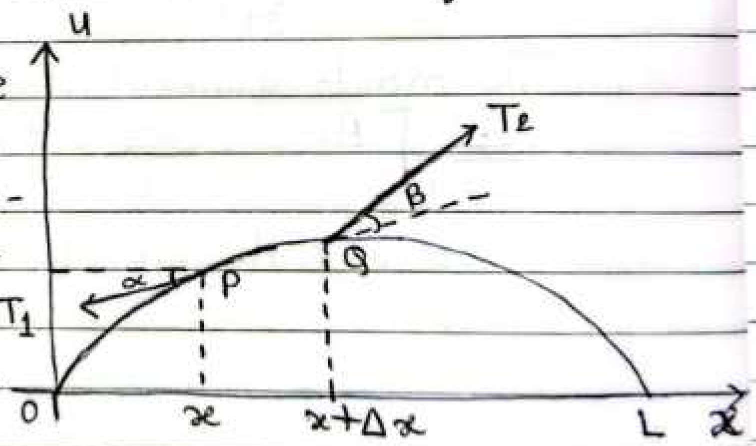
$$u_0 \sin^3 \frac{\pi}{L}x = A_1 \sin \frac{\pi}{L}x + A_2 \sin \frac{2\pi}{L}x + A_3 \sin \frac{3\pi}{L}x + \ldots$$

or, $$\frac{3}{4}u_0 \sin \frac{\pi}{L}x - \frac{1}{4}u_0 \sin \frac{3\pi}{L}x = A_1 \sin \frac{\pi}{L}x + A_2 \sin \frac{2\pi}{L}x + A_3 \sin \frac{3\pi}{L}x + \ldots$$

Equating the coeff. of like terms.
$A_1 = \frac{3}{4}u_0$, $A_2 = 0$, $A_3 = -\frac{1}{4}u_0$, $A_4 = 0$, ...

from ②,
$$u(x,t) = \frac{3}{4}u_0 \cos \frac{c\pi}{L}t \sin \frac{\pi}{L}x - \frac{1}{4}u_0 \cos \frac{3c\pi}{L} + \sin \frac{3\pi}{L}x$$

Derivation of one dimensional wave eq^n



Let us suppose that the force acting on small portion of the string since the string does not offer resistance to bending, the tensions (T_1 & T_2) is tangential to the curve of the string at each point. Let T_1 & T_2 are tensions at the ends point P & Q of that portion. Since the points of the string move vertically, there is no motion in the horizontal direction. Therefore the horizontal components of the tension must be constant.

$$\lim_{\Delta x \to 0} \frac{f_x(x+\Delta x) - f_x(x)}{\Delta x} = f''(x)$$

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we get, $T_1 \cos\alpha = T_2 \cos\beta = T$ (say) constant — ①
in the vertical direction, the forces in vertical components are $-T_1 \sin\alpha$ & $T_2 \sin\beta$ of T_1 & T_2.
Here negative sign appears become the component at p in directed downward
By Newton's 2nd Law,
Resultant force is equals to mass of the portion times the acceleration. That is

$$T_2 \sin\beta - T_1 \sin\alpha = \rho \Delta x \frac{\partial^2 u}{\partial t^2} \quad \ldots\ldots ②$$

where ρ be the mass of undetected string per unit length, Δx is the length of the portion of the undeflected string.
Dividing eq^n ② by eq^n ①

$$\frac{T_2 \sin\beta}{T_2 \cos\beta} - \frac{T_1 \sin\alpha}{T_1 \cos\alpha} = \frac{\rho \Delta x}{T} \frac{\partial^2 u}{\partial t^2}$$

$$\frac{1}{\Delta x}[\tan\beta - \tan\alpha] = \frac{\rho}{T} \frac{\partial^2 u}{\partial t^2} \quad \ldots\ldots ③$$

We know, $\tan\alpha$ & $\tan\beta$ are slopes of the string at x & $x+\Delta x$ respectively.
where $\tan\alpha = \left.\frac{\partial u}{\partial x}\right]_{at\ x}$, $\tan\beta = \left.\frac{\partial u}{\partial x}\right|_{at\ x+\Delta x}$
from ③

$$\frac{1}{\Delta x}\left[\left.\frac{\partial u}{\partial x}\right|_{at\ x+\Delta x} - \left.\frac{\partial u}{\partial x}\right|_{at\ x}\right] = \frac{\rho}{T} \frac{\partial^2 u}{\partial t^2}$$

Taking limit as $\Delta x \to 0$ on both sides,

or, $$\lim_{\Delta x \to 0}\left[\left.\frac{\partial u}{\partial x}\right|_{at\ x+\Delta x} - \left.\frac{\partial u}{\partial x}\right|_{at\ x}\right] = \frac{\rho}{T} \frac{\partial^2 u}{\partial t^2}$$

or, $$\frac{\partial^2 u}{\partial x^2} = \frac{\rho}{T} \frac{\partial^2 u}{\partial t^2}$$

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or, $\frac{\partial^2 u}{\partial t^2} = \frac{T}{\rho} \frac{\partial^2 u}{\partial x^2}$

$\therefore \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, where $c^2 = \frac{T}{\rho}$

which is one dimensional wave eq^n.

2013 fall 3b

Solution of one dimensional heat equation with given boundary condition $u(0,t)=0$, $u(l,t)=0$ for all t and initial condition $u(x,0)=f(x)$.

Sol^n:

Here, the one dimensional heat eq^n is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \cdots\cdots (1)$$

Let $u(x,t) = X(x)\,T(t)$

i.e. $u = XT \cdots\cdots (2)$ be the sol^n of (1)

Then eq^n (1) becomes $\frac{\partial (XT)}{\partial t} = c^2 \frac{\partial^2}{\partial x^2}(XT)$

or, $XT' = c^2 X''T$

or, $\frac{T'}{c^2T} = \frac{X''}{X}$, it is true for same constant.

or, $\frac{T'}{c^2T} = \frac{X''}{X} = k$ (say)

or, $\frac{T'}{c^2T} = k$ & $\frac{X''}{X} = k$

Case I: When k is negative say $k = -p^2$

$\frac{T'}{c^2T} = -p^2$

$\frac{T'}{T} = -c^2p^2$

Integrating,

$\frac{X''}{X} = -p^2$

or, $X'' + p^2X = 0$

It's auxillary eq^n is

$m^2 + p^2 = 0$

or, $\log T = -c^2p^2t + \log c_1$

or, $\log\left(\frac{T}{c_1}\right) = -c^2p^2t$

or, $\frac{T}{c_1} = e^{-c^2p^2t}$

$\therefore T = c_1 e^{-c^2p^2t}$

$m = \pm ip$

$\therefore X = (c_2 \cos px + c_3 \sin px)$

from ②,

$u(x,t) = c_1 e^{-c^2p^2t}(c_2 \cos px + c_3 \sin px) \ldots\ldots ③$

Case II : When k is positive say $k = p^2$

$\frac{T'}{c^2T} = p^2$

$\frac{X''}{X} = p^2$

$\therefore T = c_4 \cancel{e^{2}p}\, e^{c^2p^2t}$

$m^2 = \pm p$

$\therefore X = c_5 e^{px} + c_6 e^{-px}$

from ②,

$u(x,t) = c_4 e^{c^2p^2t}(c_5 e^{px} + c_6 e^{-px}) \ldots\ldots ④$

Case III : when $k = 0$

$\frac{T'}{c^2T} = 0$

$\frac{X''}{X} = 0$

$T = c_7$

$X = c_8x + c_9$,

from ②

$u(x,t) = c_7(c_8x + c_9) \ldots ⑤$

The feasible solution is:

$u(x,t) = c_1 e^{-c^2p^2t}(c_2 \cos px + c_3 \sin px) \ldots\ldots ⑥$

using boundary condition $u(0,t)$ in eq^n ⑥

$u(0,t) = c_1 e^{-c^2p^2t}(c_2)$

$c_1 e^{-c^2p^2t} c_2 = 0$

$c_2 = 0$, $c_1 e^{-c^2p^2t} \neq 0$

from ⑥,

$u(x,t) = c_1 e^{-c^2p^2t} c_3 \sin px \ldots\ldots ⑦$

using $u(l,t) = 0$, in eq^n ⑦

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$u(l,t) = c_1 e^{-c^2p^2t} \, c_3 \sin pL$

or, $0 = c_1 e^{-c^2p^2t} c_3 \sin pL$

or, $c_3 \neq 0$, $c_1 e^{-c^2p^2t} \neq 0$

$\sin pL = 0$

or, $\sin pL = \sin n\pi$, for $n = 0, 1, 2, \ldots$

or, $pL = n\pi$

or, $p = \frac{n\pi}{L}$

from ⑦,

$u(x,t) = c_1 e^{-c^2\frac{n^2\pi^2}{L^2}} c_3 \sin \frac{n\pi}{L} x$

The most general solⁿ of one dimensional heat eqⁿ is:

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} A_n e^{-c^2\frac{n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x} \quad \ldots\ldots ⑧$$

using initial condition, $u(x,0) = f(x)$

$u(x,0) = \sum_{n=1}^{\infty} A_n e^{\frac{-c^2n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x$

$$\boxed{f(x) = \sum_{n=1}^{\infty} A_n e^{\frac{-c^2n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x} \quad \ldots\ldots ⑨$$

which is half range sine series in the interval $(0, L)$

where,

$$\boxed{A_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi}{L} x}$$

substituting the value of A_n in eqⁿ ⑧ which is the solⁿ of one dimensional heat equation.

2021 (Spring)
5a) Wave
5b) Heat

Trigo : Equating
Algebra: Integration

2014 (fall)

3b. Find the temperature in a laterally insulated bar of length L whose ends are kept at a zero temperature, assuming that the initial temperature is

$$f(x) = \begin{cases} x \text{ if } 0 < x < \frac{L}{2} \\ L - x \text{ if } \frac{L}{2} < x < L \end{cases}$$

Solⁿ,

We have one dimensional heat eqⁿ

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \ldots\ldots (1)$$

The most general solⁿ of (1) is,

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-c^2 \frac{n^2\pi^2}{L^2} t} \sin \frac{n\pi}{L} x \quad \ldots\ldots (2)$$

where $A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx$

$$= \frac{2}{L} \left[\int_0^{L/2} x \sin \frac{n\pi}{L} x \, dx + \int_{L/2}^{L} (L-x) \sin \frac{n\pi}{L} x \, dx \right]$$

$$= \frac{2}{L} \left[\left[x \left(\frac{\cos \frac{n\pi}{L} x}{-n\pi/L} \right) - \frac{1}{\frac{-n\pi}{L}} * \frac{\sin \frac{n\pi}{L} x}{\frac{n\pi}{L}} \right]_0^{L/2} + \left[(L-x) \left(\frac{\cos \frac{n\pi}{L}}{\frac{-n\pi}{L}} \right) x - \frac{(-1)}{\frac{-n\pi}{L}} * \frac{\sin \frac{n\pi}{L}}{\frac{n\pi}{L}} \right]_{L/2}^{L} \right]$$

$$= \frac{2}{L} \left[\frac{x}{\frac{-n\pi}{L}} \cos \frac{n\pi}{L} + \frac{1}{\frac{n^2\pi^2}{L^2}} \sin \frac{n\pi}{L} x \right]_0^{L/2} + \left[\frac{(L-x)}{-n\pi/L} \left(\cos \frac{n\pi}{L} \right) x - \frac{1}{\frac{n^2\pi^2}{L}} \sin \frac{n\pi}{L} \right]_{L/2}^{L}$$

$$= \frac{2}{L}\left[\left\{\cancel{\frac{-l^2}{2n\pi}\cos\frac{n\pi}{2}}\ \frac{l^2}{n^2\pi^2}\sin\frac{n\pi}{2}\right\} - \{0-0\} + \{0-0\} - \left\{\cancel{\frac{L}{-2n\pi}\cos\frac{n\pi}{2}} - \frac{L}{n^2\pi^2}\sin\frac{n\pi}{2}\right\}\right]$$

$$= \frac{2}{L} \times \frac{2L^2}{n^2\pi^2}\sin\frac{n\pi}{2}$$

$$A_n = \frac{4L}{n^2\pi^2}\sin\frac{n\pi}{2}$$

from (2), $u(x,t) = \sum_{n=1}^{\infty} \frac{4L}{n^2\pi^2}\sin\frac{n\pi}{2}\, e^{-\frac{c^2n^2\pi^2}{L^2}t}\sin\frac{n\pi}{L}x$

which is required sol^n.

2015 fall

6b) A homogeneous rod of conducting material of length 100 cm has its end kept at zero temperature and temperature initially is

$$f(x) = \begin{cases} x; & 0 \le x \le 50 \\ 100-x, & 50 \le x \le 100 \end{cases}$$

Sol^n,

We have one dimensional heat eq^n.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \ \ldots\ldots (1) \quad \text{since } L = 100$$

The most general sol^n of (1) is

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-c^2\frac{n^2\pi^2}{10000}t}\sin\frac{n\pi}{100}x \ \ldots\ldots (2)$$

where $A_n = \frac{2}{100}\int_0^{100} f(x)\sin\frac{n\pi}{100}x\,dx$

$$= \frac{2}{100}\left[\int_0^{50} x\sin\frac{n\pi}{100}x\,dx + \int_{50}^{100}(100-x)\sin\frac{n\pi}{100}x\,dx\right]$$

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$$= \frac{2}{100}\left[x\left(\frac{\cos\frac{n\pi}{100}x}{\frac{-n\pi}{100}}\right) - \frac{1}{\frac{-n\pi}{100}} \times \frac{\sin\frac{n\pi}{100}x}{\frac{n\pi}{100}} \right]_0^{50}$$

$$+ \left[(100-x)\left(\frac{\cos\frac{n\pi}{100}}{\frac{-n\pi}{100}}\right)x - \frac{(-1)}{\frac{-n\pi}{100}} * \frac{\sin\frac{n\pi}{100}}{\frac{n\pi}{100}} \right]_{50}^{100}$$

$$= \frac{2}{100}\left[\left[\frac{x}{\frac{-n\pi}{100}} \cos\frac{n\pi}{100} + \frac{1}{\frac{n^2\pi^2}{l^2}} \sin\frac{n\pi}{100}x \right]_0^{50}\right.$$

$$\left. + \left[\frac{(100-x)}{-n\pi/100}\left(\cos\frac{n\pi}{100}\right)x - \frac{1}{\frac{n^2\pi^2}{100}} \sin\frac{n\pi}{100} \right]_{50}^{100}\right]$$

$$= \frac{2}{100}\left[\left\{ \frac{-10000}{2n\pi} \cos\frac{n\pi}{2} \quad \frac{10000}{n^2\pi^2} \sin\frac{n\pi}{2} \right\} - \{0-0\} + \{0-0\} - \left\{ \frac{100}{-2n\pi} \cos\frac{n\pi}{2} - \frac{100}{n^2\pi^2} \sin\frac{n\pi}{2} \right\} \right]$$

$$= \frac{2}{100} * \frac{2(100)^2}{n^2\pi^2} \sin\frac{n\pi}{2}$$

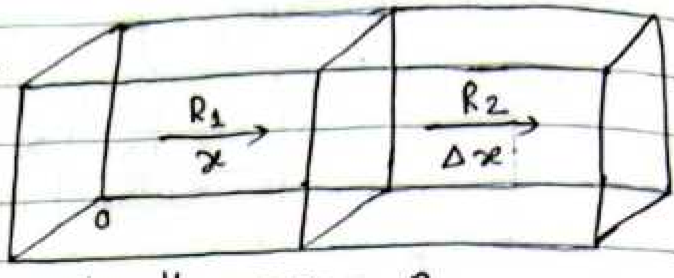
$$= \frac{400}{n^2\pi^2} \sin\frac{n\pi}{2}$$

from ②,

$$u(x,t) = \sum_{n=1}^{\infty} \frac{400}{n^2\pi^2} \sin\frac{n\pi}{2}\, e^{\frac{-c^2n^2\pi^2}{10000}t} \sin\frac{n\pi}{100}x$$

which is required sol^n.

Derivation of one dimensional heat eqⁿ:



Consider the heat flows along a homogeneous bar of uniform cross section $A\,(cm)^2$ in the direction perpendicular to the area A.

Take one end of the bar at the origin and the direction of flows along the positive x-axis.

If the heat flows along the positive x-axis, then the temperature of the bar depends on the distance from the origin and time taken.

Let u be the temperature and x be the distance from origin where the heat flows at time "t". So $u = u(x,t)$. The rate of increase of heat in this slab of area A (cm² and thickness δx in one second) = mass X specific heat capacity X Rate of change temperature in one second

$$= (\rho\, \delta x\, A) \times s \times \frac{\partial u}{\partial t} \quad \text{where } \rho \text{ is density} \quad \cdots\cdots (1)$$

Let R_1 & R_2 be the rate of inflow and outflow of heat in this slab, then the rate of heat in flow at R_1 at a distance x from the origin O.

$$= -kA\left(\frac{\partial u}{\partial x}\right)_x$$

and the rate of heat out flow at R_2 at a distance $x + \delta x$ from the origin O.

$$= -KA\left(\frac{\partial u}{\partial x}\right)_{x+\delta x}$$

where x is a component depending upon the material of the body and k is called thermal conductivity and the negative sign appearing as the heat flow from higher to the lower temperature.

Also the rate of increase of heat in this slab of Area A (cm^2 and thickness δx in one second)

$$= R_1 - R_2$$

$$= -kA\left(\frac{\partial u}{\partial x}\right)_x + kA\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} \quad \ldots\ldots (2)$$

from eq^n (1) & (2)

$$(\rho\,\delta x\,\cancel{A}) \times s \times \frac{\partial u}{\partial t} = k\cancel{A}\left[\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right]$$

$$\delta x\,\frac{\partial u}{\partial t} = \frac{k}{s\rho}\left[\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right]$$

$$\frac{\partial u}{\partial t} = \frac{k}{s\rho}\,\frac{\left[\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right]}{\delta x}$$

Taking limit on both sides as $\delta x \to 0$

$$\boxed{\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}} \quad \text{where } c^2 = \frac{k}{s\rho}$$

is the one dimensional heat eq^n.

V.V. Imp 2016 Spring (4b) or Ex 8·3 Q no. 5 (ii)

Q Find the temperature distribution in a laterally insulated thin copper bar ($c^2 = 1.158\ cm^2/sec$), 100 cm long and of constant thickness whose end points at $x = 0$ and $x = 100$ are kept at 0°c and initial temperature is $f(x) = \sin^3 (0.01)\pi x$.

Soln:

We have, one dimensional heat eqn is.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \cdots\cdots (1)$$

The most general soln of one dimensional heat eqn is: $u(x,t) = \sum_{n=1}^{\infty} A_n e^{-\frac{c^2 n^2 \pi^2}{L^2} t} \sin \frac{n\pi}{L} x \cdots\cdots (2)$

Since $c^2 = 1.158\ cm^2$ & $L = 100$

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-\frac{1.158}{10000} n^2 \pi^2 t} \sin \frac{n\pi}{100} x$$

$$= A_1 e^{-0.000115\pi^2 t} \sin 0.01\pi x$$

$$+ A_2 e^{-0.000463\pi^2 t} \sin 0.02\pi x + A_3 e^{-0.00104\pi^2 t} \cdot \sin 0.03\pi x + \cdots\cdots \quad (3)$$

Also we have,

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{100} x \qquad \boxed{f(x) = u(x,0)}$$

$$\sin^3 (0.01)\pi x = A_1 \sin 0.01\pi x + A_2 \sin 0.02\pi x + A_3 \sin 0.03\pi x + \cdots\cdots$$

or, $\frac{3}{4} \sin 0.01\pi x - \frac{1}{4} \sin 0.03\pi x = A_1 \sin 0.01\pi x + A_2 \sin 0.02\pi x + A_3 \sin 0.03\pi x + \cdots\cdots$

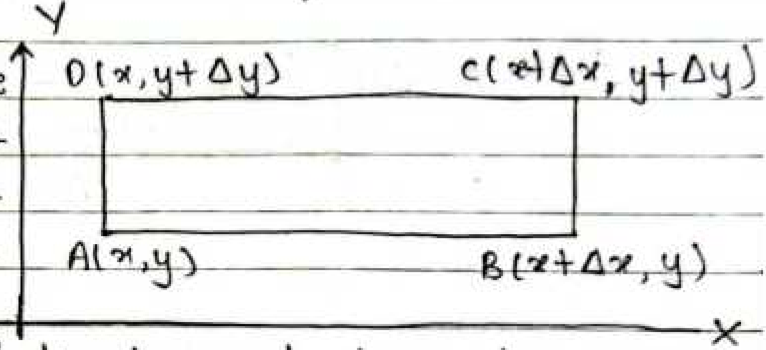
Equating the coeff. of like terms,

$A_1 = \frac{3}{4}$, $A_2 = 0$, $A_3 = -\frac{1}{4}$

Sobstitoting the value of $A_1, A_2, A_3 \dots$ in eq^n (3)

$$u(x,t) = \frac{3}{4} e^{-0.000115\pi^2 t} \sin 0.01\pi x - \frac{1}{4} e^{-0.00104 t\pi^2} \sin 0.03\pi x + \dots$$

Derive two dimensional heat equation



Let us suppose that the flow of heat in a metal plate in XOY plane if the temperature at any point independent of z-coordinate and depends on x, y and time t only, then the flow is called two dimensional heat flow lies XOY plane only and is zero along the normal to the XOY plane. Take a rectangular element of the plate with sides Δx & Δy and thickness α. Then the quantity of heat that enters the plate per second from the side AB & AD is given by $-k\alpha\, \Delta x \left(\frac{\partial u}{\partial y}\right)$ and $-k\alpha\, \Delta y \left(\frac{\partial u}{\partial x}\right)$ respectively and that which flow out through the sides CD & BC persecond is

$$-k\alpha\, \Delta x \left(\frac{\partial u}{\partial y}\right)_{y+\Delta y} \quad \& \quad -k\alpha\, \Delta y \left(\frac{\partial u}{\partial x}\right)_{x+\Delta x}$$

respectively.

Therefore, the total gain of heat by the rectangular plate ABCD per second is

$$= \text{inflow} - \text{outflow}$$

$$= -k\alpha \Delta x \left(\frac{\partial u}{\partial y}\right)_y - k\alpha \Delta y \left(\frac{\partial u}{\partial x}\right)_x + k\alpha \Delta x \left(\frac{\partial u}{\partial y}\right)_{y+\Delta y} + k\alpha \Delta y \left(\frac{\partial u}{\partial x}\right)_{x+\Delta x}$$

$$= k\alpha \Delta x \Delta y \left[\frac{\left\{\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right\}}{\Delta x} + \frac{\left\{\left(\frac{\partial u}{\partial y}\right)_{y+\Delta y} - \left(\frac{\partial u}{\partial y}\right)_y\right\}}{\Delta y}\right] \quad \text{--- ①}$$

Also, we know the rate of gain of heat by the plate is

$$s\rho \, \Delta x \, \Delta y \, \alpha \frac{\partial u}{\partial t} \cdots\cdots ②$$

where s = specific heat

ρ = density of metal plate

from ① & ②

$$s\rho \cancel{\Delta x} \cancel{\Delta y} \cancel{\alpha} \frac{\partial u}{\partial t} = k\cancel{\alpha} \cancel{\Delta x} \cancel{\Delta y} \left[\frac{\left\{\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right\}}{\Delta x} + \frac{\left\{\left(\frac{\partial u}{\partial y}\right)_{y+\Delta y} - \left(\frac{\partial u}{\partial y}\right)_y\right\}}{\Delta y}\right]$$

Taking limit as $\Delta x \to 0$, $\Delta y \to 0$ on both sides,

$$s\rho = \frac{\partial u}{\partial t} = k \; \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

or, $\frac{\partial u}{\partial t} = \frac{k}{s\rho}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$

or, $\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$

where, $c^2 = \frac{k}{s\rho}$

This eq^n is said to be two dimensional heat equation.

Imp
ant Que

2024 Spring

6b) Express $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar coordinates.

Soln

We know Laplacian in cartesian co-ordinate is

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \quad \cdots\cdots (1)$$

We have to change it into polar coordinate.

put $x = r\cos\theta$, $y = r\sin\theta$

i.e. $u = u(r, \theta)$

Diff it w.r to x

$$u_x = u_r r_x + u_\theta \theta_x$$

Again, diff it p.w.r to x

$$u_{xx} = (u_r r_x)_x + (u_\theta \theta_x)_x$$

$$= (u_r)_x r_x + u_r r_{xx} + (u_\theta)_x \theta_x + u_\theta \theta_{xx}$$

$$\frac{\partial^2 u}{\partial x^2} = (u_{rr} r_x + u_{r\theta}\theta_x) r_x + u_r r_{xx} + (u_{\theta r} r_x + u_{\theta\theta}\theta_x)\theta_x + u_\theta \theta_{xx} \quad (2)$$

We have, $r^2 = x^2 + y^2$, Diff p.w.r to x

$$2r\frac{\partial r}{\partial x} = 2x + 0$$

$$\text{or, } \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\text{i.e. } r_x = \frac{x}{r}$$

again, diff it p.w.r to x.

$$\text{or, } r_{xx} = \frac{r \cdot 1 - x \dfrac{\partial r}{\partial x}}{r^2}$$

$$= \frac{r - x \cdot \dfrac{x}{r}}{r^2}$$

$$\therefore r_{xx} = \frac{r^2 - x^2}{r^3}$$

$$= \frac{\cancel{x^2} + y^2 - \cancel{x^2}}{r^3}$$

$$= \frac{y^2}{r^3}$$

and also we have, $\theta = \tan^{-1}\dfrac{y}{x}$

$$\text{or, } \tan\theta = \frac{y}{x}$$

Diff p.w.r to x

$$\text{or, } \sec^2\theta \cdot \frac{\partial \theta}{\partial x} = \frac{-y}{x^2}$$

$$\text{or, } \frac{\partial \theta}{\partial x} = \frac{-y}{x^2} \frac{1}{\sqrt{1+\tan^2\theta}}$$

$$\text{or, } \theta_x = \frac{-y}{x^2} \frac{1}{\sqrt{1+\dfrac{y^2}{x^2}}}$$

$$\text{or, } \theta_x = \frac{-y}{\cancel{x^2}\,\frac{x^2+y^2}{\cancel{x^2}}}$$

$$\text{or, } \theta_x = \frac{-y}{r^2}$$

$$\text{or, } \theta_x = -y\left(\frac{r^{-1}}{\cancel{r}}\right)$$

Diff p.w.r to x

$$\text{or, } \theta_{xx} = -y\,\frac{\partial}{\partial x}(r^{-2})$$

$$= -y(-2)r^{-3}\,\frac{\partial r}{\partial x}$$

$$= \frac{2y}{r^3}\,\frac{x}{r}$$

$$\therefore \theta_{xx} = \frac{2xy}{r^4}$$

from ②,

$$\frac{\partial^2 u}{\partial x^2} = \left\{u_{rr}\cdot\frac{x}{r} + u_{r\theta}\left(\frac{-y}{r^2}\right)\right\}\frac{x}{r} + u_r\frac{y^2}{r^3} + \left\{u_{\theta r}\frac{x}{r} + u_{\theta\theta}\left(\frac{-y}{r^2}\right)\right\}\left(\frac{-y}{r^2}\right) + u_\theta\frac{2xy}{r^4}$$

$$\text{or, } \frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2}u_{rr} - \frac{xy}{r^3}u_{r\theta} + u_r\frac{y^2}{r^3} - \frac{xy}{r^3}u_{\theta r} + \frac{y^2}{r^4}u_{\theta\theta} + \frac{2xy}{r^4}u_\theta$$

$$\text{or, } \frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2}u_{rr} - \frac{2xy}{r^3}u_{r\theta} + u_r\frac{y^2}{r^3} + \frac{2xy}{r^4}u_\theta + \frac{y^2}{r^4}u_{\theta\theta} \quad \ldots\ldots ③$$

Similarly,

$$\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{r^2} u_{rr} + \frac{2xy}{r^3} u_{r\theta} + \frac{x^2}{r^3} u_r - \frac{2xy}{r^4} u_\theta$$
$$+ \frac{x^2}{r^4} u_{\theta\theta} \cdots\cdots (4)$$

Adding (3) & (4).

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \left(\cancel{\frac{x^2+y^2}{r^2}}\right) u_{rr} + \left(\frac{x^2+y^2}{r^3}\right) u_r +$$
$$\frac{x^2+y^2}{r^4} u_{\theta\theta}$$

$$\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta},$$

this is said to be laplacian of u in polar form

Similarly,

the laplacian of u in ~~ta~~ cylindrical ~~form~~ coordinate is $\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz}$

The two dimensional heat eq^n is $\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$ —— ①

In the steady state condition u is independent of "t" so that $\frac{\partial u}{\partial t} = 0$ then ① becomes $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, which is called two dimensional laplace's eq^n.

Solution of Laplace eq^n.

The laplace equation in two dimensions is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \ldots\ldots ①$$

Let $u = X(x)\,Y(y)$ ....② be the sol^n of eq^n ①

Then eq^n ① becomes, $\frac{\partial^2 (XY)}{\partial x^2} + \frac{\partial^2 (XY)}{\partial y^2} = 0$

$$\text{or, } X''Y + XY'' = 0$$

$$\text{or, } \frac{X''}{X} = \frac{-Y''}{Y}$$

It is true for same constant.

$$\frac{X''}{X} = \frac{Y''}{Y} = k \text{ (say)}$$

$$\text{i.e. } \frac{X''}{X} = k \quad \& \quad \frac{Y''}{Y} = -k$$

Case I: When $k < 0$ so that put $k = -p^2$.

$$\frac{X''}{X} = -p^2 \quad \& \quad \frac{Y''}{Y} = p^2$$

or, $X'' + p^2 X = 0$

Its auxillary eq^n is

$m^2 + p^2 = 0$

or, $Y'' - p^2 Y = 0$

Its auxillary eq^n is

$m^2 - p^2 = 0$

$m = \pm ip$ $\qquad\qquad$ $m = \pm p$

$\therefore X(x) = (c_1 \cos px + c_2 \sin px)$ $\qquad$ $\therefore Y(y) = c_3 e^{py} + c_4 e^{-py}$

from ② $u(x,y) = \underbrace{(c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py})}_{③}$

Case II: When $k>0$, so that put $k = p^2$

$\frac{X''}{X} = p^2$ $\qquad$ $\frac{Y''}{Y} = -p^2$

or, $m = \pm p$

$\therefore X(x) = (c_5 e^{px} + c_6 e^{-px})$ $\qquad$ or, $m = \pm ip$

$\therefore Y(y) = (c_7 \cos py + c_8 \sin py)$

from ②,

$u(x,y) = \underbrace{(c_5 e^{px} + c_6 e^{-px})(c_7 \cos py + c_8 \sin py)}_{④}$

Case III: When $k = 0$,

$\frac{X''}{X} = 0$ $\quad$ & $\quad$ $\frac{Y''}{Y} = 0$

or, $X'' = 0$ $\qquad$ or, $Y'' = 0$

or, $X' = c_9$ $\qquad$ $\therefore Y = c_{11} y + c_{12}$

$\therefore X = c_9 x + c_{10}$

from ②,

$u(x,y) = (c_9 x + c_{10})(c_{11} y + c_{12})$ —— ⑤

$\therefore$ The possible solution are ③, ④ and ⑤ of these, we have to choose one of the solⁿ ③ or ④ which is consistant with the physical nature of the problem.

Q. Solve, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the conditions

$u(0,y) = u(L,y) = u(x,0) = 0$ & $u(x,a) = \sin\frac{n\pi x}{L}$.

(Two dimensional ~~heat~~ Laplace eqn in rectangular boundary)

Soln

The Laplace equation in two dimension is:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ —— (1)}$$

Let $u = X(x)\, Y(y)$ —— (2) be the solution of eq (1),

Then eqn (1) becomes,

$$\frac{\partial^2 (XY)}{\partial x} + \frac{\partial^2 (XY)}{y^2} = 0$$

$$\text{or, } X''Y + XY'' = 0$$

$$\text{or, } \frac{X''}{X} = \frac{-Y''}{Y}$$

It is true for semi constant

$$\frac{X''}{X} = \frac{-Y''}{Y} = k \text{ (say)}$$

Let $\frac{X''}{X} = k$ & $\frac{Y''}{Y} = -k$

Case I: $k < 0$ so that put $k = -p^2$

$\frac{X''}{X} = -p^2$ & $\frac{Y''}{Y} = p^2$

or, $X'' + p^2 X = 0$ or, $Y'' - p^2 Y = 0$

Its auxillary eqn is: Its auxillary eqn is:

$m^2 + p^2 = 0$ $m^2 - p^2 = 0$

$\therefore m = \pm ip$ $\therefore m = \pm p$

$\therefore X(x) = (c_1 \cos px + c_2 \sin px)$ $\therefore Y(y) = (c_3 e^{py} + c_4 e^{-py})$

from ②,

$u(x,y) = (c_1 \cos px + c_2 \sin px)\ (c_3 e^{py} + c_4 e^{-py})$

Case II : when $k>0$, so that put $k = p^2$

$\frac{X''}{X} = p^2$ $\qquad\qquad$ $\frac{Y''}{Y} = -p^2$

$\therefore X(x) = (c_5 e^{px} + c_6 e^{-px})$ $\qquad$ $\therefore Y(y) = (c_7 \cos py + c_8 \sin py)$

from ②,

$u(x,y) = (c_5 e^{px} + c_6 e^{-px})\ (c_7 \cos py + c_8 \sin py)$ — ④

Case III : when $k=0$

$\frac{X''}{X} = 0$ $\qquad\qquad$ $\frac{Y''}{Y} = 0$

or, $X'' = 0$ $\qquad\qquad$ or, $Y'' = 0$

$\therefore X = c_9 x + c_{10}$ $\qquad\qquad$ $\therefore Y = c_{11} y + c_{12}$

from ②

$u(x,y) = (c_9 x + c_{10})\ (c_{11} y + c_{12})$ — ⑤

$\therefore$ The possible solution are ③, ④ & ⑤

of these, we have to choose one of sol^n ③ or ④, which is consistant with the physical nature of the problem.

The feasible sol^n is

$u(x,y) = (c_1 \cos px + c_2 \sin px)\ (c_3 e^{py} + c_4 e^{-py})$ — ⑥

Using the condition $u(0,y) = 0$

or, $0 = c_1 (c_3 e^{py} + c_4 e^{-py})$

$\therefore$, $c_3 e^{py} + c_4 e^{-py} \neq 0$, $c_1 = 0$

from ⑥, $u(x,y) = c_2 \sin px\ (c_3 e^{py} + c_4 e^{-py})$ — ⑦

using the condition $u(L,y) = 0$

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$u(L,y) = c_2 \sin pL \, (c_3 e^{py} + c_4 e^{-py})$

or, $0 = c_2 \sin pL \, (c_3 e^{py} + c_4 e^{-py})$

$\therefore c_2 \neq 0$, $(c_3 e^{py} + c_4 e^{-py}) \neq 0$, $\sin pL = 0$

or, $\sin pL = \sin n\pi$

for $n = 0, 1, 2, \ldots\ldots$

$pL = n\pi$

$p = \frac{n\pi}{L}$

from ⑦,

$$u(x,y) = c_2\left(c_3 e^{\frac{n\pi y}{L}} + c_4 e^{\frac{-n\pi y}{L}}\right) \sin \frac{n\pi}{L} x \quad \text{—— ⑧}$$

Again using condition $u(x,0)=0$ i.e. $u=0$, $y=0$

$0 = c_2(c_3 + c_4) \sin \frac{n\pi}{L} x$

$\therefore c_3 + c_4 = 0$ & $c_2 \neq 0$, $\sin \frac{n\pi}{L} x \neq 0$

or, $c_4 = -c_3$

from ⑧,

$$u(x,y) = c_2 c_3 \left(e^{\frac{n\pi}{L} y} - e^{\frac{-n\pi}{L} y}\right) \sin \frac{n\pi}{L} x \quad \text{—— ⑨}$$

using $u(x,a) = \sin \frac{n\pi}{L} x$ i.e. $y = a$, $u = \sin \frac{n\pi}{L} x$

or, $1 \cdot \sin \frac{n\pi}{L} x = c_2 c_3 \left(e^{\frac{n\pi}{L} a} - e^{\frac{-n\pi}{L} a}\right) \sin \frac{n\pi}{L} x$

Equating the coefficient of like term

or, $1 = \left(e^{\frac{n\pi}{L} a} - e^{\frac{-n\pi}{L} a}\right) c_2 c_3$

$$\therefore c_2 c_3 = \frac{1}{e^{\frac{n\pi}{L} a} - e^{\frac{-n\pi}{L} a}}$$

from (9), $u(x,y) = \frac{1}{e^{\frac{n\pi}{L}a} - e^{-\frac{n\pi}{L}a}} \left(e^{\frac{n\pi}{L}y} - e^{-\frac{n\pi}{L}y}\right) \sin\frac{n\pi x}{L}$

$$u(x,y) = \frac{\sinh\frac{n\pi}{L}y \quad \sin\frac{n\pi}{L}x}{\sinh\frac{n\pi}{L}a} \qquad \left[\because \sinh\frac{n\pi}{L}y = e^{\frac{n\pi}{L}y} - e^{-\frac{n\pi}{L}y}\right]$$